

The spectrum of a compact internal space. I. Gauge structure and fermion content

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ABSTRACT: This paper develops a spectral approach to the gauge structure and the mass gap of a quantum field theory, working within a specific framework. The basic object is a discrete spectrum with multiplicities, together with a positive gap parameter. Physical quantities are not defined as expectation values with respect to a measure. Instead, they are defined as convergent spectral sums over this spectrum. The spectrum is not fixed by hand. It arises from a single principle, the disorder axiom, which states that a minimal, unbiased, spin-1/2 change exists. To this principle one must add a set of explicit framework conventions. From these ingredients the internal-space geometry, the gauge algebra, and the spacetime dimension are constructed or selected. The gauge field is then built with a colour-singlet sector in which a scalar order parameter acquires a positive mass through condensation. This mass is a colour-singlet mass of the framework at the emergence scale. It is the mass-gap parameter of the framework, the positive spectral lower bound of the colour-singlet sector. The spectral framework is made explicit by treating the spectral datum, the spectral-sum definition of physical quantities, and a criterion for a datum to describe a quantum field theory as axioms. The standard quantum kinematics is then located within the formulation. This includes the state space, the Born rule, unitary evolution, the tensor-product structure, and locality. Encoding theorems connect the spectral datum to the field-theoretic structure. The non-perturbative content admits a spectral-sum representation carried by a structural bridge that does not go through a measure. The relation to the standard measure-theoretic semantics is stated precisely. On the free sector the two semantics coincide. On the interacting sector they are complementary in their bridges, not in their content. Throughout the paper the status of each ingredient is indicated explicitly. It may be a theorem, a definition, a convention, or an assumption.

KEYWORDS: Spectral datum; Scale flow; Gauge structure; Mass gap; Disorder axiom; Encoding; Osterwalder–Schrader reconstruction; Spectral sums

Contents

1	Introduction	2
2	Preliminaries	3
2.1	Wightman axioms	3
2.2	Maximum entropy principle	4
2.3	Representation theory preliminaries	4
3	The Disorder Axiom	5
4	Emergent Structures	8
4.1	The minimal internal space	8
4.2	The gauge algebra	9
4.3	Fermion content	11
4.4	Content symmetries	11
4.5	$\text{Cl}(4)$ and spacetime dimension 4	12
4.6	Scale field, equi- σ surfaces, and measure	13
5	The axioms	16
5.1	The spectral datum	17
5.2	Formulation criterion and status of the axioms	19
5.3	Gaussian measure and spacetime dimension from the axioms	20
6	Kinematics	21
6.1	Probability and the Born rule	21
6.2	State space and superposition	21
6.3	Composite systems and the tensor product	23
6.4	Field operators and non-commutativity	23
6.5	Spin and statistics	24
6.6	Time evolution	24
6.7	Energy quantisation	25
6.8	The measurement process	25
6.9	Location of the quantum postulates	28
7	Mapping relations	28
7.1	Analytic mapping: analytic control, positivity, condensation, and gap	29
7.2	Structural mapping: two-point and composite quantities	31
7.3	Effective Hilbert space of the condensed sector	33
7.4	Thermal states	34
7.5	Scattering	36

8	Construction of the gauge field	37
8.1	Gauge field and scale semigroup	38
8.2	Measure construction	39
8.3	Technical prerequisites	40
8.4	Scale flow and consistent family	42
8.5	Spectral-sum-definition conditions and reconstruction	45
8.6	Positive scalar lower bound	46
8.7	Endpoint physical quantities	51
8.8	Vacuum, spin–statistics, and summary	52
9	The formal correspondence and the non-perturbative content	54
9.1	The formal correspondence without a measure	55
9.2	Non-perturbative content as spectral-sum representation	56
10	The spectral-sum definition and the standard path integral formulation	58
10.1	The two formulations	58
10.2	Part 1: The non-trivial content	59
10.3	Part 2: The free-sector realisation	60
10.4	Relation to the standard formulation	64
11	Conclusion	65
A	Spectral analysis: long-root mode, Dirac spectrum, and the emergence window	67
A.1	$\text{Spin}(4)$ harmonics on S^3 and RP^3	67
A.2	The long-root mode	67
A.3	The Dirac spectrum of RP^3	68
A.4	The emergence window and the generation count	68
B	The product-space action: one-loop matching and Kaluza–Klein decoupling	68
B.1	The product-space action	69
B.2	The one-loop matching theorem	70
B.3	Kaluza–Klein decoupling	71
B.4	The three technical conditions	73
B.5	Status of the correspondence	73
C	Recovery of the free sector	73
C.1	The candidate Schwinger functions	73

1 Introduction

A quantum field theory admits a description in which physical content is carried by spectral data. This description is classical and uncontroversial. The Källén–Lehmann representation expresses the two-point function as an integral of free propagators against a positive spectral density [1, 2]. A theory with a discrete spectrum is therefore encoded in a set of masses together with their multiplicities. The present paper studies gauge structure and a positive scalar mass parameter from this spectral point of view.

The use of spectral data as a starting point has several precedents. In noncommutative geometry, the spectral action principle of Chamseddine and Connes [3] derives physical actions from the spectrum of a Dirac operator. The present framework differs from that approach in two ways. It uses the discrete spectrum of a compact internal space, and it defines physical quantities directly as convergent spectral sums rather than through an action functional. It also shares structural features with the functional renormalization group description of effective potentials [4, 5], although no path integral is assumed here.

The starting point is a single principle. Section 3 states the disorder axiom, which asserts the existence of a minimal, unbiased, spin-1/2 change. The axiom is deliberately minimal, and its content is made explicit through three clauses. From this axiom and a set of stated framework conventions, Section 4 constructs or selects the internal-space geometry, the gauge algebra, and the spacetime dimension. The outcome is a discrete spectrum with multiplicities together with a spectral datum and a specific gauge structure. Section 8 then constructs a gauge field in the framework. In its colour-singlet sector, a scalar order parameter acquires a positive mass through condensation. This mass is a colour-singlet mass of the framework at the emergence scale. It is the mass-gap parameter of the framework, the positive spectral lower bound of the colour-singlet sector; its relation to the standard gauge-sector mass gap is discussed in Section 10. These three sections contain the main constructions of the paper. The spectral datum is their common output and input.

The division between derivation and compatibility deserves a brief clarification. The disorder axiom supplies the spinor structure and the statistical principle from which the internal geometry and the gauge algebra are selected. The fermion representations and their hypercharge assignments, however, are not derived from the axiom alone. They are the unique representation content compatible with the spectral datum and the chiral clause of the axiom. The counting identities $N_L = N_g = 8$, $N_R = 7$, $\sum Y^2 = 10/3$ are therefore not inputs but consequences of this compatibility condition. This is the same logic that underlies the finite-algebra classification in noncommutative geometry: the algebra $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ is selected as the unique solution to a set of representation-theoretic constraints, not assumed. The present framework adopts that classification as a consistency requirement rather than as an independent derivation, and records the numerical content that follows from it.

The remainder of the paper formalises the spectral point of view that these sections exemplify. Section 5 states the axioms of the spectral framework, including the spectral datum, the definition of physical quantities as convergent spectral sums, and a criterion for a

datum to describe a quantum field theory. Section 6 locates the standard quantum kinematics within the formulation. This includes the state space, the Born rule, unitary evolution, the tensor-product structure, and locality. Section 7 states the mapping relations, which relate the spectral datum to the resulting field-theoretic structure. Section 9 records the structural results, including a spectral-sum representation of the non-perturbative content and a formal correspondence that does not pass through a measure. Section 10 states the relation of the framework to the standard measure-theoretic formulation. Section 11 concludes.

A further remark on language is useful. In four dimensions the conventional measure-based construction of an interacting scalar sector is known to have a trivial continuum limit. This is a theorem, not a conjecture. The present paper does not attempt to improve or circumvent that construction. It simply organises the relevant quantities directly at the level of the spectrum. This spectral organisation allows one to define and study certain objects without relying on a measure-based limit. Whether such definitions carry independent physical interest is left to the reader to judge.

Two remarks fix the scope and the status of the ingredients. First, the paper presents a perspective rather than a claim to have settled an outstanding problem. The gauge structure and the scalar mass parameter are examined within a spectral formulation. The status of every ingredient is indicated explicitly. A statement that is a theorem of the formulation is marked as such. A structure that is carried by the datum, by a convention, or by a standard theorem of analysis or of quantum field theory is recorded rather than re-derived. Second, the relation to the standard measure-theoretic formulation is stated as a relation between two formulations, each complete for the content it carries. The precise locus of their difference is the triviality theorem. For a modern account of that theorem in the four-dimensional scalar case, see Duminil-Copin [6]. The theorem itself is recalled in Section 10.

2 Preliminaries

This section recalls some standard mathematical facts. It includes the Wightman axioms, the maximum entropy principle, and the relevant representation theory of $\text{Spin}(4) \cong SU(2) \times SU(2)$.

2.1 Wightman axioms

Let H be a Hilbert space, let $U(a, \Lambda)$ be a unitary representation of the Poincaré group $\mathcal{P}_+^\uparrow$ on H , and let $\Omega \in H$ be a unit vector. Denote by $\mathcal{S}(\mathbb{R}^4)$ the Schwartz space.

Definition 2.1 (Wightman axioms [7]). *A family $\{\varphi_i\}_{i \in I}$ of operator-valued distributions on $\mathbb{R}^4$, with $\varphi_i : \mathcal{S}(\mathbb{R}^4) \rightarrow \mathcal{L}(H)$, is said to satisfy the Wightman axioms if the following conditions hold.*

- (W1) *There exists a strongly continuous unitary representation $U(a, \Lambda)$ of the Poincaré group on H that satisfies the spectral condition stated in (W3).*

- (W2) Each $\varphi_i(f)$ is a densely defined operator on H whose closure is a closed operator. Its domain contains a common dense subspace D such that $D \subset \text{Dom } \varphi_i(f)$, $\varphi_i(f)D \subset D$, and the map $f \mapsto \langle \Psi, \varphi_i(f)\Phi \rangle$ is a tempered distribution on $\mathcal{S}(\mathbb{R}^4)$.
- (W3) The joint spectrum of the energy–momentum operators P^μ , defined by $U(a, 1) = e^{ia_\mu P^\mu}$, is contained in the closed forward light cone $\bar{V}^+ = \{p : p^2 \geq 0, p^0 \geq 0\}$.
- (W4) There exists a unique (up to a phase) U -invariant state Ω , i.e. $U(a, \Lambda)\Omega = \Omega$. The cyclicity of Ω is stated in (W6).
- (W5) If the supports of f and g are spacelike separated, then $[\varphi_i(f), \varphi_j(g)]_\mp = 0$, where the minus sign applies to bosons and the plus sign to fermions (spin–statistics connection).
- (W6) The set $\{\varphi_{i_1}(f_1) \cdots \varphi_{i_n}(f_n)\Omega\}$ spans a dense subspace of H .

Theorem 2.2 (Osterwalder–Schrader reconstruction [8–10]). *Let $\{\phi_i\}$ be a family of Euclidean fields whose Schwinger functions $\{S^{(n)}\}$ satisfy the OS axioms OS0 through OS4 (OS0 distribution property and regularity, OS1 Euclidean invariance, OS2 reflection positivity, OS3 symmetry, OS4 clustering). Then there exist quantum fields $\{\varphi_i\}$ satisfying Wightman axioms W1 through W6 such that their Wightman functions are the analytic continuations of the Schwinger functions. The axioms W1 through W3, W5 and W6 hold without OS4, while W4 follows from OS4.*

2.2 Maximum entropy principle

Let X be a random vector taking values in $\mathbb{R}^d$ with probability density ρ . The differential entropy is defined by

$$H(\rho) = - \int_{\mathbb{R}^d} \rho(x) \log \rho(x) d\mu_x.$$

Theorem 2.3 (Jaynes [11], maximum entropy principle). *Under the constraints of fixed mean and covariance Σ , the unique maximiser of the differential entropy $H(\rho)$ is the Gaussian distribution with covariance Σ .*

A variational proof is given in [11]. For any ρ satisfying the same moment constraints one has $H(\rho) \leq H(\rho_G)$, with equality if and only if $\rho = \rho_G$ almost everywhere.

2.3 Representation theory preliminaries

Let $\text{Spin}(4) \cong SU(2)_L \times SU(2)_R$. Its irreducible representations are labelled by (j_L, j_R) and have dimension $(2j_L + 1)(2j_R + 1)$.

Lemma 2.4 (Minimality). *The minimal non-trivial irreducible representations of $\text{Spin}(4)$ are precisely $(1/2, 0)$ and $(0, 1/2)$, both of dimension two. For every non-trivial irreducible representation ρ one has $\dim \rho \geq 2$, with equality if and only if $\rho \in \{(1/2, 0), (0, 1/2)\}$.*

Proof. The irreducible representations of $SU(2)$ are labelled by $j \in \frac{1}{2}\mathbb{N}$ and have dimension $2j + 1$. The condition $\dim \geq 2$ holds exactly when $j \geq 1/2$, and $\dim = 2$ exactly when $j = 1/2$. Irreducible representations of $\text{Spin}(4)$ are of the form (j_L, j_R) with dimension $(2j_L + 1)(2j_R + 1)$. The minimal non-trivial dimension is two and occurs precisely for $(j_L, j_R) \in \{(1/2, 0), (0, 1/2)\}$. $\square$

Lemma 2.5 (Faithfulness). *The representations $(1/2, 0)$ and $(0, 1/2)$ are not faithful on $\text{Spin}(4)$: each is trivial on one of the two factors. The minimal faithful representation of $\text{Spin}(4)$ is the Dirac representation $(1/2, 0) \oplus (0, 1/2)$, of dimension four.*

3 The Disorder Axiom

This section states the disorder principle and fixes the mathematical objects that follow from it. The aim is to give a precise formulation of a minimal and unbiased change, together with its symmetry group and statistical content.

A change is first understood as an invertible map of a set to itself. The disorder principle then selects a particular kind of change by adding three requirements. The first is existence. The second fixes the quantum unit as spin-1/2. The third requires unbiasedness with respect to the symmetry group of the change.

Definition 3.1 (Change). *Let X be a set. A change on X is a bijection $f : X \rightarrow X$. The identity transformation $f = \text{id}_X$ is not regarded as a change.*

The use of a bijection means that every change remains within X and is reversible. The composition of all changes on a fixed set forms a semigroup. Since every change is a bijection, each change has an inverse, and the semigroup is a group. This is the natural symmetry group associated with the notion of change. The disorder principle stated below imposes further conditions on this group.

Definition 3.2 (Unbiasedness). *Let G be a group of transformations on X , and let μ be a probability measure on X . The measure μ is called unbiased with respect to G if it is G -invariant, that is,*

$$\mu(g^{-1}B) = \mu(B)$$

for all $g \in G$ and all measurable sets B .

Unbiasedness means that the statistical description does not favour any particular group element. When combined with the maximum entropy principle (Theorem 2.3), this requirement determines the statistical distribution of the change.

Definition 3.3 (Minimality). *Let H be a finite-dimensional complex Hilbert space, and let $\rho : G \rightarrow GL(H)$ be a faithful continuous representation of a connected compact Lie group G . The pair (G, ρ) is called minimal if*

- (i) ρ is irreducible, that is, H has no non-trivial G -invariant subspace, and
- (ii) among all such pairs with irreducible ρ on the given carrier, $\dim G$ is minimal.

A change whose symmetry group acts on its carrier as a minimal pair is called a minimal change. Condition (i) formalises indecomposability. An invariant subspace would describe a sub-change, so a reducible action factorises the change into independent parts. Condition (ii) is an economy criterion. It selects the smallest symmetry group consistent with (i). The carrier and the group are the data of the minimisation problem, and the order is the dimension of the group.

Axiom 3.4 (Disorder Principle). *There exists a minimal, unbiased, spin-1/2 change.*

To make its mathematical content explicit, the axiom is unfolded into three clauses.

(A1) **Existence.** A change exists.

(A2) **Quantum unit.** The change takes spin-1/2 as its unit. The carrier of the change is a two-dimensional complex Hilbert space. The symmetry group of the change is minimal in the sense of Definition 3.3. By Lemma 3.5 this selects $SU(2) \cong \text{Spin}(3)$ with its two-dimensional fundamental representation. The two-dimensional complex representation carries a chiral structure, meaning that left- and right-handed components are distinguished. Its mathematical realisation through the two inequivalent minimal representations of $\text{Spin}(4)$ is given in Section 4.

(A3) **Unbiasedness.** The change is unbiased with respect to its symmetry group. By the maximum entropy principle (Theorem 2.3), its statistical distribution is uniquely determined to be Gaussian.

Lemma 3.5 (Symmetry group of the change). *Let $\rho : G \rightarrow GL(2, \mathbb{C})$ be a faithful, continuous, irreducible representation of a connected compact Lie group G on the two-dimensional carrier of the change. Then G is isomorphic to $SU(2)$ or to $U(2)$. The minimal such group (Definition 3.3) is $SU(2) \cong \text{Spin}(3)$, acting by its two-dimensional fundamental representation.*

Proof. By clause (A2) the carrier is a two-dimensional complex Hilbert space, and the change preserves its unitary structure. Hence $G \subseteq U(2)$, a connected compact Lie group. Irreducibility excludes every abelian group. An abelian compact group has only one-dimensional irreducible complex representations, so a two-dimensional representation of an abelian group is necessarily reducible. In particular $U(1)$ is excluded, despite having the faithful two-dimensional representation $z \mapsto \text{diag}(1, z)$. The failing property is reducibility, not faithfulness.

For a non-abelian connected compact group with a two-dimensional faithful irreducible representation, the decomposition

$$G = (T^k \times G_1 \times \cdots \times G_r)/Z$$

into a central torus and simple factors forces $r = 1$. The irreducible representation is a tensor product of irreducible representations of the factors, whose dimensions multiply to 2. Exactly one simple factor is therefore non-trivial, and it must admit a two-dimensional

faithful irreducible representation. The only simple Lie group with a two-dimensional faithful irreducible representation is $SU(2)$. Its fundamental representation is faithful and irreducible. The group $SO(3)$ has only odd-dimensional irreducible representations ($2j+1$, $j \in \mathbb{Z}_{\geq 0}$), and every simple group of rank at least 2 has minimal representation dimension at least 3. The central torus contributes only scalars, extending $SU(2)$ to $U(2) \cong (U(1) \times SU(2))/\mathbb{Z}_2$ without reducing the dimension. The candidates are therefore $SU(2)$, with $\dim G = 3$, and $U(2)$, with $\dim G = 4$. By condition (ii) of Definition 3.3 the minimal group is $SU(2) \cong \text{Spin}(3)$, whose fundamental representation has dimension 2. $\square$

Remark 3.6 (The three clauses). The three clauses of Axiom 3.4 are mutually independent. Existence does not imply a specific symmetry group. The quantum unit does not imply a statistical description. Unbiasedness does not imply an algebraic structure of the change. The spin quantum number $1/2$ in clause (A2) is an input of the axiom. It is realised by the fundamental representation of $SU(2)$, which has spin label $1/2$ and is faithful.

The fundamental representation of $SU(2)$ is pseudoreal. It is equivalent to its complex conjugate, the equivalence being realised by the antisymmetric tensor of $SU(2)$. A left-right chiral distinction therefore cannot arise within $SU(2)$ alone. The chiral structure of clause (A2) is realised at the level of the full symmetry group of the carrier space. The carrier sphere S^3 has isometry group $O(4)$ with identity component $\text{Spin}(4) \cong SU(2)_L \times SU(2)_R$. The two factors carry the two inequivalent minimal representations $(1/2, 0)$ and $(0, 1/2)$. This is the mathematical reason for introducing $\text{Spin}(4)$ in Section 4.

Theorem 3.7 (Quantum unit theorem). *Under the disorder axiom, the symmetry group of the change is $SU(2)$, and the carrier is its two-dimensional fundamental representation. The spin label $1/2$ of clause (A2) is the spin quantum number of this representation.*

Proof. By Lemma 3.5 the minimal symmetry group of the two-dimensional carrier is $SU(2) \cong \text{Spin}(3)$, acting by its fundamental representation. The irreducible representations of $SU(2)$ are labelled by $j \in \frac{1}{2}\mathbb{N}$ and have dimension $2j+1$. The fundamental representation has $j = 1/2$, dimension 2, and is faithful (its kernel is $\{e\}$). The unit $1/2$ of clause (A2) is therefore realised as the spin label of the minimal representation. It is an input of the axiom, not a derived quantity. The angular momentum operators on the carrier space satisfy the $\mathfrak{su}(2)$ commutation relations, and their smallest absolute eigenvalue is $1/2$. $\square$

Lemma 3.8 (Clifford semispinor dimensions). *Let $n \in \mathbb{N}$. The complexification of the Euclidean Clifford algebra $\text{Cl}(n)$ satisfies*

$$\text{Cl}(n) \otimes_{\mathbb{R}} \mathbb{C} \cong \begin{cases} M_{2^{\lfloor n/2 \rfloor}}(\mathbb{C}) & \text{if } n \text{ is even,} \\ M_{2^{\lfloor n/2 \rfloor}}(\mathbb{C}) \oplus M_{2^{\lfloor n/2 \rfloor}}(\mathbb{C}) & \text{if } n \text{ is odd.} \end{cases}$$

The spinor module has complex dimension $2^{\lfloor n/2 \rfloor}$. A chiral operator $\Gamma = \gamma_1 \cdots \gamma_n$ exists if and only if n is even, $n = 2m$, and then the spinor module decomposes into two semispinor modules, each of complex dimension 2^{m-1} . For odd n no chiral decomposition exists.

Remark 3.9 (Relative consistency). The consistency of the three clauses of the axiom can be seen from an explicit model. Take $X = S^3$, the unit sphere in the two-dimensional complex carrier space. Take f a fixed rotation of S^3 . Take $G = SU(2)$ acting on X by rotations. Take μ the round maximum-entropy measure. Then (A1) holds because f is a bijection. (A2) holds because the minimal faithful representation of $SU(2)$ on the two-dimensional carrier has dimension 2. (A3) holds because μ is G -invariant and is the maximum-entropy measure on the compact carrier. The disorder axiom is consistent with ZFC.

Remark 3.10 (Origin of the moment constraints). The maximum entropy principle (Theorem 2.3) requires fixing the mean and the covariance. The mean is zero as a direct consequence of unbiasedness (Definition 3.2). When a measure is invariant under $E(4)$, its mean must be translation-invariant and therefore zero. The covariance is a constructive datum of the Gaussian structure. Its spectral representation is defined in Section 4.6. The spectral density $\rho(p^2)$ encodes the full two-point correlation information. The statements of the paper do not depend on its detailed form beyond temperedness.

4 Emergent Structures

4.1 The minimal internal space

Clause (A2) fixes the carrier of the change as a two-dimensional complex spinor space. Its unit sphere is S^3 . The chiral structure of (A2) distinguishes left-handed and right-handed components. That distinction has to be realised geometrically. The change therefore acts on a definite internal space, not on the carrier sphere alone.

The action of $\text{Spin}(4)$ on S^3 is not faithful. Its kernel is the central subgroup generated by $(-1_L, -1_R)$. The antipodal map $z \mapsto -z$ corresponds to the central element $(-1_L, +1_R)$. This element acts as -1 on the left-handed Weyl spinor and as $+1$ on the right-handed one. Thus the chiral structure of (A2) is realised precisely by the antipodal identification.

Definition 4.1 (Minimal internal space). *Let S^3 be the unit sphere in the two-dimensional complex carrier space. A free quotient S^3/Γ is called admissible if the chiral Weyl pair $(1/2, 0) \oplus (0, 1/2)$ descends to the quotient and Γ contains the antipodal element, so that the antipodal identification realises the chiral structure of clause (A2). Among all admissible free quotients, the one with the smallest order $|\Gamma|$ is called the minimal internal space.*

Proposition 4.2. *The minimal internal space is $RP^3 = S^3/\mathbb{Z}_2$.*

Proof. We first verify that $\mathbb{Z}_2$ is admissible. The antipodal map $x \mapsto -x$ acts as the $\text{Spin}(4)$ element $(-1_L, +1_R)$. Under this element, the left-handed spinor changes sign and the right-handed spinor does not. Hence the Weyl pair survives in the quotient $S^3/\mathbb{Z}_2$. Conversely, admissibility requires Γ to contain the antipodal element, which is what distinguishes the two Weyl components. Any admissible Γ therefore contains $\mathbb{Z}_2$. The smallest possible admissible quotient is accordingly $S^3/\mathbb{Z}_2 = RP^3$. The remaining symmetry statements follow from the standard action of $\text{Spin}(4)$ on S^3 . $\square$

4.2 The gauge algebra

The abstract symmetry group $SU(2)$ of axiom (A2) acquires gauge meaning through the $SU(2)_L$ factor in the isometry algebra of RP^3 . The electroweak breaking direction and the colour sector are fixed as follows. The R factor breaks to $\mathfrak{u}(1)_R$ via the condensation mechanism of Theorem 4.6, while $SU(2)_L$ remains unbroken. The minimal colour algebra is selected in Proposition 4.9.

Lemma 4.3. *The isometry algebra of RP^3 is $\mathfrak{so}(4)$.*

Proof. The isometry group of S^3 is $O(4)$. Since the antipodal map lies in the centre of $O(4)$, the isometry group of $RP^3 = S^3/\mathbb{Z}_2$ is $O(4)/\{\pm 1\}$. Its identity component has Lie algebra $\mathfrak{so}(4)$. $\square$

Lemma 4.4. $\mathfrak{so}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$.

Proof. The Hodge star on $\Lambda^2(\mathbb{R}^4)^* \cong \mathfrak{so}(4)$ decomposes the space into the ± 1 eigenspaces Λ_+^2 and Λ_-^2 , each of dimension three. Equipped with the cross-product bracket, each summand is isomorphic to $\mathfrak{su}(2)$. $\square$

Definition 4.5. *The gauge algebra $\mathfrak{g}_{\text{ch}}$ is the subalgebra of $\mathfrak{so}(4)$ retained as gauge symmetry. According to Theorem 4.6, $\mathfrak{su}(2)_L$ survives completely, while $\mathfrak{su}(2)_R$ breaks to its maximal torus $\mathfrak{u}(1)_R$. Thus*

$$\mathfrak{g}_{\text{ch}} = \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_R.$$

Theorem 4.6 (Electroweak breaking direction). *The condensation mechanism of Section 8.6 acts on the R sector. The long-root mode of this sector carries charge 2 under the unbroken torus (Appendix A). Below the critical curvature R_c it becomes tachyonic and condenses, breaking $\mathfrak{su}(2)_R$ to its maximal torus $\mathfrak{u}(1)_R$. The maximal torus theorem ensures uniqueness up to conjugation. The generator of this torus is identified with weak hypercharge.*

Proof. The long-root mode is the mode of the R -sector connection in the representation $(2, 1)$ of $\text{Spin}(4)$, described in Appendix A. Its squared mass equals

$$m_{\text{long}}^2 = \frac{8}{3}(R - R_c),$$

which is negative when $R < R_c$. Condensation of this mode breaks $\mathfrak{su}(2)_R$. The surviving subgroup is the stabiliser of the condensed direction. For a simple rank-one factor this stabiliser is a maximal torus, unique up to conjugation. The identification of the surviving $\mathfrak{u}(1)_R$ generator with weak hypercharge is fixed by the hypercharge assignment derived in Section 4.3. The condensation mechanism itself is established in Section 8.6, while the present theorem only identifies the breaking direction. $\square$

Definition 4.7 (Colour carrier). *Let E_{col} be the space of Kaluza–Klein zero modes of a connection on RP^3 with values in a simple Lie algebra $\mathfrak{g}_{\text{col}}$ distinct from the electroweak factor $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_R$. The natural representation of $\mathfrak{g}_{\text{col}}$ on E_{col} is the adjoint representation induced by the zero-mode expansion.*

Remark 4.8. The existence of a colour factor distinct from the electroweak one is a structural datum of the framework. The disorder axiom fixes the electroweak factor through the isometry algebra of RP^3 , and it does not by itself force the presence of a second simple factor. The colour carrier records this additional factor, and its algebra is fixed by the conformal-gauge duality of Proposition 4.9.

Proposition 4.9 (Colour algebra). *The colour algebra is $\mathfrak{su}(3)$.*

Proof. A scalar field on a curved space of dimension d is conformally invariant only when it couples to the scalar curvature through the term $\xi R\varphi^2$ with the conformal coupling

$$\xi = \frac{d-2}{4(d-1)}.$$

This is the standard Yamabe conformal coupling—the coefficient of the Yamabe operator $\Delta + \xi R$ —fixed uniquely by the conformal requirement, and not a convention of the present framework [12, 13]. By Theorem 4.13 the internal dimension is $d = 3$, so $\xi = 1/8$.

Let N_g denote the number of generators of the colour algebra $\mathfrak{g}_{\text{col}}$ of Definition 4.7. The scalar conformal weight is $\Delta = (d-2)/2$, and the conformal coupling is its normalisation $\xi = \Delta/(2(d-1))$. The conformal structure of the framework is carried by ξ , and the gauge structure by N_g . The two symmetries are complementary faces of one symmetry structure, and their total conformal weight is a conserved unit. This is the conformal-gauge duality

$$N_g \xi = 1,$$

equivalently $N_g \Delta = 2(d-1)$. With $\xi = 1/8$ it fixes $N_g = 8$. Among the simple Lie algebras distinct from the electroweak factor the only one with eight generators is $\mathfrak{su}(3)$, because $\dim \mathfrak{so}(5) = 10$ and $\dim G_2 = 14$. Hence $\mathfrak{g}_{\text{col}} = \mathfrak{su}(3)$, and its eight generators are the eight Kaluza–Klein zero modes of the colour sector. $\square$

Remark 4.10 (Status of the colour selection). The colour algebra is fixed by the conformal-gauge duality rather than by the economy criterion. The economy criterion of Definition 3.3 remains in force for the two other selections of the paper, namely $SU(2)$ over $U(2)$ in Lemma 3.5 and RP^3 over the larger lens spaces in Proposition 4.2. For the colour factor the conformal-gauge duality fixes the generator count $N_g = 8$, and this singles out $\mathfrak{su}(3)$ because it is the only simple Lie algebra of dimension eight.

Definition 4.11 (Constructive data). *The constructive data of the paper consist of the axioms (A1)–(A3), including the chiral structure of clause (A2). The minimality principle of Definition 3.3 selects $SU(2)$ over $U(2)$ in Lemma 3.5 and RP^3 over the larger lens spaces in Proposition 4.2. The conformal-gauge duality $N_g \xi = 1$, an emergent consistency condition on the conformal and gauge structures, fixes the colour algebra in Proposition 4.9. All further data are derived in Sections 4–8 and Appendix A.*

Theorem 4.12 (Gauge group). *The gauge algebra is*

$$\mathfrak{su}(3) \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y.$$

Proof. Lemmas 4.3 and 4.4 give $\mathfrak{so}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$. Definition 4.5 fixes the electroweak part as $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_R$. Proposition 4.9 selects the colour part $\mathfrak{su}(3)$. Taking the direct sum and identifying $\mathfrak{u}(1)_Y$ with the weak hypercharge generator yields the result. $\square$

4.3 Fermion content

The chiral clause (A2) supplies the Weyl pair $(1/2, 0) \oplus (0, 1/2)$ of $\text{Spin}(4)$, as recorded in Remark 3.6. The left-handed component $(1/2, 0)$ is an $\mathfrak{su}(2)_L$ doublet. The colour algebra $\mathfrak{su}(3)$ of Proposition 4.9 acts on the fermions through its fundamental representation $\mathbf{3}$, the smallest non-trivial irreducible representation. Colour is vectorial, so the left-handed fermions carry $\mathbf{3}$ and the right-handed ones carry the conjugate $\bar{\mathbf{3}}$, which cancels the $\text{SU}(3)^3$ anomaly.

The five representations are the minimal combination of chirality and colour. The left-handed doublet carries either colour, giving $Q_L = (3, 2)$, or no colour, giving $L_L = (1, 2)$. The right-handed singlets mirror the doublet members. The coloured members u_L, d_L give $u_R = (3, 1)$ and $d_R = (3, 1)$ in the conjugate representation, the uncoloured member e_L gives $e_R = (1, 1)$, and ν_L has no light right-handed partner. The five representations are

$$Q_L = (3, 2)_{Y_Q}, \quad u_R = (3, 1)_{Y_u}, \quad d_R = (3, 1)_{Y_d}, \quad L_L = (1, 2)_{Y_L}, \quad e_R = (1, 1)_{Y_e}.$$

The hypercharges are not free. The Higgs doublet carries the standard electroweak hypercharge $Y_H = 1/2$, which is taken as the standard electroweak input; the remaining hypercharges are then fixed by anomaly cancellation and the Yukawa structure. The charge convention $Q = T_3 + Y$ gives the Yukawa relations

$$Y_u = Y_Q + Y_H, \quad Y_d = Y_Q - Y_H, \quad Y_e = Y_L - Y_H.$$

The mixed $\text{SU}(2)^2\text{U}(1)$ anomaly vanishes when

$$3Y_Q + Y_L = 0,$$

and the mixed gravitational $\text{U}(1)$ anomaly vanishes when

$$6Y_Q + 3Y_u + 3Y_d + 2Y_L + Y_e = 0.$$

Substituting the Yukawa relations into these two equations gives $3Y_Q - Y_H = 0$, hence $Y_Q = 1/6$, and then $Y_u = 2/3$, $Y_d = -1/3$, $Y_L = -1/2$, $Y_e = -1$. The hypercharge assignment is therefore a consequence of anomaly cancellation and the Yukawa structure, with $Y_H = 1/2$ the single electroweak input.

4.4 Content symmetries

The five representations of Section 4.3 carry a counting symmetry. The left-handed doublets contribute Q_L with $3 \times 2 = 6$ components and L_L with $1 \times 2 = 2$ components, so the left-handed count per generation is

$$N_L = 6 + 2 = 8.$$

The right-handed singlets contribute u_R with 3, d_R with 3, and e_R with 1, so

$$N_R = 3 + 3 + 1 = 7.$$

The left-handed count equals the number of colour generators $N_g = 8$, and the right-handed count is one less. This content-gauge symmetry

$$N_L = N_g, \quad N_R = N_g - 1$$

relates the fermion content to the gauge structure. The chiral asymmetry

$$N_L - N_R = 1$$

is odd, and its parity is the non-trivial element of $\pi_1(RP^3) = \mathbb{Z}_2$. An odd chiral asymmetry selects the non-trivial spin structure of the internal space, whose first cohomology $H^1(RP^3, \mathbb{Z}_2) = \mathbb{Z}_2$ has a single non-trivial class. The hypercharges of Section 4.3 satisfy

$$\sum Y = 0,$$

which is the vanishing of the mixed gravitational anomaly, and their first non-zero moment is

$$\sum Y^2 = \frac{10}{3}.$$

These content identities are exact consequences of the representation content of Section 4.3.

4.5 $\text{Cl}(4)$ and spacetime dimension 4

Theorem 4.13 (Dimension). *Under the disorder axiom, the Clifford algebra carrying the chiral two-dimensional representation of clause (A2) is $\text{Cl}(4)$. Its generators span a real space V of dimension four.*

Proof. Clause (A2) fixes the representation to be the two-dimensional fundamental representation of $SU(2)$ with chiral structure. By Lemma 3.8, a chiral decomposition exists only in even dimensions $n = 2m$. Hence the representation is a semispinor module of some $\text{Cl}(2m)$. The semispinor dimension equation then gives $2^{m-1} = 2$, from which $m = 2$ and $n = 4$. The generators $\gamma_1, \dots, \gamma_4$, satisfying $\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}$, therefore span the four-dimensional real space

$$V = \text{span}\{\gamma_1, \dots, \gamma_4\}.$$

The spin operators

$$S_{\mu\nu} = \frac{i}{4}[\gamma_\mu, \gamma_\nu] \in \text{Cl}(4)$$

generate $\text{Spin}(4)$. Its vector representation on V gives $SO(4)$. The quadratic form defined by the Clifford relation is the Euclidean metric on V . This metric is used in the Osterwalder–Schrader reconstruction of Section 8. Combined with the $3 + 1$ decomposition of Corollary 4.14, it supplies the metric structure of the emergent spacetime. The Lorentzian metric is obtained by analytic continuation. $\square$

Corollary 4.14 (Spacetime dimension $4 = 3 + 1$). *The dimension is $3 + 1$, where three dimensions come from the internal space RP^3 and one from the scale flow parameter σ .*

Proof. At every regular point of σ , the level set Σ_σ is a smooth submanifold of V of codimension one, and its normal direction is spanned by $\nabla\sigma$. Hence at such a point,

$$V = T_x \Sigma_\sigma \oplus \mathbb{R} \nabla\sigma(x),$$

with $\dim T_x \Sigma_\sigma = 3$ and $\dim \mathbb{R} \nabla\sigma = 1$. Therefore $\dim V = 4$, consistent with Theorem 4.13. By Theorem 4.25 and Proposition 4.27, the ensemble geometry of Σ_σ is RP^3 . The three internal dimensions arise there, and the remaining dimension is the scale direction. $\square$

4.6 Scale field, equi- σ surfaces, and measure

The previous subsections fixed the algebraic structure of the theory. They determined the symmetry group, the internal space, the gauge algebra, and the spacetime dimension. None of this fixes the dynamical content. It does not tell us how the change acts at each point and at each scale. The remaining clause of the axiom, unbiasedness (A3), supplies this content statistically. The statistical distribution of the change is the Gaussian fixed by the maximum entropy principle (Theorem 2.3). Realised on the emergent spacetime V , this statistical content becomes a random field. The change sweeps out the scale direction, which is the fourth dimension of Corollary 4.14. The random field is therefore the scale field $\sigma(x)$. The coherence scale of σ is denoted L ; it is both the radius of the internal space RP^3 and the Gaussian integral width of σ , related by $L = \sqrt{2\pi} \sigma$ at the condensation threshold (Theorem 8.27). At a point x , the value $\sigma(x)$ is the local value of the scale parameter. The statistics of σ are described by the unbiased Gaussian structure of (A3). The equi- σ surfaces are the sets $\Sigma_\sigma = \{x : \sigma(x) = \sigma\}$. They are the loci of constant local scale.

Theorem 4.15 (Measure). *A Gaussian probability measure $d\mu_{0,\sigma}$ exists on the configuration space $\mathcal{E} = \mathcal{S}'(V)$ with mean zero and covariance Σ_σ .*

Proof. By (A3) and Definition 3.2 the distribution is G -invariant. With moment constraints, Theorem 2.3 gives the Gaussian distribution. Minlos' theorem [14] extends the Gaussian cylinder measure to $\mathcal{E}$ when Σ_σ is a tempered kernel. $\square$

Lemma 4.16 (Covariance structure). *The covariance kernel of the σ field is radial, $C(x, y) = G(|x - y|)$.*

Proof. The measure is $E(4)$ -invariant by (A3). Translation invariance gives $C(x, y) = G(x - y)$. Rotation invariance gives $G(x - y) = G(R(x - y))$ for every $R \in SO(4)$. Thus G depends only on $|x - y|$. $\square$

Definition 4.17 (Spectral representation of the Gaussian structure). *The covariance kernel $C(x, y) = G(|x - y|)$ admits, by the Bochner–Schwartz theorem, the spectral representation*

$$C(x, y) = \int e^{ip \cdot (x - y)} \rho(p^2) \frac{d^4 p}{(2\pi)^4},$$

where $\rho(p^2) \geq 0$ is a tempered spectral density. Rotational invariance makes ρ a function of p^2 alone. Its explicit form is fixed up to a normalisation constant by the scale-invariant fixed point of Theorem 4.18.

Theorem 4.18 (Conformal two-point structure at a fixed point). *At a scale-invariant fixed point the two-point function takes the conformal form*

$$C(x, y) = \frac{c}{|x - y|^2},$$

with $c > 0$. Equivalently, the momentum-space covariance is proportional to p^{-2} . In the Källén–Lehmann spectral representation the spectral measure is $c\delta(\mu^2)$, describing a single massless mode. The constant c cancels in all ratios of physical quantities.

Proof. At a scale-invariant fixed point the two-point function is scale covariant, $C(\lambda x, \lambda y) = \lambda^{-2\Delta}C(x, y)$, where Δ is the scaling dimension of the free field. For the free Gaussian field associated with the Clifford structure of Theorem 4.13, the dimension is $\Delta = 1$. Scale and rotation invariance then fix the two-point function up to a constant, giving $C(x, y) = c|x - y|^{-2}$. The Fourier transform of $|x - y|^{-2}$ in four dimensions is proportional to p^{-2} . The mass spectral measure is therefore $c\delta(\mu^2)$. The constant c multiplies the covariance linearly and drops out of all ratios defined by the spectral sums of Definition 8.1. $\square$

Lemma 4.19 (Level set dimension). *At a regular point of σ , the level set Σ_σ is a three-dimensional smooth submanifold of V .*

Proof. By the implicit function theorem, $\Sigma_\sigma = \sigma^{-1}(\sigma)$ has codimension one. Since $\dim V = 4$ by Theorem 4.13, $\dim \Sigma_\sigma = 3$. $\square$

The internal space is fixed twice in this paper, and the two routes meet at the same object. The first route is algebraic. In Section 4.1 the chiral structure of clause (A2) required the internal space to be an admissible free quotient of S^3 . The minimal admissible quotient is $S^3/\mathbb{Z}_2$. The second route is statistical. The unbiased statistics of the scale field fixes the effective geometry to be a compact constant-curvature space S^3/Γ . The chiral structure requires Γ to contain the antipodal element. The same economy criterion then selects the smallest admissible Γ , namely $\mathbb{Z}_2$. Thus both routes lead to RP^3 . They are not two independent assumptions. They are two applications of the same economy principle, one to the chiral quotient and one to the effective geometry.

Definition 4.20 (Ensemble geometry). *The ensemble geometry of $\{\Sigma_\sigma\}$ is the spectral geometry of the family. It is the three-manifold whose Laplace spectrum is fixed by the correlation structure of the σ field on Σ_σ . The relevant sector is selected by the chiral structure of (A2). This gives the spectrum $\lambda_l = l(l+2)/L^2$ with multiplicities $d_l = (l+1)^2$ for even l . The ensemble geometry is a statistical object. The individual level sets of the Gaussian field are irregular random surfaces in general. No statement is made about the geometry of any single realisation.*

Proposition 4.21 (Maximal symmetry). *The ensemble geometry is maximally symmetric. Its spectral data are the Laplace spectrum of a constant-curvature space.*

Proof. The spectral data $\lambda_l = l(l+2)/L^2$ with multiplicities $d_l = (l+1)^2$ are the Laplace spectrum of the round sphere S^3 of radius L . The ensemble data are the sector of this spectrum selected by the quotient structure. Every quotient of the round sphere has constant curvature and is therefore maximally symmetric. The specific quotient is selected by the chiral structure and verified by the spectral data. The maximal symmetry is a property of the ensemble geometry, not of the individual realisations. $\square$

Theorem 4.22 (Effective geometry of the level sets at the flow endpoint). *Let σ be the Gaussian scale field of $(A3)$, $E(4)$ -invariant with coherence scale L . Let the coarse-graining flow erase short-distance fluctuations of Σ_σ down to the scale L . Then the effective geometry of the level set Σ_σ at the flow endpoint is isotropic and homogeneous. It has constant curvature, with curvature scale $R \sim 1/L^2$. Its spectral data are those of the round sphere S^3 (Proposition 4.21), so the curvature is positive. The effective geometry is therefore a connected compact maximally symmetric space S^3/Γ [15].*

Proof. The $E(4)$ -invariance of the measure fixes the distributional symmetry of the scale field. After integrating out fluctuations below the coherence scale L , the effective geometry inherits translational and rotational invariance. Hence it is homogeneous and isotropic. A homogeneous and isotropic Riemannian metric has constant curvature. The curvature scale is set by the coherence scale L , so by dimensional analysis $R = c/L^2$. The internal space is compact. A compact constant-curvature three-manifold is spherical, flat, or hyperbolic. The spectral data are those of the round sphere S^3 (Proposition 4.21), so the curvature is positive, which excludes the flat and hyperbolic cases. The effective geometry is therefore a spherical quotient S^3/Γ [15]. $\square$

Remark 4.23 (From the correlation structure to the spectrum). The chain is now explicit. The effective geometry of the level sets is determined by the statistics of the scale field. The spectral data are the Laplace spectrum of this geometry. The chain is

statistics of the scale field $\rightarrow$ effective geometry $\rightarrow$ Laplace spectrum $\rightarrow$ spectral data.

The spectral data are the spectrum of the effective geometry fixed by the statistics.

Lemma 4.24 (Compactness of the internal space). *The internal space is compact.*

Proof. In the spectral-sum definition of Definition 8.1, physical quantities are defined by absolutely convergent spectral sums over the spectrum of the internal Laplacian. This requires the spectrum to be purely discrete. A complete non-compact Riemannian manifold has non-empty essential spectrum. Hence a purely discrete spectrum implies compactness. Compactness also follows directly from Proposition 4.2, since $RP^3 = S^3/\mathbb{Z}_2$ is compact. The present argument shows consistency of the two routes. $\square$

Theorem 4.25 (Classification). *The ensemble geometry is a three-dimensional connected compact maximally symmetric space. It is therefore isomorphic to S^3/Γ , where $\Gamma \subset SO(4)$ is a finite subgroup acting freely. The quotient is selected by the chiral structure and the economy criterion, giving $RP^3 = S^3/\mathbb{Z}_2$. Its spectrum is the even- l sector of the sphere spectrum.*

Proof. Dimension three is Lemma 4.19. Compactness is Lemma 4.24. Maximal symmetry is Proposition 4.21. A connected compact maximally symmetric three-manifold has constant curvature and is of the form S^3/Γ [15]. The chiral structure requires Γ to contain the antipodal element. The economy criterion of Definition 4.1 selects the minimal quotient $\Gamma = \mathbb{Z}_2$. The resulting space is RP^3 . Its spectrum is the even- l sector of the sphere spectrum. $\square$

Remark 4.26 (Ensemble geometry as a spectral object). The ensemble geometry is defined by the spectral data, not by the geometry of individual level sets. The level sets of a stationary isotropic Gaussian field are irregular in general. The maximal symmetry of Proposition 4.21 is a property of the ensemble geometry, not of the realisations. The construction uses only the spectral data of the ensemble. These data are fixed by the $E(4)$ -invariance of the measure through the correlation structure.

Proposition 4.27 (Chiral selection). *The chirality of (A2) together with the economy criterion selects $\Gamma = \mathbb{Z}_2$, so the ensemble geometry is $RP^3 = S^3/\mathbb{Z}_2$.*

Proof. The chiral Weyl pair $(1/2, 0) \oplus (0, 1/2)$ decomposes under the antipodal map on S^3 as ψ_L odd and ψ_R even. This parity distinguishes left from right. For S^3/Γ to carry this chiral pair, the antipodal map must act trivially on the quotient. Hence Γ must contain the antipodal element. Among the free quotients S^3/Γ containing the antipodal identification, the economy criterion of Definition 4.1 selects the smallest order $|\Gamma|$. This gives $\Gamma = \mathbb{Z}_2$. Larger quotients such as the lens spaces $S^3/\mathbb{Z}_{2m}$ with $m \geq 2$ also contain the antipodal element but are excluded by the economy criterion. The resulting space RP^3 has the even- l sector of the sphere spectrum as its Laplace spectrum. $\square$

Remark 4.28 (Status of the chiral quotient selection). The selection of $\Gamma = \mathbb{Z}_2$ uses the same economy criterion that selects $SU(2)$ over $U(2)$ in Lemma 3.5 and $\mathfrak{su}(3)$ over B_2, G_2 in Proposition 4.9. In the present case the criterion is applied to the finite quotient, as stated in Definition 4.1. The statistics alone determine a constant-curvature compact space S^3/Γ . The reduction to RP^3 is driven by clause (A2), which requires Γ to contain the antipodal element, and by the economy criterion, which selects the smallest admissible $|\Gamma|$. The chirality requirement comes from (A2). The minimality step is the economy criterion. No dynamical origin of the economy criterion is claimed here.

Corollary 4.29. *The equi- σ surface ensemble geometry is RP^3 .*

Proposition 4.2 and Corollary 4.29 both give RP^3 . This internal space determines the gauge algebra of Section 4.2 and supplies the spectral data used in Section 8.6.

5 The axioms

This section collects the axioms of the formulation in a self-contained form and states the status of each. Three axioms are definitional. They specify the primitive data, the rule for physical quantities, and the state space. Four further axioms describe the kinematic

structure. They locate the standard rules of quantum mechanics within the formulation. These rules are probability, dynamics, composition, and locality. On the free sector each of them reduces to a standard theorem for free fields. On the interacting sector each of them is a conditional statement within the framework. The final axiom is the mapping axiom. It defines the relation between the spectral datum and the physical structure.

5.1 The spectral datum

Throughout, M denotes a compact three-dimensional Riemannian manifold without boundary, and Δ_M the non-negative scalar Laplacian on M . By the standard spectral theory of elliptic operators on compact manifolds, Δ_M has a discrete spectrum $\{\lambda_l\}_{l \geq 0}$ with finite multiplicities $\{d_l\}_{l \geq 0}$, accumulating only at infinity, and the counting function $N(\lambda) = \#\{l : \lambda_l \leq \lambda\}$ satisfies the Weyl asymptotics $N(\lambda) \sim C \lambda^{3/2}$ as $\lambda \rightarrow \infty$.

Axiom 5.1 (Spectral datum). *The primitive object of the formulation is the spectral datum*

$$S = (M, \{\lambda_l\}, \{d_l\}), \quad (5.1)$$

consisting of the compact three-dimensional Riemannian manifold M and the spectrum $\{\lambda_l\}$ of its scalar Laplacian with multiplicities $\{d_l\}$. No further parameters are introduced by hand.

From the spectrum and the derived parameter m_δ^2 the mass tower is defined as

$$M_l^2 = \lambda_l + m_\delta^2. \quad (5.2)$$

The spectral projector kernels $K_l(y, y')$ of the l -th eigenspace of Δ_M are used throughout. The derivation of m_δ^2 from the spectral datum is given in Theorem 8.32. It does not require any additional input.

Remark 5.2 (Derived parameters). The positive gap parameter m_δ , the quartic coupling λ , and the condensate $\langle E \rangle$ are not independent inputs. They are constructed from the datum in Sections 7 and 8, where their existence and positivity are established. Throughout, m_δ denotes the mass and m_δ^2 its square, the two being used consistently.

Remark 5.3 (Example: the projective space). A concrete instance of the datum is obtained by taking $M = \mathbb{R}P^3$, the real projective three-space of radius L . The scalar Laplacian has spectrum $\lambda_l = l(l+2)/L^2$ with multiplicities $d_l = (l+1)^2$ for even l [16]. The mass tower is then $M_l^2 = l(l+2)/L^2 + m_\delta^2$, with m_δ^2 derived as in Section 8. This example is used throughout the paper to illustrate the general definitions. Nothing in the axioms requires it. The reconstruction theorem of Appendix C holds for any datum satisfying the stated conditions.

Axiom 5.4 (Physical quantities). *A physical quantity is a function of the spectral datum of Axiom 5.1 of the form*

$$Q = \sum_l d_l f(\lambda_l), \quad (5.3)$$

where f is such that the series converges absolutely, or a closed analytic expression in the datum. A closed analytic expression is one obtained from the spectrum and the derived parameters of the datum by finitely many elementary operations and known special functions. Physical quantities are defined by these spectral sums. They are not defined as expectations with respect to a measure. This rule is the constitutive definition of the formulation. It specifies the meaning of the term physical quantity within the formulation. It is not an independent hypothesis.

Remark 5.5 (Convergence). For the mass-tower resolvent sums that appear below, the Weyl law gives $\sum_l d_l (\lambda_l + m^2)^{-p} \sim \int (\lambda + m^2)^{-p} \lambda^{1/2} d\lambda$, so absolute convergence requires $p > 3/2$. This is the Weyl exponent of the three-dimensional factor M . The exponent $3/2 = \frac{1}{2} \dim M$ is the source of the convergence threshold.

Axiom 5.6 (State space). *The states of the theory are the normalised vectors and density operators of the physical Hilbert space $\mathcal{H}$, the reflection-positive reconstruction space of the Gaussian structure supplied by the datum (Appendix C). On the free sector the reconstruction is the Fock construction $\mathcal{H} = \mathcal{F}(h)$ over the mass tower $\{M_l\}$. This holds unconditionally. On the interacting sector the standard Osterwalder–Schrader reconstruction is conditional. The formulation defines the kinematic carrier of the interacting sector through the effective Fock space of Theorem 7.10 and the formal correspondence of Theorem 9.2.*

Axiom 5.7 (Probability). *A measurement of a self-adjoint observable is a projection-valued measure. The probability of an outcome is the Born rule $p = \langle \psi | P \psi \rangle$. This is the kinematic rule of quantum mechanics on a Hilbert space. It is the unique probability assignment on the projection lattice by Gleason’s theorem [17]. It is not derived from the Gaussian structure alone.*

Axiom 5.8 (Dynamics). *Time evolution is the unitary group supplied by the reflection-positive reconstruction and Stone’s theorem, $U(t) = e^{-iHt}$ with H self-adjoint. The states satisfy the Schrödinger equation. On the free sector $H = d\Gamma(\omega)$ with $\omega = \sqrt{-\Delta + M_l^2}$ on each mode of the mass tower. This holds unconditionally. The scale parameter of the formulation and the emergent time t are independent directions.*

Axiom 5.9 (Composition). *The composite of two systems is the tensor product of their physical Hilbert spaces. The independence of preparations is expressed by product initial states. On the free sector the tensor product is the Fock-space tensor product $\mathcal{F}(h_1 \oplus h_2) \cong \mathcal{F}(h_1) \otimes \mathcal{F}(h_2)$. This holds unconditionally.*

Axiom 5.10 (Locality). *The field algebra satisfies the spacetime locality conditions. Fields at spacelike separation commute for bosons and anticommute for fermions. The fermion anticommutation relations are supplied by the spin–statistics connection.*

Axiom 5.11 (Mapping). *The physical structure of the formulation is defined to be the reconstruction of the spectral datum. This includes the Hilbert space, the field algebra, the dynamics, and the vacuum. The free-sector reconstruction is a theorem (Appendix C). The*

interacting-sector reconstruction is stipulated by the spectral-sum definition. It is defined by the formal correspondence of Section 9 and the spectral-sum definition of physical quantities. It is not claimed as a theorem in the standard measure-theoretic sense.

Remark 5.12 (Status of the axioms). The axioms fall into three classes. Axioms 5.1, 5.4 and 5.6 are definitional. They specify the primitive data, the rule for physical quantities, and the state space. Axioms 5.7–5.10 are kinematic structure. On the free sector they reduce to standard theorems for free fields. They are not new results of this paper. On the interacting sector they are conditional statements within the framework. Axiom 5.11 is the mapping axiom. It is a definitional statement specifying the reconstruction relation between the spectral datum and the physical structure. The free-sector instance is proved. The interacting instance is defined within the spectral-sum definition.

5.2 Formulation criterion and status of the axioms

Definition 5.13 (Formulation criterion). *A system of axioms constitutes a formulation of quantum field theory if it determines the following physical structures as well-defined mathematical objects.*

- (i) *State space, a Hilbert space with a vacuum.*
- (ii) *Observables, an algebra of self-adjoint operators.*
- (iii) *Probability, the Born rule of projective measurement.*
- (iv) *Dynamics, unitary time evolution with a self-adjoint Hamiltonian.*
- (v) *Composite systems, the tensor-product structure.*
- (vi) *Locality, spacelike commutation or anticommutation.*
- (vii) *Spectrum, the mass spectrum and the spectral gap.*
- (viii) *Correlation functions, the n -point spectral functions.*
- (ix) *Scattering, the scattering amplitudes.*
- (x) *Thermal states, the KMS states.*
- (xi) *Measurement, the system–apparatus coupling and decoherence.*

Theorem 5.14 (Status of the physical structures). *The axioms of Section 5.1 locate the structures required by Definition 5.13 as follows. Each entry is marked as derived from the spectral datum, defined within the spectral-sum definition, or axiomatic standard kinematics.*

- (i) *State space. Derived on the free sector. Defined on the interacting sector by the effective Fock space of Theorem 7.10 and the formal correspondence of Theorem 9.2.*

- (ii) *Observables. Derived on the free sector from the Fock construction. Defined on the interacting sector by the mapping dictionary of Theorem 9.2.*
- (iii) *Probability. Axiomatic standard kinematics. The Born rule is part of the kinematic axioms and follows from Gleason’s theorem.*
- (iv) *Dynamics. Axiomatic standard kinematics. Unitary evolution follows from the kinematic axioms and Stone’s theorem applied to the reconstructed Hamiltonian.*
- (v) *Composite systems. Axiomatic standard kinematics. The tensor product is the standard Fock-space tensor product on the free sector.*
- (vi) *Locality. Derived on the free sector from the commutator of free fields. On the effective sector it follows from Proposition 7.12.*
- (vii) *Spectrum. Derived from the spectral datum. The mass tower $\{M_l\}$ and the gap parameter m_δ are obtained from the spectrum through the constructions of Sections 7 and 8.*
- (viii) *Correlation functions. Derived on the free sector from the spectral kernel. On the interacting sector they are spectral-sum defined.*
- (ix) *Scattering. Defined within the framework by spectral-sum amplitudes. The particle content is identified by the spectral poles.*
- (x) *Thermal states. Derived on the free and effective sectors from the KMS construction.*
- (xi) *Measurement. Axiomatic standard kinematics. The system–apparatus coupling and decoherence are standard.*

On the free sector each structure reduces to a standard free-field statement. On the interacting sector the structures are defined within the spectral-sum definition. No external physical parameters are used beyond the spectral datum.

Remark 5.15 (Relation to standard axioms). The correspondence of these axioms to the standard Wightman, Osterwalder–Schrader and Haag–Kastler formulations is recorded in Section 10, with the free-sector instance in Appendix C.

5.3 Gaussian measure and spacetime dimension from the axioms

The axioms of Section 5.1 are stated for an arbitrary compact three-manifold M . The four-dimensional factor $\mathbb{R}^4$ of the reconstruction enters through the Osterwalder–Schrader construction of Appendix C. Two standard mathematical facts then fix the Gaussian form of the measure and the dimension of spacetime. Both facts follow from the axioms themselves. They are not additional inputs.

The first fact is the maximum entropy principle, Theorem 2.3 of Section 2. The mean is zero as a direct consequence of unbiasedness (A3). A non-zero mean would single out a preferred configuration, while the constraints carry no such preference. The Gaussian measure is therefore the least-information distribution fixed by the unbiased spectral datum.

The maximum entropy theorem is a standard mathematical result used inside axiom (A3). It introduces no additional physical parameter. The Gaussian measure is not an additional postulate.

The second fact is the Clifford algebra dimension. The chiral Weyl pair $(1/2, 0) \oplus (0, 1/2)$ is a semispinor module of complex dimension two (Lemma 3.8 of Section 3).

Remark 5.16 (Dimension of spacetime). By Lemma 3.8 a chiral decomposition exists only in even dimensions $n = 2m$, and the semispinor dimension equation $2^{m-1} = 2$ gives $m = 2$, hence $n = 4$. This is the content of Theorem 4.13 of Section 4: the Clifford algebra carrying the chiral pair is $\text{Cl}(4)$, whose generators span a four-dimensional real vector space, so $\dim \mathbb{R}^4 = 4 = 3 + 1$. The identification of the carrier as a semispinor module is part of the kinematic content of the chiral clause (A2). The Gaussian form and the ambient dimension four therefore follow from the axioms together with the standard facts of Sections 2 and 3, without additional input.

6 Kinematics

This section locates the standard kinematic rules of quantum mechanics within the formulation of Section 5. The rules treated are probability, the state space, the Born rule, non-commutativity, spin and statistics, unitary time evolution, energy quantisation, and the measurement process. Each rule is marked as axiomatic standard kinematics or as derived from the spectral datum. On the free sector the Fock space of the reconstruction theorem is unconditional. On the interacting sector the corresponding statements are addressed by the mapping relations of Section 7.

6.1 Probability and the Born rule

Definition 6.1 (Probability measure). *Let $d\mu$ be the Gaussian measure on the configuration space $\mathcal{E} = \mathcal{S}'(\mathbb{R}^4)$. Its covariance is the two-point kernel determined by the spectral datum through the mass tower (5.2) of Axiom 5.1. For every measurable set $B \subset \mathcal{E}$, the number $P(B) = \mu(B)$ is the probability that the field configuration lies in B .*

The Gaussian measure is the reconstruction input. The physical probabilities are those computed in the reconstructed Hilbert space by the Born rule. On the free sector the two descriptions agree.

Remark 6.2 (Gaussian measure from the maximum entropy principle). Among all distributions consistent with a fixed mean and covariance, the unique maximum-entropy distribution is the Gaussian (Theorem 2.3 of Section 2). The mean vanishes by unbiasedness (A3), since a non-zero mean would single out a preferred configuration. The covariance is fixed by the spectral datum through the mass tower (5.2). Hence the Gaussian measure of Definition 6.1 is the unique probability measure consistent with the datum.

6.2 State space and superposition

Definition 6.3 (Euclidean and physical Hilbert spaces). *Let $\mathcal{E} = \mathcal{S}'(\mathbb{R}^4)$ be the configuration space with the Gaussian measure $d\mu$ of Definition 6.1. The Euclidean Hilbert space*

is $\mathcal{H}_E = L^2(\mathcal{E}, d\mu)$, the space of square-integrable functionals on $\mathcal{E}$. The physical Hilbert space $\mathcal{H}$ is the reflection-positive reconstruction space of the Osterwalder–Schrader construction [8, 10]. It is the completion of the positive-time functionals modulo the null space of the reflection-positive sesquilinear form. On the free sector this gives the bosonic Fock space $\mathcal{H} = \mathcal{F}(h)$ over the one-particle space h of the mass tower $\{M_l\}$. This holds unconditionally. On the interacting sector the standard reconstruction is conditional. The formulation defines the kinematic carrier through the effective Fock space of Theorem 7.10 and the formal correspondence of Theorem 9.2.

The physical Hilbert space $\mathcal{H}$ is not the Euclidean space $\mathcal{H}_E$. It is a quotient and completion of the positive-time subspace. It carries the vacuum and the relativistic dynamics that the Euclidean space does not. The states of the theory are the normalised vectors and density operators of $\mathcal{H}$.

Theorem 6.4 (Born rule, standard kinematic rule). *Let ψ, ϕ be normalised states of $\mathcal{H}$. The probability that a system prepared in the state ψ is found in the state ϕ is*

$$p([\psi], [\phi]) = |\langle \psi | \phi \rangle|^2. \quad (6.1)$$

This rule is standard in quantum mechanics. For completeness we include an elementary uniqueness proof on finite-dimensional subspaces. The full infinite-dimensional statement is provided by Gleason’s theorem [17].

Proof. It suffices to prove the statement on an arbitrary finite-dimensional subspace, since any pair of unit rays lies in such a subspace. The unitary group acts transitively on pairs of unit rays with a fixed value of $|\langle \psi | \phi \rangle|$. A unitary-invariant transition probability therefore depends only on this value. Write $p([\psi], [\phi]) = f(|\langle \psi | \phi \rangle|)$ with f continuous. The resolution-of-unity condition fixes f . A basis containing $\phi = \psi$ gives $f(1) = 1$ and $f(0) = 0$. Restricting to a two-dimensional subspace, write $\psi = a\phi_1 + b\phi_2$ with $|a|^2 + |b|^2 = 1$. This gives $f(|a|) + f(|b|) = 1$. In terms of $g(t) = f(\sqrt{t})$, this is $g(t) + g(1-t) = 1$. Restricting to a three-dimensional subspace with $|a|^2 = |b|^2 = t$ gives $2g(t) + g(1-2t) = 1$. The two-dimensional relation gives $g(1-2t) = 1 - g(2t)$. Combining these yields $g(2t) = 2g(t)$. By continuity $g(t) = ct$ on $[0, 1]$, and $g(1) = 1$ fixes $c = 1$. Hence $g(t) = t$ and $p([\psi], [\phi]) = |\langle \psi | \phi \rangle|^2$. $\square$

Theorem 6.5 (Superposition). *The state space $\mathcal{H}$ is a complex Hilbert space. Every complex linear combination of two non-zero vectors in $\mathcal{H}$ is a legitimate state of the theory.*

Proof. The complex structure of $\mathcal{H}$ follows from the complex-valued nature of the square-integrable functionals. The algebraic closure of a complex vector space guarantees that every complex linear combination of two state vectors belongs to $\mathcal{H}$. Such a combination defines a consistent probability assignment through the transition probability of Theorem 6.4. It is therefore a legitimate state of the theory. $\square$

Remark 6.6 (Status of the Born rule). The Born rule is the standard kinematic rule of quantum mechanics on a Hilbert space. The preceding proof shows uniqueness on finite-dimensional subspaces under unitary invariance, continuity, and resolution of unity. The

full infinite-dimensional statement is Gleason's theorem [17]. The Born rule is not derived from the Gaussian measure alone. It belongs to the Hilbert-space structure of the physical space $\mathcal{H}$.

6.3 Composite systems and the tensor product

Definition 6.7 (Entangled state). *A state $\Psi \in \mathcal{H}_1 \otimes \mathcal{H}_2$ of a bipartite system is called entangled if it cannot be written as a product state $\psi_1 \otimes \psi_2$ with $\psi_1 \in \mathcal{H}_1$ and $\psi_2 \in \mathcal{H}_2$.*

Remark 6.8 (Composite systems and the tensor product). Two layers of the composite-system structure are distinguished. At the level of Euclidean configuration spaces, the factorisation identity

$$L^2(\mathcal{E}_1 \times \mathcal{E}_2, d\mu_1 \otimes d\mu_2) \cong L^2(\mathcal{E}_1, d\mu_1) \otimes L^2(\mathcal{E}_2, d\mu_2) \quad (6.2)$$

holds by the standard factorisation theorem for independent measures. The independence of the two systems is expressed by the product measure. At the level of physical Hilbert spaces, the tensor product is supplied by the reconstruction. On the free sector the Fock-space tensor product $\mathcal{F}(h_1 \oplus h_2) \cong \mathcal{F}(h_1) \otimes \mathcal{F}(h_2)$ is the standard multi-particle structure and holds unconditionally. On the interacting sector the tensor product is not derived from a standard reconstruction. It is stipulated as part of the spectral-sum definition through the effective construction of Theorem 7.10 and the formal correspondence of Theorem 9.2. The composite-system rule is therefore the independence of preparations together with the reconstruction on the free sector, and a framework stipulation on the interacting sector.

6.4 Field operators and non-commutativity

Definition 6.9 (Field operators). *Let $\phi(f)$ be the Wightman field operators supplied by the Osterwalder–Schrader reconstruction theorem [8, 10, 18]. They are defined on a dense subspace of the physical Hilbert space $\mathcal{H}$.*

Theorem 6.10 (Non-commutativity of the fermionic sector). *The chiral fermion fields of spin 1/2 satisfy canonical anticommutation relations, so the fermionic sector of the field algebra is non-commutative. The non-Abelian gauge structure contributes the classical Lie-algebraic bracket $[A_\mu, A_\nu] \neq 0$. Quantum commutators of the gauge-field operators are not asserted here.*

Proof. The field operators of Definition 6.9 are operator-valued distributions acting on the physical Hilbert space $\mathcal{H}$ of Definition 6.3. Non-commutativity is not automatic for operators. A commutative algebra of multiplication operators exists on L^2 . In the present framework the non-commutativity arises from the fermionic sector and the gauge sector. The chiral Weyl fields of spin 1/2 satisfy canonical anticommutation relations. For suitable test functions f, g with overlapping supports,

$$\psi(f)\psi^\dagger(g) + \psi^\dagger(g)\psi(f) = \langle f, g \rangle,$$

and the same pair satisfies $\psi(f)\psi^\dagger(g) \neq \psi^\dagger(g)\psi(f)$. Therefore the algebra generated by the fermionic fields is not commutative in the ordinary sense. Scalar fields satisfy commutation relations. The non-Abelian gauge interaction enters through the field strength

$$F = dA + g[A, A].$$

The bracket $[A_\mu, A_\nu] \neq 0$ is a direct consequence of the non-vanishing structure constants $f^{abc} \neq 0$ of the gauge algebra. This is the Lie-algebraic self-interaction content of the gauge field. Quantum commutators of gauge-field operators are not asserted here. The non-commutativity of the field algebra rests on the fermionic anticommutators and the classical non-Abelian structure. $\square$

Remark 6.11 (Self-adjoint observables). The reconstructed field operators are self-adjoint on a dense domain. This is a standard property of the Osterwalder–Schrader reconstruction [8]. Physical observables built from the fields are therefore self-adjoint operators, and their spectra are real by the spectral theorem.

6.5 Spin and statistics

Theorem 6.12 (Spin–statistics, standard theorem). *The quantum fields corresponding to the chiral Weyl pair $(1/2, 0) \oplus (0, 1/2)$ satisfy Fermi anticommutation relations. Scalar fields satisfy Bose commutation relations.*

Proof. This is the standard spin–statistics theorem for relativistic quantum fields [7]. It states that fields of half-odd-integer spin satisfy anticommutation relations and fields of integer spin satisfy commutation relations. The theorem requires the standard hypotheses of relativistic quantum field theory, including relativistic covariance, locality, the spectral condition, and positivity of the Hilbert-space structure. The spin value is one ingredient, not the sole input. The theorem is used here as a standard result. It is not re-derived from the spin value alone. $\square$

6.6 Time evolution

Theorem 6.13 (Unitary time evolution). *The Osterwalder–Schrader reconstruction supplies a strongly continuous unitary representation of the time-translation group on $\mathcal{H}$. By Stone’s theorem this representation is generated by a self-adjoint Hamiltonian H ,*

$$U(t) = e^{-iHt},$$

and the states satisfy the Schrödinger equation

$$i\partial_t\Psi = H\Psi.$$

Proof. The Euclidean measure satisfies the Osterwalder–Schrader axioms [8, 9]. In particular it is invariant under Euclidean time translations. The reconstruction therefore supplies a strongly continuous unitary representation of the time-translation group on the physical Hilbert space $\mathcal{H}$ of Definition 6.3. By Stone’s theorem this representation is generated by

a self-adjoint Hamiltonian H . On the free sector the Hamiltonian is the Fock operator $H = d\Gamma(\omega)$, where

$$\omega = \sqrt{-\Delta + M_t^2}$$

on each mode of the mass tower. This holds unconditionally. The scale parameters of the formulation and the emergent time t are independent quantities. The scale parameters belong to the spectral datum, while t is the time direction of the reconstructed spacetime. $\square$

6.7 Energy quantisation

Theorem 6.14 (Energy quantum and Bohr frequency condition). *A stationary state of the reconstructed theory has energy $E = \hbar\omega$, where ω is its angular frequency. A transition between two stationary states of energies E_i and E_f emits or absorbs radiation at frequency $\Delta E/h$, where $\Delta E = E_f - E_i$ and $h = 2\pi\hbar$.*

Proof. Theorem 6.13 supplies the Schrödinger equation. A stationary state of energy E evolves as

$$\Psi(t) = e^{-iEt/\hbar}\psi.$$

Its phase rotates at angular frequency $\omega = E/\hbar$. Thus $E = \hbar\omega$. The transition rule follows from the spectral representation of the coupling to radiation. Let the system couple to a radiation field by

$$H_{\text{int}} = gO \otimes A,$$

where O is a system observable and A is the radiation field. In first-order time-dependent perturbation theory the transition amplitude from ψ_i to ψ_f with emission of a photon of energy $\hbar\omega_\gamma$ is proportional to

$$\int_0^t dt' e^{i(E_f - E_i - \hbar\omega_\gamma)t'/\hbar} \langle \psi_f | O | \psi_i \rangle \langle \gamma_f | A | \gamma_i \rangle. \quad (6.3)$$

This integral is resonant at $\hbar\omega_\gamma = E_f - E_i$. Transitions therefore occur only when the emitted photon carries the energy difference. This is the Fermi golden rule. Hence the transition frequency satisfies $\Delta E = h\nu$ with $h = 2\pi\hbar$. $\square$

Remark 6.15 (Status of $\hbar$). The Planck constant $\hbar$ is the conversion scale of the formulation's units. It converts dimensionless representation data into dimensionful observables, such as $J^2 = \hbar^2 j(j+1)$ and $E = \hbar\omega$. Its numerical value fixes the unit system. The present section only records the kinematic relation $E = \hbar\omega$ and the Bohr frequency condition derived from the radiation coupling. The discreteness of the energy spectrum is a separate property of the Hamiltonian spectrum.

6.8 The measurement process

The measurement process is described by the standard quantum-mechanical treatment. In the present formulation this treatment is consistent with the axioms of Section 5 and the kinematic Born rule of Theorem 6.4.

Definition 6.16 (Measurement apparatus). *A measurement apparatus $\mathcal{A}$ is a quantum system with a macroscopically large number of degrees of freedom. Its Hilbert space $\mathcal{H}_{\mathcal{A}}$ is constructed in the same way as the physical Hilbert space of Definition 6.3. On the free sector this is the Fock construction. In general it is given by the spectral-sum-definition construction of the corresponding sector. The pointer states $|\phi_k\rangle$ are the macroscopically distinct configurations of the read-out. They form a mutually orthogonal basis to a good approximation. Their stability is guaranteed by a positive gap of the apparatus sector.*

Theorem 6.17 (Interaction establishes correlation). *Let $\mathcal{S}$ be a quantum system with Hilbert space $\mathcal{H}_{\mathcal{S}}$, and let $\mathcal{A}$ be a measurement apparatus with Hilbert space $\mathcal{H}_{\mathcal{A}}$. Let the interaction between $\mathcal{S}$ and $\mathcal{A}$ be the von Neumann coupling [19]*

$$H_{\text{int}} = g O \otimes Q,$$

where O is the measured observable and Q is the pointer observable. Let the initial system state have non-zero components in at least two eigenspaces of O , and let the displaced pointer states

$$\phi_k(t) = e^{-ig o_k Q t} \phi_{\mathcal{A}}$$

be pairwise distinct for distinct eigenvalues o_k . Then the initial product state evolves into an entangled state of the composite system.

Proof. The evolution is generated by the total Hamiltonian

$$H = H_{\mathcal{S}} + H_{\mathcal{A}} + H_{\text{int}}.$$

Writing the initial state in the eigenbasis of O ,

$$\psi_{\mathcal{S}} = \sum_k c_k \hat{\psi}_k,$$

the evolution under H_{int} gives

$$\Psi(t) = e^{-iH_{\text{int}}t}(\psi_{\mathcal{S}} \otimes \phi_{\mathcal{A}}) = \sum_k c_k \hat{\psi}_k \otimes \phi_k(t),$$

where $\phi_k(t)$ is the pointer state displaced by the eigenvalue o_k . The displaced pointer states are pairwise distinct by hypothesis, and the eigenbranches $\hat{\psi}_k$ are orthogonal. A product decomposition $\Psi(t) = \psi \otimes \phi$ would require a common apparatus factor. This is impossible when at least two distinct displaced pointer states occur. Hence $\Psi(t)$ is entangled. $\square$

Theorem 6.18 (Decoherence from many internal modes, physical argument). *Let $\mathcal{A}$ be a macroscopic apparatus whose pointer states $|\phi_k\rangle$ are the macroscopically distinct configurations of Definition 6.16. Assume that the internal mode frequencies and couplings are sufficiently regular, so that the phases of different modes are effectively uncorrelated. Then the off-diagonal elements of the reduced density matrix of the system, expressed in the pointer basis, decay exponentially. The dephasing rate is determined by the couplings of the system observable to the internal modes of the apparatus.*

Proof. After tracing out the apparatus, the reduced density matrix of the system is

$$\rho_S = \text{Tr}_{\mathcal{A}} |\Psi(t)\rangle\langle\Psi(t)|.$$

In the eigenbranch basis $\{\hat{\psi}_k\}$ its off-diagonal elements are

$$\rho_{kl}(t) = c_k \bar{c}_l f_{kl}(t),$$

where the suppression factor is

$$f_{kl}(t) = \langle\phi_l(t)|\phi_k(t)\rangle.$$

The interaction couples the observable O to a collective pointer observable

$$Q = \sum_m g_m Q_m,$$

the sum running over the N internal modes of the apparatus. Each mode contributes an independent phase factor to $f_{kl}(t)$. Under the regularity assumption on the internal modes, the phases are uncorrelated for large N , and the product is exponentially suppressed,

$$|f_{kl}(t)| \leq e^{-\Gamma t},$$

with a dephasing rate Γ determined by the couplings and fluctuations of the modes. The positive gap of the apparatus sector ensures stability of the pointer states. Decoherence is therefore a dynamical consequence of the macroscopic number of internal modes under the stated assumptions [20]. It is not an additional assumption, but the exponential suppression is a physical argument rather than a rigorous theorem without further hypotheses. $\square$

Theorem 6.19 (Born rule as projective measurement). *Let the apparatus $\mathcal{A}$ measure an observable O of the system $\mathcal{S}$ on the physical Hilbert space $\mathcal{H}_{\mathcal{S}}$. Let*

$$O = \sum_k o_k P_k$$

be the spectral decomposition, where P_k is the projector onto the eigenspace with eigenvalue o_k . The probability of obtaining the outcome o_k when the system is in the state $\psi_{\mathcal{S}}$ is

$$p_k = \|P_k \psi_{\mathcal{S}}\|^2.$$

Proof. On the physical Hilbert space, a measurement of a self-adjoint observable is a projection-valued measure. By the spectral theorem, $O = \sum_k o_k P_k$. The probability of the outcome o_k is the standard kinematic rule

$$p_k = \langle\psi_{\mathcal{S}}|P_k\psi_{\mathcal{S}}\rangle = \|P_k\psi_{\mathcal{S}}\|^2.$$

This is the transition probability of Theorem 6.4 between the initial state and the eigenbranch $\hat{\psi}_k = P_k \psi_{\mathcal{S}} / \|P_k \psi_{\mathcal{S}}\|$. The measurement interaction realises this kinematic rule dynamically. Before the measurement the system and apparatus are independent. The state of the composite system is $\Psi(0) = \psi_{\mathcal{S}} \otimes \phi_{\mathcal{A}}$. The von Neumann coupling [19] correlates each eigenbranch with a distinct pointer state. After the coupling the reduced density matrix of the apparatus is diagonal in the pointer basis, consistent with the kinematic rule $p_k = |c_k|^2 = \|P_k \psi_{\mathcal{S}}\|^2$. This is the Born rule. Its status is kinematic. $\square$

6.9 Location of the quantum postulates

The standard postulates of quantum mechanics are located in the formulation as follows.

Table 1. Location of the quantum postulates within the spectral formulation.

Standard postulate	Located in
State space is a complex Hilbert space	Physical Hilbert space of Definition 6.3. Fock space on the free sector
Superposition principle	Complex Hilbert-space structure, Theorem 6.5
Probability via the Born rule	Projective measurement, Theorem 6.19
Observables are Hermitian operators	Self-adjoint field operators, Remark 6.11
Spin–statistics relation	Standard spin–statistics theorem, Theorem 6.12
Time evolution by the Schrödinger equation	Reconstruction and Stone’s theorem, Theorem 6.13
Energy quantisation	Phase dynamics of stationary states, Theorem 6.14
Measurement is an interaction	System–apparatus coupling, Theorem 6.17
Decoherence suppresses interference	Many-mode dephasing of the apparatus, Theorem 6.18

7 Mapping relations

The mapping axiom of Section 5 states the intended relation between the spectral datum and the physical structure. This section collects the mapping statements. The free-sector instance is proved in Appendix C. The interacting-sector instance is defined within the spectral-sum definition by the analytic, structural, effective, and non-perturbative statements of the present section.

Theorem 7.1 (Mapping within the framework). *Let S be the spectral datum of Axiom 5.1. The reconstruction of Axioms 5.6–5.10 yields a physical structure $(\mathcal{H}, \mathcal{A}, U, \Omega)$. On the free sector every physical quantity of Axiom 5.4 is a spectral-data quantity of the reconstructed structure, as a trace of the internal Green operator of the reconstructed propagator (Theorem 7.8). On the interacting sector every physical quantity is spectral-sum defined, with formal correspondence to the field algebra given by Theorems 9.2–9.6. The spectrum of the Hamiltonian contains the mass tower $\{M_l\}$ with lower bound m_δ . The free-sector statement is proved by Theorem C.2. The interacting-sector statement is defined within the spectral-sum definition. It is not claimed as a theorem in the standard measure-theoretic sense.*

Remark 7.2 (Interacting mapping and the standard formulation). The interacting content of the formulation has no realisation as a standard four-dimensional scalar field theory, because the continuum limits of four-dimensional lattice scalar models are Gaussian. The interacting mapping is therefore a statement about the formulation itself. It is not a claim about a standard-formulation continuum limit. This point is discussed in Section 10.4.

7.1 Analytic mapping: analytic control, positivity, condensation, and gap

The analytic part of the mapping records the analytic properties of the spectral sums. The input is the spectral datum of Axiom 5.1, namely the compact three-manifold M and its spectrum $\{\lambda_l\}$ with multiplicities $\{d_l\}$. The parameters m_δ , λ , and $\langle E \rangle$ are derived from this datum in Sections 7 and 8.

Theorem 7.3 (Well-definedness and analytic control). *For $p > 3/2$ the spectral sum*

$$Q_p(m^2) = \sum_l d_l (\lambda_l + m^2)^{-p}$$

is absolutely convergent, positive, and analytic in m^2 on $(0, \infty)$, with the bound

$$Q_p(m^2) \leq C_p \text{vol}(M) m^{3-2p}$$

for $m^2 \geq m_\delta^2$.

Proof. By the Weyl law for the scalar Laplacian on the compact three-manifold M , the counting function $N(\lambda) = \#\{l : \lambda_l \leq \lambda\}$ satisfies $N(\lambda) \sim C \text{vol}(M) \lambda^{3/2}$ as $\lambda \rightarrow \infty$. The series is therefore summable if and only if the integral

$$\int (\lambda + m^2)^{-p} dN(\lambda) \sim \int (\lambda + m^2)^{-p} \lambda^{1/2} d\lambda$$

converges at infinity. This requires $p > 3/2 = \frac{1}{2} \dim M$. The series converges absolutely and uniformly on compact subsets of $(0, \infty)$. Each term is positive and analytic in m^2 . The sum is therefore positive and analytic. For the bound, split the sum at $\lambda_* \sim m^2$. The low modes give

$$Q_p^{(\text{low})} \leq m^{-2p} \sum_{\lambda_l \leq m^2} d_l \sim m^{-2p} N(m^2) \sim \text{vol}(M) m^{3-2p}.$$

The high modes give

$$Q_p^{(\text{high})} \leq \sum_{\lambda_l > m^2} d_l \lambda_l^{-p} \sim \int_{m^2}^{\infty} \lambda^{-p} \lambda^{1/2} d\lambda \sim m^{3-2p}.$$

Both parts are of the stated form. This is the analytic control of the physical quantities. Every physical quantity of Axiom 5.4 is a well-defined analytic function of its parameters, with no infrared singularities for $m^2 \geq m_\delta^2 > 0$. $\square$

Theorem 7.4 (Spectral positivity). *For every non-negative function f , the physical quantity*

$$Q = \sum_l d_l f(\lambda_l)$$

of Axiom 5.4 is non-negative. In particular, the two-point spectral quantities are positive, and the spectral measure of the mass tower

$$\rho(\mu^2) = \sum_l d_l \delta(\mu^2 - M_l^2)$$

is non-negative with support in $[m_\delta^2, \infty)$.

Proof. Every multiplicity $d_l \geq 1$ is positive, and every eigenvalue satisfies

$$M_l^2 = \lambda_l + m_\delta^2 \geq m_\delta^2 > 0.$$

Every term is therefore non-negative, and the support statement follows. The positivity of the two-point spectral quantities is a direct consequence of the spectral data. It is not an additional input. $\square$

Theorem 7.5 (Condensation). *Let λ and $\langle E \rangle$ be the derived parameters of Sections 7 and 8. Define the long-root mass parameter by*

$$m_{\text{long}}^2 = -\lambda \langle E \rangle^2 < 0. \quad (7.1)$$

Then the Landau potential

$$V(E) = \frac{1}{2} m_{\text{long}}^2 E^2 + \frac{\lambda}{4} E^4 \quad (7.2)$$

has a unique non-trivial minimum at $E = \langle E \rangle$. The curvature at the minimum is

$$m_\delta^2 = V''(\langle E \rangle) = 2\lambda \langle E \rangle^2 = -2m_{\text{long}}^2 > 0. \quad (7.3)$$

Proof. The Landau potential (7.2) is even and has derivative

$$V'(E) = E (m_{\text{long}}^2 + \lambda E^2).$$

With $m_{\text{long}}^2 < 0$ (by (7.1)), the critical points are $E = 0$ and $E = \pm \sqrt{-m_{\text{long}}^2/\lambda} = \pm |\langle E \rangle|$. The point $E = 0$ is a local maximum because $V''(0) = m_{\text{long}}^2 < 0$. The non-zero critical points are the two minima. The condensate $\langle E \rangle \neq 0$ is the non-trivial minimum. The curvature there is

$$V''(\langle E \rangle) = m_{\text{long}}^2 + 3\lambda \langle E \rangle^2 = -2m_{\text{long}}^2 = 2\lambda \langle E \rangle^2 = m_\delta^2.$$

The condensation is therefore a consequence of the derived parameters λ and $\langle E \rangle$. $\square$

Theorem 7.6 (Positive spectral lower bound). *Let m_δ be the derived parameter of Theorem 7.5. Then every spectral mass satisfies*

$$M_l^2 = \lambda_l + m_\delta^2 \geq m_\delta^2 > 0.$$

No massless mode occurs in the spectrum. The physical structure of Axioms 5.6–5.10 carries the lower bound $m_\delta > 0$.

Proof. At the level of the spectral data, every spectral mass satisfies $M_l \geq m_\delta > 0$ by Theorem 7.4. No massless mode occurs in the spectrum. At the operator level the lower bound is explicit on the free sector through the Fock spectrum of the mass tower (Theorem C.2). On the interacting sector it is defined by the effective construction of Theorem 7.10. The parameter m_δ is not an input. It is derived from the spectral datum through Theorem 7.5. This theorem records the propagation of the lower bound from the spectral datum to the physical structure. $\square$

Remark 7.7. Theorems 7.3–7.6 record the analytic part of the mapping. They establish the analytic control, the positivity, the condensation, and the positive lower bound of the physical quantities. Their input is the spectral datum of Axiom 5.1. Together with the free-sector mapping of Theorem C.2, they constitute the body of the mapping statement of Theorem 7.1.

7.2 Structural mapping: two-point and composite quantities

The structural mapping records the free-sector instance of Axiom 5.11. It shows that the spectral sums of Axiom 5.4 are traces of the reconstructed propagator. The input is the free sector of the recovery theorem (Theorem C.2), namely the Fock space $\mathcal{H} = \mathcal{F}(h)$ over the mass tower $\{M_l\}$ with field operators $\varphi(f \otimes g)$, where $f \in \mathcal{S}(\mathbb{R}^4)$ and $g \in C^\infty(M)$.

Theorem 7.8 (Mapping of two-point and composite quantities). *Let the free sector of the recovery theorem be given. Then the following hold.*

- (a) *The vacuum two-point function of the field operators is the full spectral kernel. For internal test functions g, g' ,*

$$\langle \Omega, \varphi(f \otimes g) \varphi(f' \otimes g') \Omega \rangle = \sum_l (g, K_l g')_{L^2} \langle \Omega, \varphi_l(f) \varphi_l(f') \Omega \rangle,$$

where K_l is the spectral projector kernel of the l -th eigenspace and φ_l is the mode field of mass M_l .

- (b) *The spectral sum*

$$Q_p(m^2) = \sum_l d_l (\lambda_l + m^2)^{-p}$$

with $p > 3/2$ is the trace of the p -th power of the internal Green operator,

$$Q_p(m^2) = \text{Tr}_{L^2(M)} [(\Delta_M + m^2)^{-p}] = \sum_l \frac{d_l}{(\lambda_l + m^2)^p}.$$

For $p = 2$ the trace is the integral of the square of the internal Green function,

$$Q_2(m^2) = \iint_{M \times M} G_{m^2}(y, y')^2 dy dy',$$

where

$$G_{m^2}(y, y') = \sum_l \frac{K_l(y, y')}{\lambda_l + m^2}.$$

This is the zero-momentum internal trace of the full propagator of the recovery theorem. It is not the vacuum expectation of the four-dimensional normal-ordered field $:\varphi^2:$, whose expectation involves the square of the full propagator over all momenta and is not the quantity Q_2 .

- (c) *Every physical quantity of Axiom 5.4 with $f(\lambda) = (\lambda + m^2)^{-p}$ and $p > 3/2$ is therefore a trace of the spectral data. It is linked to the field structure of the free sector through the zero-momentum component of the full spectral kernel.*

Proof. (a) This is the standard Fock two-point function. The modes φ_l are independent free fields of mass M_l by Theorem C.2, and the internal projector kernels implement the orthogonality of the modes. (b) By the spectral decomposition of the internal Green function and the orthogonality of the projector kernels,

$$\int_M K_l K_{l'} = \delta_{ll'} d_l,$$

one obtains

$$\iint_{M \times M} G_{m^2}^2 = \sum_{l, l'} \frac{\int_M K_l K_{l'}}{(\lambda_l + m^2)(\lambda_{l'} + m^2)} = \sum_l \frac{d_l}{(\lambda_l + m^2)^2} = Q_2(m^2).$$

The general p is the trace of the p -th power,

$$\text{Tr}[(\Delta_M + m^2)^{-p}] = \sum_l \frac{d_l}{(\lambda_l + m^2)^p}.$$

The link to the reconstructed field structure is through the zero-momentum component of the full spectral kernel. The full propagator of the recovery theorem is $G_{\text{full}}((x, y), (x', y')) = \sum_l K_l(y, y') \Delta_{M_l}(x - x')$ with $M_l^2 = \lambda_l + m_\delta^2$, and its zero-momentum component is

$$\int_{\mathbb{R}^4} G_{\text{full}}((x, y), (0, y')) dx = \sum_l \frac{K_l(y, y')}{M_l^2} = G_{m_\delta^2}(y, y').$$

Thus the resolvent kernel G_{m^2} is the zero-momentum component of the reconstructed propagator at the physical value $m^2 = m_\delta^2$; for general m^2 it is the auxiliary analytic continuation used for the convergence control of Theorem 7.3. The quantity Q_p is the internal trace of the reconstructed propagator, and it is not a vacuum expectation of the four-dimensional fields. (c) Each Q_p is a spectral-data trace by (b), absolutely convergent by Theorem 7.3. The correspondence to the reconstructed field structure is through the zero-momentum component of the full spectral kernel. $\square$

Remark 7.9. The correspondence between the spectral sums and the field-theoretic quantities is summarised as follows.

Quantity	Spectral-data realisation	Status
$Q_1(m^2) = \sum_l d_l (\lambda_l + m^2)^{-1}$	multiplicity diagonal of the two-point function (divergent; the well-defined object is the full spectral kernel)	Theorem C.2
$Q_2(m^2) = \sum_l d_l (\lambda_l + m^2)^{-2}$	$\text{Tr}[(\Delta_M + m^2)^{-2}] = \iint G_{m^2}^2$ (internal Green trace)	Theorem 7.8
$Q_p, p > 3/2$	$\text{Tr}[(\Delta_M + m^2)^{-p}]$, the p -th power of the internal Green operator	Theorem 7.8

The structural mapping of Axiom 5.11 is thereby recorded on the free sector. The spectral sums of Axiom 5.4 are spectral-data traces of the reconstructed propagator, specifically its zero-momentum internal components. They are not vacuum expectations of the four-dimensional fields.

7.3 Effective Hilbert space of the condensed sector

The free-sector mapping of Theorem 7.8 records the free content of Axiom 5.11. The present subsection defines the kinematic carrier of the interacting content within the framework. The condensed sector defines an effective theory whose parameters are closed functions of the spectral data. Its Hilbert space is the Fock space of the fluctuation field of mass m_δ . The construction is the effective-level instance of the mapping axiom. It is not a standard-formulation continuum limit.

Theorem 7.10 (Effective Hilbert space of the condensed sector). *Let λ , $\langle E \rangle$, and m_δ be the derived parameters of Sections 7 and 8. Define the effective field*

$$e = E - \langle E \rangle.$$

Then the following hold.

- (a) *The condensate is non-vanishing, and the fluctuation mass is the curvature of the effective potential at its minimum,*

$$m_\delta^2 = V''(\langle E \rangle) = 2\lambda \langle E \rangle^2.$$

Both are closed expressions in the spectral data.

- (b) *The effective Hilbert space of the condensed sector is the Fock space $\mathcal{H}_{\text{eff}} = \mathcal{F}(h_\delta)$ of the fluctuation field e of mass m_δ . The single-particle space is*

$$h_\delta = L^2 \left(\mathbb{R}^3, \frac{d^3 p}{2\omega_\delta} \right),$$

with

$$\omega_\delta = \sqrt{p^2 + m_\delta^2}.$$

The free Hamiltonian is the Fock operator $d\Gamma(\omega_\delta)$, which is a strict operator. The interaction term λe^4 is an effective object of the formulation. Its status is given by the mapping dictionary and the spectral-sum amplitudes of Theorems 7.19, 9.2 and 9.6. It is not a strict Fock-space operator. The parameters of the effective theory are all closed functions of the spectral data.

- (c) *The effective theory carries a positive lower bound $m_\delta > 0$. The spectrum of $d\Gamma(\omega_\delta)$ has no weight below m_δ above the vacuum.*

Proof. (a) The Landau form has its unique non-trivial minimum at

$$E_*^2 = -\frac{m_{\text{long}}^2}{\lambda} = \langle E \rangle^2 > 0$$

by Theorem 7.5. The curvature there is m_δ^2 by (7.3). (b) The effective field e is a scalar of mass m_δ . Its Fock space is constructed by standard second quantisation from the spectral data. The single-particle space is determined by m_δ . The quartic term is the effective interaction with the spectral-data coupling λ . (c) The single-particle spectrum of $d\Gamma(\omega_\delta)$ starts at $m_\delta > 0$. The vacuum has no weight below m_δ . $\square$

Remark 7.11. The effective Hilbert space of Theorem 7.10 is the kinematic carrier of the interacting sector within the framework. Its parameters are closed functions of the spectral data. The interaction term is defined by its physical content and structural status through Theorems 9.4 and 9.6.

Proposition 7.12 (Locality of the effective sector). *The fluctuation field e of the effective Hilbert space satisfies the locality condition of Axiom 5.10. For test functions f, g with spacelike-separated supports,*

$$[e(f), e(g)] = 0.$$

Proof. The effective field e is a free scalar field of mass m_δ on its Fock space. The commutator of a free scalar field is

$$[e(x), e(y)] = i\Delta(x - y),$$

where Δ is the Pauli–Jordan function, which vanishes for spacelike separations. Hence the locality condition holds on the effective sector. $\square$

Remark 7.13. The field structure of the free sector and the effective sector satisfies the locality condition. The spectral-sum quantities of Axiom 5.4 are global functions of the spectral data. They are not local operators, so spacelike commutation is not defined for them. Locality is a property of the field structure, not of the numerical spectral sums.

7.4 Thermal states

The temperature β is an external physical condition. It is not an additional parameter of the model. On the free sector the thermal states are the standard KMS states, which are Gaussian and determined by the temperature-parameterised spectral functions.

Theorem 7.14 (KMS states of the free sector). *Let the free sector of the recovery theorem be given. For every inverse temperature $\beta > 0$ there exists a unique quasi-free KMS state ω_β in the Gaussian class on the field algebra. Its two-point function is the per-mode thermal propagator*

$$\langle \varphi(f \otimes g) \varphi(f' \otimes g') \rangle_{\omega_\beta} = \sum_l (g, K_l g') \int \int dx dx' f(x) f'(x') \Delta_\beta^{(M_l)}(x - x'),$$

where

$$\Delta_\beta^{(M)}(x) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2\omega_M(p)} \left[(1 + n_\beta(\omega_M(p))) e^{-i(\omega_M(p)x^0 - \mathbf{p} \cdot \mathbf{x})} + n_\beta(\omega_M(p)) e^{+i(\omega_M(p)x^0 - \mathbf{p} \cdot \mathbf{x})} \right],$$

with $\omega_M(p) = \sqrt{\mathbf{p}^2 + M^2}$ and the Bose occupation number

$$n_\beta(\omega) = (e^{\beta\omega} - 1)^{-1}.$$

This is the standard thermal two-point function. The state satisfies the KMS condition

$$\omega_\beta(A \varphi_t(B)) = \omega_\beta(\varphi_{t+i\beta}(B) A)$$

for all polynomials A, B in the fields. The thermal expectation of the composite quantities of Theorem 7.8 is the temperature-parameterised spectral sum

$$Q_p(\beta, m^2) = \sum_l d_l T_p(\beta, M_l),$$

where T_p is the thermal spectral function.

Proof. The Gaussian state with the thermal two-point function is the standard KMS state of the free field. The two-point function satisfies the KMS condition in the distributional sense. The higher correlations are Wick products of the two-point function. The KMS condition for polynomials follows from the two-point case. Uniqueness holds within the Gaussian class, whose two-point function is fixed by the KMS condition. The state is determined by the spectral data of the mass tower $\{M_l\}$ and the temperature β . $\square$

Theorem 7.15 (KMS states of the effective sector). *Let the effective sector of Theorem 7.10 be given. For every inverse temperature $\beta > 0$ there exists a unique quasi-free KMS state of the free part, whose two-point function is the thermal propagator $\Delta_\beta^{(m_\delta)}$ of mass m_δ . The correction by the interaction is the perturbative series of the effective theory. The n -th order correction is a function of the spectral data λ and m_δ , computed with the thermal Feynman rules and the dispersion subtraction of Theorem 9.4.*

Proof. The free part of the effective Hamiltonian is the Fock operator of the field e of mass m_δ . By the construction of Theorem 7.14 there is a unique quasi-free KMS state with the thermal propagator $\Delta_\beta^{(m_\delta)}$. The interaction correction is the perturbative thermal series of the effective theory. At each order the thermal amplitudes are integrals of thermal propagators and the vertex λ . The vertex λ is a spectral sum by Theorem 7.3. Each order is therefore a definite function of the spectral data. $\square$

Theorem 7.16 (Asymptotic states of the effective sector). *The asymptotic states of the effective sector are the Fock multi-particle states of the free field e . The scattering amplitudes of Theorem 9.4 are the transition amplitudes between these asymptotic states.*

Proof. The effective Hilbert space is the Fock space of the free field e , by Theorem 7.10. The asymptotic states are therefore the Fock multi-particle states. The scattering amplitudes of Theorem 9.4 are the transition amplitudes between these states, computed by the Feynman rules of the effective theory with the spectral-data parameters λ and m_δ . $\square$

Remark 7.17. The definition of scattering within the formulation uses the spectral pole criterion together with the effective amplitudes. The isolated poles of the two-point spectral functions correspond to physical single-particle states within the formulation. In the Källén–Lehmann representation the atomic part of the spectral measure is the single-particle spectrum, as in Theorem 7.4. On the free sector the single-particle states are the Fock one-particle states of the mass tower, and the S-matrix of the free theory is trivial. The interacting S-matrix is defined by the amplitudes of the effective theory, which are explicit functions of the spectral data. The pole criterion identifies the particles of the theory from the spectral data.

7.5 Scattering

Scattering within the formulation is defined from the same spectral data as the other physical quantities. The particles are identified by the single-particle states of the spectral pole criterion. The scattering amplitudes are the perturbative amplitudes of the effective theory of Theorem 7.10. Their parameters, the mass m_δ and the coupling λ , are closed functions of the spectral data. The amplitudes are therefore explicit functions of the spectral data.

Theorem 7.18 (Particle spectrum from spectral poles). *Let the two-point spectral function have an isolated pole at $p^2 = M_l^2$. Then the spectral measure has an atom at M_l , corresponding to a single-particle state of mass M_l within the formulation. On the free sector the single-particle states are the Fock one-particle states of the mass tower $\{M_l\}$ by Theorem C.2. On the effective sector the single particle is the fluctuation quantum of mass m_δ , corresponding to the pole of the e -field propagator at $p^2 = m_\delta^2$.*

Proof. In the Källén–Lehmann representation of the two-point function, the atomic part of the spectral measure is the single-particle spectrum. An isolated pole at $p^2 = M_l^2$ is an atom of the spectral measure at M_l . On the free sector this atom is the mass of the l -th Fock mode by Theorem C.2. On the effective sector the propagator of the free field e has a single pole at $p^2 = m_\delta^2$, which identifies the single particle of the condensed sector. $\square$

Theorem 7.19 (Tree-level amplitude of the effective sector). *Let the effective theory of Theorem 7.10 be written as*

$$\mathcal{L}_{\text{eff}} = \frac{1}{2}(\partial e)^2 + \frac{1}{2}m_\delta^2 e^2 + \frac{\lambda}{4}e^4.$$

The parameters m_δ and λ are closed functions of the spectral data by Theorems 7.3 and 7.6. The tree-level $2 \rightarrow 2$ scattering amplitude of the effective quanta is

$$\mathcal{M}(s, t) = -6\lambda.$$

The differential cross-section is

$$\frac{d\sigma}{d\Omega} = \frac{9\lambda^2}{16\pi^2 s},$$

where s is the Mandelstam variable.

Proof. The Feynman rules of the effective theory are standard. The propagator is

$$\frac{i}{p^2 - m_\delta^2 + i\varepsilon},$$

and the e^4 vertex is $-6i\lambda$ (the quartic coupling is normalised as in the Landau potential $V = (\lambda/4)E^4$ of Theorem 8.32, so the vertex carries the combinatorial factor $4!$ of the four fields). At tree level the $2 \rightarrow 2$ amplitude receives only the contact contribution $-6i\lambda$. There is no s -, t -, or u -channel exchange because the effective theory is stipulated quartic,

with no e^3 term. Hence $\mathcal{M} = -6\lambda$. The cross-section for equal-mass $2 \rightarrow 2$ scattering with a contact interaction is

$$\frac{d\sigma}{d\Omega} = \frac{|\mathcal{M}|^2}{64\pi^2 s} = \frac{9\lambda^2}{16\pi^2 s},$$

which is angular-independent. The coupling λ is a convergent spectral sum of the spectral data by Proposition 8.31 and Theorem 7.3. The amplitude is therefore an explicit function of the spectral data. $\square$

Remark 7.20. The scattering construction of this subsection is the perturbative scattering structure of the effective sector. The particles are identified by the spectral poles of Theorem 7.18. The amplitudes are the perturbative amplitudes of the effective theory, which are explicit functions of the spectral data. The asymptotic states of the effective sector are the Fock multi-particle states of the free field e by Theorem 7.16. The amplitudes are defined order by order through the Feynman rules of the effective theory. The relation to the standard continuum formulation is discussed in Section 10.4.

8 Construction of the gauge field

Section 4 identified the gauge field A_μ as a connection on the four-dimensional spacetime V , realised by Kaluza–Klein zero modes. Its algebraic structure was given in Theorem 4.12. The present section develops the dynamical description of this gauge field within the spectral-sum definition.

The constructions of Section 4 produced the gauge algebra, the internal space, the spacetime dimension, and the Gaussian measure. The present section assembles these building blocks into a quantum field theory. The assembly is carried out within the spectral-sum definition of Definition 8.1. All subsequent statements are made within this definition.

Definition 8.1 (Spectral-sum definition). *The spectral-sum definition of the present paper is the evaluation scheme defined as follows.*

(i) *The basic datum is the spectral data of the emergent internal space RP^3 , namely the scalar Laplacian spectrum $\lambda_l = l(l+2)/L^2$ with multiplicities $d_l = (l+1)^2$ for even l , together with the scale flow $\{\Phi_\sigma\}$ defined in Section 8.4 (Theorem 8.20).*

(ii) *A physical quantity is an expression of the form*

$$Q = \sum_{l \text{ even}} d_l \frac{f(\lambda_l)}{(\lambda_l + m^2)^p}, \quad p > 3/2,$$

where f is determined by the quantity in question, or a closed analytic expression obtained from the scale flow.

(iii) *The value of Q is its endpoint evaluation at σ_C . By Lemma 8.21 the series converges absolutely, and the closed forms of Theorem 8.35 are explicit.*

- (iv) *The spectral-sum-definition analogue of the Osterwalder–Schrader axioms is defined as follows. Regularity corresponds to the absolute convergence of the spectral sums. Euclidean invariance corresponds to the $SO(4)$ -invariance of the spectral data under the scale flow, together with the $E(4)$ -invariance of the Gaussian part. Reflection positivity corresponds to the reflection positivity of the lattice-regularised measure, closed under weak limits. Symmetry corresponds to the Euclidean covariance of the action. Clustering corresponds to the exponential decay implied by the positive lower bound m_δ . The spectral-sum-definition analogue of the Wightman axioms is understood similarly.*
- (v) *The relation between the spectral-sum definition and the standard measure-theoretic formulation is analysed in Section 10. The non-trivial content of the framework has no known realisation in the standard continuum formulation. Only the free part is translatable, and its standard-formulation construction is given there.*

These conditions are analogues of the standard axioms, not equivalent statements. They are the conditions under which the spectral data define a quantum field theory within the present formulation.

Remark 8.2 (Terminology). Throughout this paper, the term “spectral-sum definition” refers to the evaluation scheme in which physical quantities are defined as convergent spectral sums over a discrete spectrum, rather than as expectations with respect to a measure. This terminology is used only to name the rule stated in Axiom 5.4; it does not assert an equivalence with any other formulation.

The objects appearing in this definition, namely the scale flow Φ_σ , the endpoint σ_C , and the spectral sums, are constructed in the following subsections.

8.1 Gauge field and scale semigroup

The scale field σ was introduced in Section 4.6 as a coarse-graining scale. Its statistics are governed by the unbiased Gaussian measure. A natural evolution of the scale parameter σ is generated by the heat-kernel smoothing semigroup $e^{-\sigma^2 \Delta/2}$, where Δ is the Laplacian on V . This semigroup decomposes a field into momentum modes and defines the effective degrees of freedom at each scale.

Definition 8.3 (Slow mode). *Let $\{e^{-\sigma^2 \Delta/2}\}_{\sigma>0}$ be the scale semigroup, and let*

$$\Pi_\sigma = \mathbf{1}_{[0, \sigma^{-2}]}(-\Delta)$$

be the spectral projection. A field ϕ is called a slow mode at scale σ if $\phi = \Pi_\sigma \phi$, that is, if its momentum support lies in the region $|k| \lesssim 1/\sigma$.

The gauge field A_μ of Theorem 4.12 is reduced from the connection on RP^3 to the spacetime V by Kaluza–Klein expansion. Its internal space components are carried by zero modes whose momentum support is concentrated at the low end. The gauge field is therefore automatically a slow mode under the scale semigroup. The statistics of the gauge field are jointly determined by the unbiased Gaussian measure of axiom (A3) and the spectral representation of the Gaussian structure.

8.2 Measure construction

As a quantum field, the statistical properties of the gauge field are described by a probability measure on the space of field configurations. Axiom (A3) supplies a Gaussian measure for the free fields. The measure for the gauge fields must incorporate the F^2 nonlinear self-interaction, whose form is dictated by the emergent gauge algebra.

Theorem 8.4 (Interaction from the group structure). *Let A_μ be the four-dimensional gauge field emerging from Axiom 3.4 via Theorem 4.12. Its construction is given by the Kaluza–Klein mechanism. The connection on the principal $G_{SM} = SU(3) \times SU(2) \times U(1)$ bundle over RP^3 is expanded in zero modes. Their coefficients are the gauge fields $A_\mu(x)$ on the four-dimensional spacetime V . The colour part is carried by the eight zero modes of the colour algebra $\mathfrak{su}(3)$. The electroweak part is carried by the four isometric Killing zero modes of the retained gauge symmetries $SU(2)_L \times U(1)_R \subset SO(4)$. This gives twelve gauge bosons. The field strength of A_μ is*

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g f^{abc} A_\mu^b A_\nu^c,$$

where the f^{abc} are the structure constants of $\mathfrak{g}_{SM}$. The expansion of F^2 yields three-gluon vertices proportional to $g f^{abc}$ and four-gluon vertices proportional to $g^2 f^{abe} f^{cde}$. The chiral fermions of axiom (A2), forming the Weyl pair $(1/2, 0) \oplus (0, 1/2)$, couple to the gauge field through the covariant derivative

$$D_\mu \psi = (\partial_\mu + g A_\mu^a T^a) \psi.$$

The factor $SU(2)_L$ acts on the left-handed component, and $U(1)_Y$ acts with different weights on the two chiral components.

Proof. The curvature formula $F = dA + g[A, A]$ is a standard consequence of principal bundle geometry. The non-Abelian character $[A_\mu, A_\nu] \neq 0$ follows from the non-vanishing structure constants $f^{abc} \neq 0$ of the non-Abelian factors $\mathfrak{su}(3)$ and $\mathfrak{su}(2)$. The three-gluon and four-gluon vertices are algebraic consequences of the expansion of $\text{Tr } F^2$. The fermion–gauge coupling is given by the minimal coupling through the covariant derivative, together with the chiral content (A2). $\square$

Theorem 8.5 (Measure construction). *Let A_μ be the gauge field emerging from Axiom 3.4 via Theorem 4.12. Its Euclidean measure is defined after emergence and consists of two parts.*

For the matter fields, the base space is the continuous Euclidean space V . The matter fields are the chiral fermions of axiom (A2). Their free measure $d\mu_{\text{mat}}$ is taken as a Grassmann Gaussian measure. The free covariance is given by the spectral representation of the Gaussian structure. The Berezin integral is well defined in finite volume, and the infinite-volume limit is realised through a quasi-free state construction.

For the gauge fields, the field strength F contains nonlinear self-interactions, and its measure is non-Gaussian. The total measure is defined by the gauge-invariant action

$$S = S_{\text{gauge}}[F] + S_{\text{ferm}}[\psi, A],$$

where S_{gauge} is the $\text{Tr } F^2$ term and S_{ferm} is the minimal coupling term. The forms of these terms are determined by the requirement of gauge invariance and the minimal coupling structure. The scale dependence is governed by the scale semigroup.

Proof. The gauge group G_{SM} is given by Theorem 4.12. The field strength and the action are determined by gauge invariance and minimal coupling, as stated in Theorem 8.4. The measure for the matter fields is supplied by axiom (A3) and the spectral representation of the Gaussian structure. $\square$

8.3 Technical prerequisites

Before entering the scale-flow dynamics, this section records the basic technical prerequisites for the measure of the gauge and matter fields. These points concern lattice regularisation, fermion doubling, the chiral determinant, and the $U(1)$ sector. They are independent of the dynamical analysis in the subsequent sections.

Remark 8.6 (Lattice regularisation). The gauge-field measure contains the nonlinear F^2 term. In continuous form it is a formal path integral. A standard way to define the gauge-field measure is lattice regularisation, with the lattice spacing a serving purely as a regularisation parameter. The emergent geometry is continuous, and no finite spacing acts as a physical minimal length. A hypercubic lattice breaks $SO(4)$ to its hyperoctahedral subgroup at finite spacing. The full $SO(4)$ invariance of the emergent geometry is recovered in the continuum limit.

Remark 8.7 (Compatibility of the lattice with the emergent gauge group). The lattice action is taken as the standard plaquette action for the gauge theory with group G_{SM} , with link variables $U_p \in G_{\text{SM}}$. The plaquette expansion gives

$$S_{\text{plaq}} = \frac{a^4}{4g^2} \int \text{Tr } F^2 + O(a^6)$$

for any compact Lie group G . At the formal level the continuum limit recovers the Lie-algebra structure $SU(3) \times SU(2) \times U(1)$ of Theorem 4.12. The lattice gauge group coincides with the emergent gauge group.

Remark 8.8 (Measure in the spectral-sum definition). In the spectral-sum definition the free part of the measure is defined directly by the maximum-entropy principle of axiom (A3) together with the spectral representation of the Gaussian structure. No cutoff parameter appears. The interaction is subsumed into the scale-flow evolution, and physical quantities are given by spectral sums at the flow endpoint. The lattice construction of Remark 8.6 serves only as a technical device for establishing reflection positivity in the gauge sector. It does not belong to the definition of the measure itself.

Proposition 8.9 (No doubling). *The formulation does not produce a fermion doubling problem.*

Proof. The Nielsen–Ninomiya theorem constrains local, translation-invariant, Hermitian, chiral fermion actions on a lattice. Such actions necessarily produce doublers. The matter-field measure in Theorem 8.5 is a continuous Grassmann Gaussian measure. It does not

introduce a lattice fermion action. The hypothesis of the Nielsen–Ninomiya theorem is therefore not satisfied. The fermion content is fixed at the operator level by the representation theory of axiom (A2) and the spectral data on RP^3 . $\square$

Proposition 8.10 (Well-definedness of the chiral determinant). *Let D_+ be the Weyl operator coupled to the gauge field A . Its determinant $\det D_+$ is well defined.*

Proof. One has

$$|\det D_+|^2 = \det(D_+^\dagger D_+),$$

where $D_+^\dagger D_+$ is a self-adjoint positive elliptic operator with a discrete non-negative spectrum. Its spectral zeta function

$$\zeta(s) = \sum_j \lambda_j^{-s}$$

admits a meromorphic continuation, and the determinant is defined by

$$\det = \exp(-\zeta'(0))$$

after regularisation. On the compact internal space this is the standard zeta-function construction. On the non-compact base V the determinant is regularised within the spectral-sum definition of Definition 8.1.

The phase of $\det D_+$ is well defined if and only if all gauge anomaly coefficients vanish. The fermion content used for this verification is the standard model content, one generation Q_L, u_R, d_R, L_L, e_R with the standard hypercharge assignment $Y = (1/6, 2/3, -1/3, -1/2, -1)$. This content is taken as a phenomenological input of the framework at this stage.

One verifies term by term. The $SU(3)^2 U(1)$ anomaly vanishes, $2y_Q - y_u - y_d = 0$. The $SU(2)^2 U(1)$ anomaly vanishes, $3y_Q + y_L = 0$. The $U(1)^3$ anomaly vanishes, $\sum Y^3 = 0$. The gravitational $U(1)$ anomaly vanishes, $\sum Y = 0$. The $SU(3)^3$ anomaly vanishes because the colour representation is vector-like. The Witten $SU(2)$ anomaly is avoided because the number of doublet states is four, which is even.

In the sums $\sum Y^3$ and $\sum Y$ all fermion fields are counted as left-handed Weyl components. The right-handed fields enter with the opposite hypercharges. The cancellation holds per generation, independently of the total number of generations. All anomaly coefficients of the standard model content are zero. The Weyl determinant is therefore well defined for this content, and the chiral fermion measure exists. The anomaly freedom of any further fermion content of the framework itself is not asserted here. $\square$

Remark 8.11 (Status of the fermion content). The fermion content used in Proposition 8.10 is a phenomenological input. The gauge group G_{SM} determines the group structure but not the representation content. The colour and weak representations and the hypercharge assignments are representation-theoretic data fixed by experiment. The framework verifies the consistency of this input through anomaly freedom but does not derive the representations themselves.

Remark 8.12 (The $U(1)$ sector). Triviality of four-dimensional pure $U(1)$ gauge theory is at present only a Landau-pole conjecture. Rigorous triviality theorems exist only for ϕ_4^4 theory and do not apply to $U(1)$ gauge theory. In this paper the scale evolution is defined on a finite interval, and the $U(1)_Y$ coupling does not encounter a Landau pole within that interval. The statements of the paper concerning the positive lower bound in the colour-singlet sector and the spectral-sum-definition analogue of the Wightman axioms do not depend on the non-triviality of the $U(1)$ sector in the continuum limit.

8.4 Scale flow and consistent family

When the scale parameter σ varies continuously, the effective theory evolves accordingly. This evolution is generated by the scale semigroup. The present section develops the mathematical description of this evolution and records the consistency and analytic properties of the scale family.

Lemma 8.13 (β -function). *Let V_{eff} be the effective potential and let $k = 1/\sigma$ be the scale parameter. Then*

$$V_{\text{eff}}(k) = -\frac{C}{2}(k^4 - M_P^4), \quad \beta(\sigma) = -\frac{dV_{\text{eff}}}{d\sigma} = -2C\sigma^{-5}, \quad C < 0.$$

The reference scale M_P is a fixed boundary value of the scale parameter. Its numerical value is not used in the present paper.

Proof. The renormalisation group source is a pure power law $\text{source}(k) = Ck^4$. Its form is fixed by dimensional analysis and homogeneity. The unbiased structure of axiom (A3) forces the scale-dependent part of the effective potential to be a single power proportional to k^4 . The ultraviolet divergence structure of the coefficient C is supplied by the heat-kernel expansion of the spectral sums on the internal space [13], with leading Seeley–DeWitt coefficient $a_0 \propto \text{Vol Str}(1)$. Hence $C \propto a_0 \propto \text{Str}(1)$. The scale derivative is $dV/dk = -2\text{source}(k)/k = -2Ck^3$. With the boundary condition $V_{\text{eff}}(M_P) = 0$, integration gives

$$V_{\text{eff}}(k) = -\frac{C}{2}(k^4 - M_P^4).$$

From $d\sigma/dk = -1/k^2$ one obtains

$$\beta(\sigma) = -\frac{dV}{d\sigma} = -2C\sigma^{-5}.$$

The sign $C < 0$ follows from fermion dominance in the spectral representation. A sufficient condition is given in Proposition 8.14. The scope of this β -function is the free Gaussian part within the local potential approximation, as discussed in Remark 8.18. $\square$

Proposition 8.14 (Condition for $C < 0$). *Let C be the source coefficient of Lemma 8.13,*

$$C \propto \text{Str}(1) = \sum_b \text{Tr } 1 - \sum_f \text{Tr } 1.$$

For the emergent content of the present paper, with n_g generations, 15 Weyl fermions per generation, 12 gauge bosons, one massless spin-2 excitation, and n_s scalars, one has

$$C < 0 \iff 30n_g > 29 + n_s.$$

The condition holds for $n_g \geq 2$. The scalar content satisfies $n_s \lesssim 10$, so $C < 0$ is satisfied.

Proof. The homogeneity of the spectral representation implies $\text{source}(k) \propto k^4 \text{Str}(1)$. The fermionic contribution is $-15n_g \cdot 2 = -30n_g$, and the bosonic contribution is $12 \cdot 2 + 5 + n_s = 29 + n_s$. The weights count physical field degrees of freedom in the heat-kernel trace. Two transverse polarisations per gauge boson are counted, with the remaining components removed by the Faddeev–Popov procedure. The transverse-traceless spin-2 field contributes $10 - 4 - 1 = 5$ independent components. Each Weyl fermion contributes weight 2 relative to a real scalar, and each real scalar contributes weight 1. Thus $C < 0$ is equivalent to $\text{Str}(1) < 0$, which is $30n_g > 29 + n_s$. $\square$

Theorem 8.15 (Spin-2 sector). *The spin-2 content of the $\text{Str}(1)$ counting arises from the fermionic content. The conserved symmetric tensor of the chiral fermions of axiom (A2) under minimal coupling has a transverse-traceless part with $10 - 4 - 1 = 5$ independent components. The weight 5 is therefore the component count of the spin-2 sector, not an independent input.*

Proof. Translation invariance of the emergent action of Theorem 8.5 gives, by Noether’s theorem, a conserved symmetric tensor $T_{\mu\nu}$. Its transverse-traceless part is the spin-2 sector of the framework. The component count is standard, $10 - 4 - 1 = 5$. $\square$

Proposition 8.16 (Generation count). *The number of retained spinor modes of RP^3 is fixed by the window capacity of the scale flow. The coarse graining retains the spinor modes whose internal momentum lies below the inverse radius $1/L$, and the retained modes are those whose dimensionless Dirac eigenvalue $n + 3/2$ lies below the window capacity $(kL)^2$. Here k is the flow momentum scale and L is the internal radius. The window capacity is fixed self-consistently by the spectral sums of the framework, not chosen by hand.*

Proof. The Dirac spectrum of the selected spin structure of RP^3 is the tower $\pm(n + 3/2)/L$ for $n = 0, 2, 4, \dots$, computed in Appendix A. The coarse graining of the scale flow retains the modes whose wavelength exceeds the internal radius L , which is the condition $n + 3/2 < (kL)^2$. The window capacity $(kL)^2$ is the value at which the spectral sums close, and its determination belongs to the numerical evaluation of the framework. The extrusion order $n = \{0, 2, 4\}$ of the scale flow gives the generation mass hierarchy. $\square$

Remark 8.17. The generation count follows from the window capacity rather than from an independent threshold. The window capacity $(kL)^2$ is fixed self-consistently by the spectral sums, and its numerical value determines how many spinor modes are retained. Its determination belongs to the numerical evaluation of the framework.

Remark 8.18 (Status of the β -function). The β -function of Lemma 8.13 is the derivative of the effective potential along the scale semigroup, obtained within the local potential approximation. The derivation uses only the homogeneity of axiom (A3) and the spectral representation of the Gaussian structure. For a Gaussian measure the heat-kernel smoothing $e^{-\sigma^2 \Delta/2}$ is an exact conditional expectation, and the scale dependence of the free part of the effective potential can be computed exactly from the spectral zeta function. The β -function therefore governs the evolution of the free Gaussian part of the effective potential. The non-Abelian interaction content of Theorem 8.4 contributes to the effective potential beyond the free part. Its back-reaction on the spectral data is controlled by Theorem 8.19, and its standard-formulation realisation is analysed in Section 10. All statements that use the β -function are statements about the spectral-sum definition (Definition 8.1).

Theorem 8.19 (One-loop stability of the scalar lower bound). *The interaction back-reaction enters the scale flow through the running of the quartic coupling λ . In the local potential approximation its one-loop β -function is non-negative for $\lambda \geq 0$, with $\beta_\lambda = c\lambda^2$ and $c > 0$. The analytic positivity $\lambda > 0$ of Proposition 8.31 is therefore preserved along the flow. The condensation condition $m_{\text{long}}^2 < 0$ for $R < R_c$ and the lower bound $m_\delta^2 = 2\lambda\langle E \rangle^2 > 0$ are stable under this back-reaction.*

Proof. Within the local potential approximation the running of the quartic coupling of a scalar with $\lambda \geq 0$ is $\beta_\lambda = c\lambda^2$ with $c > 0$. Hence λ is non-decreasing along the flow, and its positivity, established independently by Proposition 8.31, is preserved at every scale. The condensation threshold and the lower bound formula depend on λ only through its positivity. They are therefore unaffected by the running of λ . $\square$

Theorem 8.20 (Scale evolution and consistent family). *The evolution of the family of effective potentials along the scale semigroup is driven by $\beta(\sigma) = -2C\sigma^{-5}$ and admits a unique global solution on the interval $\sigma \in [M_P^{-1}, \sigma_C]$, where σ_C is the evaluation endpoint of Definition 8.1, lying in the condensed phase beyond the Gaussian breakdown scale of Section 8.6. The flow map Φ_σ forms a semigroup, so the family of effective actions is compatible along the scale semigroup. The flow preserves the $SO(4)$ -invariance of the spectral data.*

Proof. The function $\beta(\sigma) = -2C\sigma^{-5}$ is analytic and Lipschitz continuous on the interval. The ordinary differential equation $dV_{\text{eff}}/d\sigma = -\beta(\sigma)$ has a unique solution by the Picard–Lindelöf theorem. The semigroup property follows from uniqueness of the initial value problem. Because β depends only on $k = 1/\sigma$, it is isotropic, and therefore $SO(4)$ -invariance of the initial spectral data is preserved at all scales. $\square$

Lemma 8.21 (Convergence of spectral sums). *The physical quantities, including the effective potential and the quartic coupling λ , are sums over the spectrum of RP^3 , with $d_l = (l+1)^2$ and $\lambda_l = l(l+2)/L^2$ for even l . For $p > 3/2$ the series $\sum_l d_l(\lambda_l + m^2)^{-p}$ converges absolutely. The two-point function is not the multiplicity-only sum. It is the full spectral kernel of Section 10, the resolvent of $\Delta_{\mathbb{R}^4} + \Delta_{RP^3} + m_\delta^2$, a tempered distribution on*

$\mathbb{R}^4 \times RP^3$. The multiplicity-only sum, the case $p = 1$, diverges linearly and is the divergent diagonal of the RP^3 Green kernel, not itself a distribution.

Proof. Since $\lambda_l \sim l^2/L^2$ and $d_l \sim l^2$, the general term behaves as $L^{2p}l^{2-2p}$, which converges absolutely when $2p - 2 > 1$, that is, $p > 3/2$. For $p = 1$ the multiplicity sum diverges linearly. It is the diagonal $G(y, y) = \sum_l d_l(\lambda_l + m^2)^{-1}$ of the RP^3 Green kernel $G(y, y') = \sum_l K_l(y, y')(\lambda_l + m^2)^{-1}$, where K_l is the spectral projector kernel of the l -th eigenspace. The kernel is a distribution with a $1/r$ short-distance singularity, while its diagonal is divergent. The distributional objects used by the construction are the kernel itself and the product-space covariance of Section 10. A positive lower bound on the spectral mass removes possible infrared singularities. $\square$

Lemma 8.22 (Reflection positivity as a closed condition). *If a family of measures $\{\mu_a\}$ satisfies reflection positivity and converges weakly to a measure μ , then μ also satisfies reflection positivity.*

Proof. Weak convergence means $\int G d\mu_a \rightarrow \int G d\mu$ for every bounded continuous G . For reflection positivity one takes $G = F(\theta F)$. For bounded continuous F the non-negativity passes to the limit directly. For the unbounded polynomial functionals of the present construction, the closure is the standard result for measures with uniform moment bounds. $\square$

8.5 Spectral-sum-definition conditions and reconstruction

The Osterwalder–Schrader axioms provide sufficient conditions for a Euclidean quantum field theory to yield a relativistic quantum field theory satisfying the Wightman axioms. The present section records the spectral-sum-definition analogues of these conditions for the measure $d\mu$. The clustering analogue is a consequence of the positive scalar lower bound established in Section 8.6.

Theorem 8.23 (Reflection positivity). *Let $\theta : \tau \mapsto -\tau$ denote Euclidean time reflection. For the measure $d\mu$ constructed in Theorem 8.5, the associated Schwinger functions satisfy reflection positivity,*

$$\mathbb{E}[F(\theta F)] \geq 0$$

for every polynomial F generated by fields $\phi(f)$ with $\text{supp } f \subset \{\tau > 0\}$.

Proof. The total measure splits into a matter part and a gauge part. The matter measure is a continuous Grassmann Gaussian measure. Its reflection positivity is a standard result. For the gauge measure, which contains the nonlinear F^2 term, we use the lattice regularisation of Remark 8.6. At finite lattice spacing, reflection positivity for lattice gauge theory is a rigorous theorem. The F^2 and minimal coupling terms are real and positive definite under time reflection. If a continuum limit is taken, reflection positivity is preserved by Lemma 8.22. $\square$

Theorem 8.24 (Spectral-sum-definition conditions). *The measure $d\mu$ satisfies the spectral-sum-definition analogue of regularity, reflection positivity, and $SO(4)$ -invariance at the level of spectral data.*

Proof. Regularity corresponds to the temperedness of the spectral kernel and the analyticity of the β -function of Lemma 8.13. Reflection positivity is Theorem 8.23. Symmetry corresponds to the Euclidean covariance of the matter and gauge field actions. The $SO(4)$ -invariance of spectral data follows from Theorem 8.20, which shows that the scale flow preserves the $SO(4)$ -equivariance of the spectral quantities. For the free part the distribution property and reflection positivity hold exactly. For the gauge part they hold at finite lattice spacing, and closure under weak limits preserves them in any limit taken. $\square$

8.6 Positive scalar lower bound

This section records the mechanism by which the colour-singlet scalar sector acquires a positive lower bound. The mechanism has three steps. First, the spectral value of the long-root mode decreases as the internal space radius grows. Beyond a critical value it becomes tachyonic and triggers condensation of the glueball field. Second, the strict positivity of the conformal Laplacian on RP^3 guarantees that the quartic coupling of the effective potential is positive, so condensation occurs at a non-zero vacuum expectation value. Third, the explicit RP^3 spectrum excludes massless modes in the colour-singlet sector.

Lemma 8.25 (Long-root mode). *The long-root mode of the colour connection is the $J = 2$ deformation mode of the R -sector, transforming in the representation $(2, 1)$ of $\text{Spin}(4)$. Its internal eigenvalue is*

$$\lambda_{\text{long}} = \frac{C_2(2, 1)}{L^2} = \frac{16}{L^2},$$

where L is the radius of the internal space RP^3 and $C_2(j_L, j_R) = 2[j_L(j_L + 1) + j_R(j_R + 1)]$ is the quadratic Casimir.

Proof. The quantum numbers $(2, 1)$ follow from the harmonic analysis of the R -sector connection modes on RP^3 in Appendix A. The long root of $\mathfrak{su}(3)$ carries charge 2 under the unbroken torus $\mathfrak{u}(1)_R$, hence $j_R = 1$. The first non-zero internal mode of the R -sector connection carrying this charge is the $l = 3$ harmonic $(j_L, j_R) = (2, 1)$. The $l = 1$ sector is the Killing zero-mode sector, and the 1-form sector of RP^3 retains odd l . The internal eigenvalue in the framework normalisation is the Casimir value C_2/L^2 , and for $(2, 1)$ one has $C_2 = 2[2 \cdot 3 + 1 \cdot 2] = 16$. The mode survives in the even antipodal sector of RP^3 . $\square$

The onset radius of condensation $L_c = \sqrt{\pi}$ is determined by Theorem 8.27. The corresponding curvature scalar is $R = 6/L^2$, and the critical curvature is $R_c = 6/L_c^2 = 6/\pi$. The condensation onsets at the Gaussian breakdown scale $\sigma_c = 1/\sqrt{2}$, equivalently $L = L_c$. The evaluation endpoint σ_C of the framework lies beyond σ_c , in the condensed phase. The scale flow continues past the Gaussian breakdown, so that $L(\sigma_C) > L_c$ and $R(\sigma_C) < R_c$. The lower bound $m_\delta > 0$ of Theorem 8.32 is strictly positive at the evaluation endpoint.

Theorem 8.26 (Long-root tachyon). *The squared mass of the long-root mode is*

$$m_{\text{long}}^2(L) = \lambda_{\text{long}} - \lambda_{\text{long}}(L_c) = \frac{8}{3}(R - R_c).$$

When $R < R_c$, $m_{\text{long}}^2 < 0$. The long-root mode is a tachyon and triggers condensation of the glueball field.

Proof. We have $\lambda_{\text{long}} = 16/L^2$, and at $L_c = \sqrt{\pi}$ one has $\lambda_{\text{long}}(L_c) = 16/\pi$. Taking the difference and rewriting in terms of the curvature scalar $R = 6/L^2$ and its critical value $R_c = 6/\pi$ yields the stated formula. $\square$

The origin of the tachyon lies in the breakdown of the Gaussian statistical description at the scale σ_c . Beyond this scale the theory evolves in the condensed phase toward the evaluation endpoint σ_C .

Theorem 8.27 (Condensation threshold). *The onset radius of condensation is $L_c = \sqrt{\pi}$, equivalently $R_c = 6/\pi$.*

Proof. At the Gaussian breakdown scale of the scale flow the unbiased Gaussian structure of (A3) saturates the entropic uncertainty bound. For a one-dimensional Gaussian the Beckner–Hirschman inequality reads $H_x + H_p \geq 1 + \ln \pi$, with equality if and only if the distribution is Gaussian. The unbiased structure of (A3) is Gaussian, so equality holds. Unbiasedness between position and momentum spreads gives $H_x = H_p$, hence $H_x = \frac{1}{2}(1 + \ln \pi)$. For a Gaussian $N(0, \sigma^2)$ the position entropy is $H_x = \frac{1}{2} \ln(2\pi e \sigma^2)$. Equating gives $2\pi e \sigma^2 = \pi e$, so $\sigma^2 = 1/2$, and the breakdown scale is $\sigma_c = 1/\sqrt{2}$. The internal radius is defined as the Gaussian integral width, $L = \sqrt{2\pi} \sigma = \sqrt{\pi}$. The constant curvature of RP^3 then gives $R_c = 6/L^2 = 6/\pi$. $\square$

Lemma 8.28 (Strict positivity of the conformal Laplacian). *On RP^3 with curvature scalar $R = 6/L^2$, the conformal Laplacian $\mathcal{L} = \Delta + \frac{1}{8}R$ has spectrum*

$$\text{spec}(\mathcal{L}) = \left\{ \frac{l(l+2) + 3/4}{L^2} : l = 0, 2, 4, \dots \right\} \subset \left[\frac{3}{4L^2}, +\infty \right).$$

Hence $\mathcal{L} \geq 3/(4L^2) > 0$, and it is strictly positive.

Proof. The scalar Laplacian spectrum on RP^3 is $\lambda_l = l(l+2)/L^2$ for even l . The three-dimensional conformal coupling is $\xi = 1/8$. The eigenvalues of $\mathcal{L}$ are $\lambda_l + \xi R$, and the minimum at $l = 0$ is $3/(4L^2)$. $\square$

The strict positivity of the conformal Laplacian is the central analytic input for the positive lower bound. It guarantees that the quartic coupling of the glueball effective potential is non-perturbatively strictly positive.

Theorem 8.29 (Effective potential from spectral sums). *Let E be the colour-singlet scalar, the glueball field, coupled to the spectral data through a quadratic coupling entering the spectrum of the conformal Laplacian, $\lambda_l + \xi R \mapsto \lambda_l + \xi R + gE^2$, where $g > 0$ is the quadratic coupling (a framework coupling constant, not fixed in absolute value in this paper) and ξR the conformal shift of Proposition 8.31. The effective potential of E is the flow-endpoint spectral sum*

$$V(E) = \sum_l d_l f(\lambda_l + \xi R + gE^2),$$

with f the one-loop function of the framework. Expanding in E to fourth order gives

$$V(E) = V_0 + \frac{1}{2} m_{\text{long}}^2 E^2 + \frac{\lambda}{4} E^4 + O(E^6),$$

with spectral-sum coefficients

$$m_{\text{long}}^2 = 2g \sum_l d_l f'(\lambda_l + \xi R), \quad \lambda = 6g^2 \sum_l d_l f''(\lambda_l + \xi R).$$

The sums converge absolutely by Lemma 8.21. With f the conformal-Laplacian logarithm the coefficient λ reproduces the quartic coupling of Proposition 8.31, as shown in the proof. The Landau form of the effective potential is the fourth-order truncation of the spectral-sum expansion, not an independent input.

Proof. The Taylor expansion of the spectral sum in E is

$$f(\lambda_l + \xi R + gE^2) = f(\lambda_l + \xi R) + gE^2 f'(\lambda_l + \xi R) + \frac{1}{2}g^2 E^4 f''(\lambda_l + \xi R) + O(E^6).$$

Summing with multiplicities d_l gives $V_0 = \sum_l d_l f(\lambda_l + \xi R)$, $m_{\text{long}}^2 = 2g \sum_l d_l f'(\lambda_l + \xi R)$, and $\lambda = 6g^2 \sum_l d_l f''(\lambda_l + \xi R)$. The convergence follows from Lemma 8.21. For the conformal-Laplacian logarithm $f(\lambda) = -\ln \lambda$, one has $f''(\lambda_l + \xi R) = (\lambda_l + \xi R)^{-2}$, so that

$$\lambda = 6g^2 \sum_l d_l (\lambda_l + \xi R)^{-2},$$

the quartic coupling of Proposition 8.31 with normalisation $\kappa = 6g^2$. The sign is positive by the analytic positivity of Proposition 8.31. The higher-order terms $O(E^6)$ are part of the full spectral-sum potential. The Landau form is the truncation at fourth order. $\square$

Remark 8.30 (Two derivations of the long-root mass). The long-root mass m_{long}^2 is determined twice, geometrically and spectrally. Theorem 8.26 gives the explicit geometric closed form $\frac{8}{3}(R - R_c)$ from the Casimir eigenvalue of the long-root mode. The spectral-sum coefficient $m_{\text{long}}^2 = 2g \sum_l d_l f'(\lambda_l + \xi R)$ of Theorem 8.29 is the general form; with f the conformal-Laplacian logarithm it is the $p = 1$ divergent diagonal $-2g \sum_l d_l (\lambda_l + \xi R)^{-1}$ of Lemma 8.21, whose renormalised value is identified with the geometric closed form. The two derivations therefore determine the same long-root mass, the geometric form supplying the explicit value used in Theorem 8.32.

Proposition 8.31 (Analytic positivity of the quartic coupling). *Let E be the glueball field, a colour-singlet scalar. Its effective potential $V(E) = \frac{1}{2}m_{\text{long}}^2 E^2 + \frac{\lambda}{4}E^4$ has the quartic coupling*

$$\lambda = \kappa \sum_{l=0,2,4,\dots} \frac{d_l}{(\lambda_l + \xi R)^2}, \quad d_l = (l+1)^2, \quad \kappa > 0.$$

The series converges and every term is strictly positive. Hence $\lambda > 0$ holds analytically.

Proof. In the spectral-sum definition the quartic coupling is defined by the spectral sum of the statement. The expression $\lambda \propto \text{Tr } \mathcal{L}^{-2}$ is its one-loop form obtained from the effective potential of the glueball field. The normalisation constant is $\kappa = 6g^2 > 0$, where g is the quadratic coupling of Theorem 8.29; this identification is established in the proof of that theorem. Lemma 8.28 guarantees $\lambda_l + \xi R \geq 3/(4L^2) > 0$ termwise. The general term behaves asymptotically as L^4/l^2 , so the series converges absolutely. Its closed form is given by the polygamma function. $\square$

Theorem 8.32 (Positive scalar mass in the Landau potential). *Let E be the glueball field with an effective potential of Landau form and $\lambda > 0$. When $R < R_c$ the potential attains its minimum at*

$$\langle E \rangle^2 = \frac{8}{3} \frac{R_c - R}{\lambda} > 0,$$

so the glueball field condenses. The fluctuation $\delta E = E - \langle E \rangle$ has squared mass

$$m_\delta^2 = V''(\langle E \rangle) = 2\lambda \langle E \rangle^2 = \frac{16}{3}(R_c - R) > 0.$$

The lowest colour-singlet excitation therefore has a positive scalar mass $m_\delta > 0$.

Proof. The Landau form is the lowest-order symmetric allowed potential for a colour-singlet scalar. From $m_{\text{long}}^2 < 0$ and $\lambda > 0$ the minimum lies at a non-zero field value. The glueball field $E = \text{Tr } F_{\mu\nu} F^{\mu\nu}$ is a colour singlet. Its condensation breaks no continuous symmetry, so no Goldstone boson appears. The fluctuation mass follows from the second derivative evaluated at the minimum. $\square$

Theorem 8.33 (Structure of the effective potential). *Let E be the glueball field. The one-loop effective potential of the framework has the form*

$$V(E) = \frac{1}{2} m_{\text{long}}^2 E^2 + \sum_{k \geq 2} c_k E^{2k}, \quad c_k > 0,$$

where the coefficients c_k are convergent spectral sums by Definition 8.1 and Lemma 8.21. For $m_{\text{long}}^2 < 0$ the potential has a unique minimum at a non-zero field value, and the fluctuation mass at the minimum is strictly positive. The Landau truncation of Theorem 8.32 preserves this structure. The higher-order terms do not affect the existence or the positivity of the lower bound.

Proof. The one-loop effective potential is the spectral sum over the glueball modes,

$$V(E) = \frac{1}{2} m_{\text{long}}^2 E^2 + \frac{1}{2} \sum_l d_l h(\lambda_l + \xi R + g E^2),$$

where h is the one-loop function of the framework. Expanding in E^2 gives $V(E) = \frac{1}{2} m_{\text{long}}^2 E^2 + \sum_{k \geq 2} c_k E^{2k}$ with

$$c_k \propto \sum_l d_l (\lambda_l + \xi R)^{-k}.$$

Every term of every coefficient is positive, and the sums converge for $k \geq 2$. Thus $c_k > 0$. For $m_{\text{long}}^2 < 0$ the derivative

$$V'(E) = E \left(m_{\text{long}}^2 + \sum_{k \geq 2} 2k c_k E^{2k-2} \right)$$

has a bracket that is strictly increasing in E^2 , negative at $E = 0$ and positive for large E . Hence there is exactly one positive root $E_*^2 > 0$, the unique minimum. The fluctuation mass

$$V''(E_*) = \sum_{k \geq 2} 2k(2k-1)c_k E_*^{2k-2} > 0$$

is strictly positive. The Landau truncation keeps $c_2 = \lambda/4 > 0$ and the same structure. The conclusion is therefore independent of the higher-order terms. $\square$

The existence of the positive lower bound is insensitive to the precise value of the long-root spectral coefficient. The condensation condition $m_{\text{long}}^2 < 0$ is the inequality $L > L_c$, and $\lambda_{\text{long}}(L) = 16/L^2$ decreases with L . The absolute scale of the lower bound is set by the internal radius L . Within the present construction L is determined only up to the condensation threshold and the evaluation endpoint σ_C , so the numerical value of m_δ in physical units is not fixed here. The scalar glueball masses obtained in lattice QCD simulations [21] provide a phenomenological reference scale. Fixing the absolute scale of L is not part of the present construction. The lower bound m_δ is a statement about the spectral-sum definition. The relation of this bound to the standard continuum formulation is analysed in Section 10. The standard-formulation realisation of the colour-singlet data carries the same lower bound m_δ .

Theorem 8.34 (Absence of massless modes). *On RP^3 the scalar spectrum contains a zero mode that carries the glueball field. The first de Rham cohomology vanishes,*

$$H_{dR}^1(RP^3; \mathbb{R}) = 0,$$

so there are no harmonic 1-forms. The Laplacian on transverse-traceless symmetric tensors obeys the Bochner formula

$$\square_L \omega = \nabla^* \nabla \omega + \text{Ric} \cdot \omega$$

with $\text{Ric} = \frac{2}{L^2}g > 0$, hence is strictly positive and admits no massless TT modes. Together with $m_\delta^2 > 0$ from Theorem 8.32, the colour-singlet spectrum contains no massless mode. The positive lower bound is consistent within the spectral-sum definition.

Proof. The scalar Laplacian spectrum on RP^3 is explicit, with $\lambda_l = l(l+2)/L^2$ and multiplicities $d_l = (l+1)^2$ for even l . The $l = 0$ mode is the constant function, which carries the glueball field E . The vanishing of the first cohomology and the strict positivity of the TT Laplacian are standard results in spectral geometry [13]. The Kaluza–Klein zero modes of the gauge fields belong to the 1-form sector and lie outside the colour-singlet spectrum. The term zero mode here refers to modes with vanishing internal momentum. The sole scalar zero mode carries the glueball field, and its fluctuation mass $m_\delta > 0$ is supplied by Theorem 8.32. $\square$

A positive lower bound $m_\delta > 0$ implies exponential clustering of the Schwinger functions, which is the clustering analogue in the spectral-sum definition. Within the spectral-sum definition the positive scalar mass $m_\delta > 0$ implies exponential decay of the colour-singlet Schwinger functions. The spectral-sum-definition analogue of clustering therefore holds, and the vacuum is unique.

8.7 Endpoint physical quantities

The evolution of the effective potential along the scale semigroup leads to condensation in the colour-singlet scalar sector and to a positive scalar mass parameter m_δ . It remains to record how the physical quantities at the flow endpoint are defined and to state their well-definedness.

In the present construction the basic object is the scale flow itself. The effective theory evolves as the scale parameter σ varies continuously. Physical quantities are evaluated at the endpoint σ_C . They are not defined by integrals over a measure. They are defined by spectral sums at the endpoint of the flow, as stated in Definition 8.1. These spectral sums are exact summations over the spectrum of the geometric Laplacian on the compact internal space RP^3 . They do not rely on a perturbative expansion.

Theorem 8.35 (Endpoint physical quantities). *In the spectral-sum definition the physical quantities at the flow endpoint σ_C are defined by spectral sums and closed analytic expressions. The effective potential and the squared masses are given by*

$$V_{\text{eff}}(k) = -\frac{C}{2} (k^4 - M_P^4), \quad m_{\text{long}}^2 = \frac{8}{3}(R - R_c), \quad m_\delta^2 = \frac{16}{3}(R_c - R).$$

The quartic coupling is given by the convergent spectral sum

$$\lambda = \kappa \sum_{l=0,2,4,\dots} \frac{d_l}{(\lambda_l + \xi R)^2}, \quad d_l = (l+1)^2, \quad \kappa > 0.$$

Its closed form is a polygamma function. Every term is strictly positive and the series converges absolutely.

Proof. The effective potential V_{eff} is given by Lemma 8.13. The squared mass m_{long}^2 is given by Theorem 8.26, and m_δ^2 by Theorem 8.32. The expression for the quartic coupling λ is given by Proposition 8.31, and its absolute convergence is guaranteed by Lemma 8.21. Each formula is explicit and does not rely on any limiting procedure. $\square$

The closed analytic expressions for the endpoint physical quantities are summarised in the following table.

Table 2. Endpoint physical quantities in the spectral-sum definition	
Physical quantity	Closed analytic expression
$V_{\text{eff}}(k)$	$-\frac{C}{2}(k^4 - M_P^4)$
$\beta(\sigma)$	$-2C\sigma^{-5}$
m_{long}^2	$\frac{8}{3}(R - R_c)$
$\langle E \rangle^2$	$\frac{8}{3}(R_c - R)/\lambda$
m_δ^2	$\frac{16}{3}(R_c - R)$
λ	$\kappa \sum_l d_l (\lambda_l + \xi R)^{-2}$, polygamma closed form

The squared masses m_{long}^2 and m_δ^2 follow directly from the spectral data and are exact closed forms. The effective potential V_{eff} is obtained within the local potential approximation, and the vacuum expectation value $\langle E \rangle^2$ is obtained within the Landau form of

the effective potential. These approximations do not affect the existence of the formulas. The formulas themselves exist explicitly, and their numerical accuracy is controlled by the respective truncations.

The endpoint physical quantities are therefore well defined within the spectral-sum definition. The relation of this construction to the standard path integral formulation is analysed in Section 10.

8.8 Vacuum, spin–statistics, and summary

The scale flow construction has supplied a Euclidean measure, the spectral-sum-definition conditions for the free and gauge sectors, a positive scalar lower bound, and closed-form expressions for the endpoint physical quantities. It remains to record the vacuum structure, the spin–statistics relation, and the overall status of the construction.

Theorem 8.36 (Vacuum). *(i) In the free sector the free matter field measure $d\mu_0$, Gaussian or Grassmann Gaussian, possesses a unique vacuum state Ω_0 , the Fock vacuum.*

(ii) Within the spectral-sum definition, the vacuum state Ω of the colour-singlet sector is the cyclic vector of the reconstruction of the spectral data. The condensation $\langle E \rangle \neq 0$ of Theorem 8.32 is a property of this state. In general Ω is not the free Fock vacuum.

Proof. Part (i) is standard. A Gaussian measure corresponds to a free field whose Hilbert space is a Fock space, and the Fock vacuum is unique. For part (ii), the reconstruction of the spectral data within the spectral-sum definition gives a cyclic vacuum vector Ω . Because the spectral data include the non-Gaussian content of the condensate and the quartic coupling, Ω is not the free Fock vacuum Ω_0 . The non-vanishing vacuum expectation value $\langle E \rangle \neq 0$ is a property of Ω . In the free theory one has $\langle E \rangle_{\Omega_0} = 0$. No contradiction arises because the two statements refer to different Hilbert spaces. $\square$

The positive scalar mass $m_\delta > 0$ implies exponential decay of the relevant correlation functions. This is the spectral-sum-definition analogue of clustering. It is consistent with the uniqueness of the vacuum within the framework, but a full derivation of vacuum uniqueness would require additional assumptions. For the purposes of this paper, the vacuum of the free sector is unique, and the interacting vacuum is characterised as the cyclic vector of the reconstruction.

Theorem 8.37 (Spin–statistics, standard theorem). *Under axiom (A2) the chiral Weyl pair $(1/2, 0) \oplus (0, 1/2)$ corresponds to quantum fields obeying Fermi anticommutation relations. Scalar fields obey Bose commutation relations.*

Proof. The chiral spinor fields carry spin $1/2$. The Pauli spin–statistics theorem states that any field with half-odd-integer spin must satisfy anticommutation relations. Axiom (A2) supplies the half-odd-integer spin content, and the conclusion follows. $\square$

Theorem 8.38 (Locality). *Let x and y be spacelike separated points. Then*

$$[\varphi_i(f), \varphi_j(g)]_{\mp} = 0$$

whenever the supports of f and g lie in spacelike separated neighbourhoods of x and y .

Proof. The reconstruction from reflection positivity gives the locality property of the fields. This part of the reconstruction does not require the clustering condition. $\square$

Theorem 8.39 (Summary of results within the spectral-sum definition). *Let A_μ be the gauge field of Theorem 4.12, equipped with the Euclidean measure $d\mu$ defined after emergence in Theorem 8.5. Then the following hold.*

- (i) *Within the spectral-sum definition, the physical quantities of the colour-singlet sector defined in Theorem 8.35 are well defined through spectral sums at the flow endpoint. The spectral data satisfy the spectral-sum-definition analogues of regularity and reflection positivity, as stated in Theorem 8.24, and are $SO(4)$ -invariant by Theorem 8.20. The colour-singlet sector carries a positive scalar mass $m_\delta > 0$ by Theorems 8.32 and 8.34.*
- (ii) *The relation to the standard continuum formulation is as follows. The non-trivial content of the framework, the positive quartic coupling and the condensate, has no known realisation in the standard continuum formulation. Only the free part is translatable. Its standard-formulation construction is given in Section 10.*

Proof. (i) The endpoint physical quantities are defined by Theorem 8.35 through closed analytic expressions whose convergence is guaranteed by Lemma 8.21. The positive scalar mass $m_\delta > 0$ follows from Theorems 8.32 and 8.34, which show that the colour-singlet scalar condenses without breaking any continuous symmetry and that no massless modes survive in the colour-singlet spectrum. (ii) The standard-formulation realisation of the free part is the free scalar field on $\mathbb{R}^4 \times RP^3$ of Theorem 10.6. Its Schwinger functions satisfy the standard OS axioms, and the OS reconstruction theorem yields a Wightman field theory satisfying W1–W6, with the same scalar mass m_δ . The field A_μ itself is not gauge-invariant. Treating it as an operator-valued distribution would require a gauge fixing that spoils locality. The Wightman fields are therefore the gauge-invariant local observables of the colour-singlet sector. $\square$

Within the spectral-sum definition the Wightman axioms are understood in the sense of Definition 8.1(iv). The spectral-sum-definition analogues of regularity, Euclidean invariance, reflection positivity, and clustering hold by Theorem 8.24, Theorem 8.20, and the positive scalar mass $m_\delta > 0$.

The construction is therefore internally consistent within the spectral-sum definition. It is not claimed that the construction satisfies the standard Wightman axioms in the measure-theoretic sense beyond the free sector.

The construction of this section supplies a concrete instance of the spectral datum of Section 5: the internal space is RP^3 with its Laplace spectrum (Section 4), and the physical quantities at the flow endpoint are the spectral sums of the framework. The following theorem records the resulting correspondence between the two axiom systems of this paper, the disorder axiom of Section 3 and the spectral-framework axioms of Section 5.

Theorem 8.40 (The gauge sector satisfies the framework axioms). *The gauge field A_μ constructed in this section, the $G_{\text{SM}} = SU(3) \times SU(2) \times U(1)$ connection of Theorem 4.12, on the internal space RP^3 with colour-singlet spectrum $\{\lambda_l = l(l+2)/L^2, d_l = (l+1)^2\}$ (Section 4), together with the endpoint physical quantities of the scale flow (Theorem 8.35) and the positive scalar mass m_δ (Theorem 8.32), constitutes a spectral datum satisfying Axioms 5.1–5.11 of Section 5. It therefore satisfies the formulation criterion of Theorem 5.14: the gauge sector is a quantum field theory within the spectral-sum definition, carrying the positive scalar mass $m_\delta > 0$.*

Proof. The axioms are verified item by item. (FSA-1) The spectral datum of the colour-singlet sector is the RP^3 spectrum with the flow-endpoint parameters, all constructed in Sections 4 and 8 (Axiom 5.1). (FSA-2) The physical quantities are the endpoint spectral sums of Theorem 8.35, of the form of Axiom 5.4. (FSA-3) The physical Hilbert space is the reconstruction space of Definition 6.3: on the free sector the Fock space (unconditional), and on the interacting sector the kinematic carrier and physical content established by the mapping relations (Theorems 7.10 and 9.6). (FSA-4) The Born rule is the kinematic rule of projective measurement (Theorems 6.4 and 6.19). (FSA-5) The dynamics is the reconstruction and Stone’s theorem (Theorem 6.13), with the Fock Hamiltonian on the free sector. (FSA-6) The composite structure is the tensor product of the reconstruction (Remark 6.8). (FSA-7) Locality holds on the free sector and on the effective sector (Theorem 6.10, Proposition 7.12). (FSA-8) The mapping holds: the free sector by the recovery theorem (Theorem C.2), the interacting sector by the analytic, structural, effective and non-perturbative mapping relations (Theorems 7.3, 7.8, 7.10, 9.6 and 9.4). The formulation criterion of Theorem 5.14 is therefore satisfied, item by item. The positive scalar mass is $m_\delta > 0$ (Theorem 8.32), with no massless mode in the colour-singlet spectrum (Theorem 8.34). $\square$

Remark 8.41 (Status). Theorem 8.40 is the correspondence between the two axiom systems of this paper: the disorder axiom of Section 3 and the spectral-framework axioms of Section 5. Within the spectral-sum definition the gauge sector is complete by Theorem 8.40, carrying the positive scalar mass m_δ . The standard-formulation counterpart of the colour-singlet scalar content is excluded by the triviality theorem (Remark 7.2); the standard-formulation counterpart of the gauge-field sector is the standard problem of Section 10, which the framework does not address in the standard formulation.

9 The formal correspondence and the non-perturbative content

The free-sector mapping of Appendix C proceeds through the Gaussian measure. Its covariance is the full spectral kernel, and Minlos’ theorem together with the Osterwalder–Schrader reconstruction produces the Fock space. The interacting content of the formulation, namely the derived parameters $\lambda > 0$ and $\langle E \rangle \neq 0$, has no standard-formulation continuum realisation. Its correspondence to the physical structure therefore follows the structure of the formulation rather than the Gaussian route.

9.1 The formal correspondence without a measure

The physical quantities of the formulation are defined without a measure. The measure appears only as the statistical content of the unbiasedness clause and as the optional correspondence of the free sector. The formal correspondence of the interacting sector is therefore constructed directly from the algebraic relations of the spectral data.

Definition 9.1 (Mapping dictionary). *The mapping dictionary is the correspondence that assigns to each closed form of the spectral datum its structural role in the physical structure of Theorem 9.2. It consists of the following assignment rules.*

- (i) *The quartic coupling λ , a convergent spectral sum, is assigned to the e^4 vertex.*
- (ii) *The scalar mass parameter m_δ is assigned to the single-particle mass, located at the spectral pole $p^2 = m_\delta^2$.*
- (iii) *The condensate $\langle E \rangle$ is assigned to the Weyl displacement parameter of the physical vacuum.*
- (iv) *The mass tower $\{M_l\}$ is assigned to the single-particle states of the free sector.*

Theorem 9.2 (Formal correspondence without a measure). *Let λ , m_δ , and $\langle E \rangle$ be the derived parameters of Sections 7 and 8. The following correspondence holds without the use of a measure.*

- (a) *Condensate as a Weyl displacement. The physical vacuum of the condensed sector is the coherent-state displacement of the Fock vacuum,*

$$|\Omega_{\text{eff}}\rangle = W(\langle E \rangle)|\Omega\rangle \in \mathcal{F}(h_\delta),$$

where W is the Weyl operator and the displacement $\langle E \rangle$ is the spectral-data closed form. The effective Hilbert space of Theorem 7.10 is the same Fock space $\mathcal{H}_{\text{eff}} = \mathcal{F}(h_\delta)$, whose physical vacuum is the displaced coherent state.

- (b) *Mapping dictionary. Each closed form of the spectral data corresponds to an element of the operator structure. The coupling λ , a convergent spectral sum, corresponds to the e^4 vertex. The parameter m_δ corresponds to the single-particle mass through the spectral pole. The condensate $\langle E \rangle$ corresponds to the Weyl displacement. The mass tower $\{M_l\}$ corresponds to the single-particle states.*
- (c) *No measure is used. Every element of the correspondence is constructed from the algebraic relations of the spectral data. No measure integral, no Minlos theorem, and no Osterwalder–Schrader reconstruction enters the interacting correspondence. This is consistent with the definition of physical quantities in Axiom 5.4, which does not depend on a measure.*

Proof. (a) The Weyl displacement of the Fock vacuum is the standard coherent-state construction. The displacement parameter is the condensate, which is a closed expression of

the spectral data. The effective Hilbert space of Theorem 7.10 is the Fock space of the fluctuation field, equivalently the displaced Fock space of the field E . (b) Each entry of the dictionary refers to the theorem that establishes the corresponding structure from the spectral data. (c) By construction, the dictionary and the displacement invoke only the closed forms of the spectral data and the standard algebraic structures of the Fock space. None of these requires a measure. $\square$

Remark 9.3. The formal correspondence of Theorem 9.2 maps the closed forms of the spectral data to elements of the operator structure. The perturbative amplitudes of Theorem 7.19 are explicit functions of these data. The non-perturbative content is spectral-sum defined, as described in Section 9.2.

9.2 Non-perturbative content as spectral-sum representation

The non-perturbative content of the formulation does not require resummation of a perturbative series. The physical quantities are spectral sums and closed forms of the spectral data by Axiom 5.4. In particular, the quartic coupling λ is itself a convergent spectral sum, not the leading order of an expansion. Every amplitude at any order of the effective expansion is therefore an explicit function of the spectral data.

Theorem 9.4 (Amplitudes by spectral-sum representation). *Let the effective theory of Theorem 7.10 be given, with the spectral data $(\lambda, m_\delta, \{M_l\})$. The following definitions are made within the spectral-sum definition. Then the following hold.*

- (a) Every order. *Every order of the effective expansion of the scattering amplitude is an explicit function of the spectral data. The Feynman rules of the effective theory depend only on λ and m_δ . The renormalisation is the dispersion subtraction of the formulation. The one-loop bubble has the dispersion representation*

$$B(s) = \int_{4m_\delta^2}^{\infty} \frac{ds'}{\pi} \frac{\text{Im } B(s')}{s' - s},$$

with the convention-independent spectral density

$$\text{Im } B(s') = \frac{1}{16\pi} \sqrt{1 - \frac{4m_\delta^2}{s'}} \theta(s' - 4m_\delta^2).$$

The dispersion integral is subtracted at the subtraction point s_0 , giving the convergent expression

$$B_{\text{ren}}(s) = \int_{4m_\delta^2}^{\infty} \frac{ds'}{\pi} \text{Im } B(s') \left(\frac{1}{s' - s} - \frac{1}{s' - s_0} \right).$$

Higher orders are renormalised by the same dispersion subtraction of their spectral representations. Hence every order is a definite function of the spectral data. The tree-level $2 \rightarrow 2$ amplitude is $\mathcal{M}^{(0)} = -6\lambda$. The renormalised one-loop contribution is $\mathcal{M}^{(1)}(s) = -18\lambda^2 B_{\text{ren}}(s)$, the factor $18 = 6^2/2$ arising from the quartic vertex $-6i\lambda$ and the symmetry factor $1/2$ of the bubble.

- (b) The non-perturbative character is carried by the coupling. *The coupling λ is a convergent spectral sum of the spectral data, independent of any expansion. The amplitude is therefore determined by the exact spectral data.*
- (c) No resummation is required. *The physical quantities are spectral sums and closed forms directly. The effective expansion is an optional description of the effective theory, not the definition of the physical quantities.*

Proof. (a) The Feynman rules give the tree amplitude -6λ and the one-loop bubble

$$B(s) = \int \frac{d^4k}{(2\pi)^4} \frac{1}{(k^2 - m_\delta^2)((k-p)^2 - m_\delta^2)},$$

with $s = p^2$. Its imaginary part is the standard two-particle phase space, and its real part follows from the dispersion relation up to the subtraction convention. The bubble depends on the spectral data only through m_δ . (b) The coupling λ is a convergent spectral sum, not the leading order of an expansion. (c) This is the constitutive definition of the formulation, Axiom 5.4. $\square$

Remark 9.5. Within the formulation, non-perturbative means spectral-sum defined. A quantity is non-perturbative if it is a spectral sum or a closed form of the spectral data. This differs from the standard-formulation notion, where non-perturbative refers to the limit or resummation of a continuum construction. The spectral-sum representation is the formulation's answer to the non-perturbative problem. The physical quantities were never defined by a series, so no resummation is needed.

Theorem 9.6 (Interaction operator within the spectral-sum definition). *Let the effective sector of Theorem 7.10 carry the interaction $\lambda : e^4$. Within the formulation the following hold.*

- (a) Physical content. *Every physical consequence of the interaction operator is an exact function of the spectral data. The amplitudes and loop corrections are given by Theorem 9.4 with the dispersion subtraction. The positive scalar mass m_δ and the condensate $\langle E \rangle$ are closed forms of the spectral data.*
- (b) Structural status. *The operator is the mapping-dictionary entry $\lambda \mapsto e^4$ vertex of Theorem 9.2(b). Its structural role is fixed by the spectral-data closed form of the coupling.*
- (c) Definitional scope. *Within the formulation the definition of the interaction operator is its physical content and formal correspondence. The strict Fock-space operator definition of $: e^4 :$ belongs to the standard formulation and is excluded by the triviality obstruction. The formulation does not require that definition, since its physical quantities are spectral sums.*
- (d) Closure. *The operator-level definition of $\lambda : e^4 :$ is therefore closed within the formulation. The physical content and structural status are fully determined by the spectral data. The standard-formulation counterpart is excluded by the triviality theorem.*

Proof. (a) The amplitudes are explicit spectral-data functions by Theorem 9.4. The positive scalar mass m_δ and the condensate $\langle E \rangle$ are closed forms. (b) This is the mapping dictionary of Theorem 9.2(b). (c) The definitional scope is the constitutive definition of the formulation, Axiom 5.4. (d) Closure follows from (a)–(c). $\square$

Remark 9.7. Theorem 9.6 records that the interaction operator is complete within the formulation through its physical content and structural status.

10 The spectral-sum definition and the standard path integral formulation

This section records the relation between the spectral-sum definition of the present paper and the standard path integral formulation of quantum gauge theory. The analysis has two parts. Part 1 discusses the non-trivial content of the framework. Part 2 constructs the standard-formulation realisation of the free part.

10.1 The two formulations

In the spectral-sum definition, physical quantities are defined directly by spectral sums over the spectrum of the internal space RP^3 , evaluated at the endpoint of the scale flow. In the standard path integral formulation, physical quantities are defined as expectations with respect to a Euclidean measure. The measure is constructed in practice as the weak limit of lattice measures with the plaquette action [22]. Reflection positivity of the lattice theory is a rigorous theorem [23], and the continuum limit is the object of the standard constructive programme.

The correspondence between the standard requirements and the framework statements is summarised in Table 3.

Table 3. Dictionary between standard requirements and their spectral-sum-definition analogues

Standard requirement	Framework statement	Location
Wightman axioms W1–W6	Spectral-sum-definition analogue of the Wightman axioms	Definition 8.1, Theorem 8.39
Mass gap	Positive scalar lower bound $m_\delta > 0$	Theorems 8.32–8.34
Non-triviality	Spectral sums, $\lambda > 0$, condensate	Theorems 8.31, 8.32
Gauge group G_{SM}	$SU(3) \times SU(2) \times U(1)$	Theorem 4.12
Spacetime dimension 4	Emergent $4 = 3 + 1$	Theorem 4.13, Corollary 4.14
Euclidean invariance	Gaussian part $E(4)$, flow $SO(4)$	Theorems 4.15, 8.20
Reflection positivity	Free field on $\mathbb{R}^4 \times RP^3$	Theorems 10.6, 10.7
Short-distance asymptotic freedom	β -function closed form, interaction stability	Lemma 8.13, Theorem 8.19

Each row of Table 3 records a spectral-sum-definition analogue of the corresponding standard requirement, not an equivalence (Definition 8.1).

10.2 Part 1: The non-trivial content

10.2.1 The scalar sector in the standard continuum formulation

The colour-singlet sector of the framework is a real scalar field with a quartic self-interaction. Its coupling is strictly positive by Proposition 8.31, and its potential develops a condensate by Theorem 8.32. Its standard-formulation counterpart is a four-dimensional scalar field theory with quartic interaction.

Theorem 10.1 (Triviality obstruction). *Let $\{S^{(n)}\}$ be the Schwinger functions of a standard-formulation continuum theory whose colour-singlet data match the framework's spectral data with quartic coupling $\lambda > 0$ and condensate $\langle E \rangle \neq 0$. Then the theory is a free field. In the continuum limit the quartic coupling vanishes, and no non-trivial interaction survives.*

Proof. The colour-singlet sector is a four-dimensional real scalar field with quartic self-interaction. Four-dimensional scalar field theory with quartic interaction is trivial. The continuum limits of the four-dimensional lattice scalar models are Gaussian. The rigorous statement for the critical four-dimensional Ising and φ^4 lattice models is the scaling-limit theorem of Aizenman and Duminil-Copin [24]. The earlier analyses of Aizenman [25] and Fröhlich [26] established free and generalised-free scaling limits in five or more dimensions and, in four dimensions, the triviality of the families described by renormalised perturbation theory. Hence no standard-formulation continuum theory of this type can carry the non-trivial content $\lambda > 0$ of the framework. $\square$

The difference between the two formulations is therefore structural. The spectral-sum definition carries the non-trivial content $\lambda > 0$ by construction through Proposition 8.31. The standard continuum formulation cannot.

10.2.2 The gauge sector and the standard formulation

Proposition 10.2. *The standard-formulation continuum limit of the non-Abelian gauge sector is the unresolved content of the standard formulation itself. The framework construction does not depend on it.*

Proof. The standard formulation requires the existence of a weak limit of the lattice measures [22, 23] and the control of the continuum limit by multi-scale cluster expansion [27–29]. The completion of this programme is an open problem of the standard formulation. Recent progress includes direct decay estimates for finite lattice gauge theories at weak coupling [30]. The framework construction defines its physical quantities directly by spectral sums. It presupposes neither the lattice measures nor their limit. $\square$

Remark 10.3 (Scalar order parameter and the gauge sector). The framework's m_δ is the mass of the colour-singlet scalar order parameter. Within the spectral-sum definition this

is the positive scalar lower bound of the colour-singlet sector, the mass-gap parameter of the framework. The corresponding object in the standard continuum formulation is the mass gap of the gauge-field sector; the two are the same physical quantity reached through different constructions. The framework’s scalar mass is complete within the spectral-sum definition, and its standard-formulation counterpart is the subject of the standard programme.

The framework’s m_δ can enter the standard programme’s estimates only as an input improving the decay constants of the cluster expansion. It cannot replace the expansion itself. Defining Wilson loops by spectral sums would be a redefinition of the observable, not a translation of it. The question whether such a definition agrees with the standard Wilson loop is part of the standard programme.

10.2.3 Conclusion of Part 1

The relation between the two formulations is not a translation in the literal sense. The non-trivial content of the framework, namely the positive quartic coupling $\lambda > 0$, the condensate $\langle E \rangle \neq 0$, and the positive scalar lower bound $m_\delta > 0$, is defined in the spectral-sum definition. No counterpart is known in the standard continuum formulation.

The standard continuum formulation can support at most the free content, which is the subject of Part 2. The two formulations are complementary in their methods. Both start from the same spectral data. The standard continuum formulation carries the measure construction, the Gaussian realisation of the free sector. The spectral-sum definition carries the formal correspondence, the spectral-sum definition of the physical quantities. The free sector is common to both constructions, and the non-trivial content is carried by the spectral-sum definition.

10.3 Part 2: The free-sector realisation

The only content that admits a standard-formulation realisation is the free part of the framework’s colour-singlet data. This includes the spectral masses M_l , the scalar mass parameter m_δ , and the condensate $\langle E \rangle$. This part constructs the standard-formulation realisation as the free scalar field on $\mathbb{R}^4 \times RP^3$ with a constant shift. The internal spectral modes of this field form the mass tower $\{M_l\}$, which is exactly the framework’s spectral data. The zero mode is the four-dimensional scalar field of mass m_δ . The construction uses only the spectral data and standard theorems. This is the concrete $M = RP^3$ instance of the abstract recovery of Appendix C (Theorems C.1 and C.2); the construction is spelled out in detail here because the standard-formulation comparison is made in this section.

The route is as follows. If a family of candidate Schwinger functions satisfies the standard OS axioms, then the Osterwalder–Schrader reconstruction yields the Hilbert space and the Wightman fields directly. For the Gaussian candidates constructed below, Minlos’ theorem produces the Euclidean measure explicitly. No lattice regularisation, no cluster expansion, and no analysis of a continuum limit is required. The question is therefore reduced to constructing candidate Schwinger functions from the framework’s spectral data that satisfy OS0–OS4. For the free part this is possible. For the non-trivial part it is impossible by Theorem 10.1.

10.3.1 Step 1: The candidate Schwinger functions

Let $\lambda_l = l(l+2)/L^2$ with multiplicities $d_l = (l+1)^2$ for even l be the scalar spectrum of RP^3 , and set $M_l^2 = \lambda_l + m_\delta^2$, where $m_\delta^2 = \frac{16}{3}(R_c - R) > 0$ is the fluctuation mass of Theorem 8.32. Let $K_l(y, y')$ denote the spectral projector kernel of the l -th eigenspace of the scalar Laplacian on RP^3 . With $\{Y_{lm}\}_{m=1}^{d_l}$ an orthonormal eigenbasis,

$$K_l(y, y') = \sum_{m=1}^{d_l} Y_{lm}(y) Y_{lm}(y').$$

Let Δ_M denote the free massive propagator of mass M in four dimensions. Define the two-point function of the colour-singlet sector by the full spectral kernel

$$S_2((x, y), (x', y')) = \sum_{l \text{ even}} K_l(y, y') \Delta_{M_l}(x - x') + \langle E \rangle^2,$$

where $x, x' \in \mathbb{R}^4$ and $y, y' \in RP^3$, and the constant $\langle E \rangle \neq 0$ is the condensate of Theorem 8.32. The field is $\phi = \psi + \langle E \rangle$, where ψ is the free scalar field on $\mathbb{R}^4 \times RP^3$ with two-point function

$$S_{2,\psi}((x, y), (x', y')) = \sum_{l \text{ even}} K_l(y, y') \Delta_{M_l}(x - x').$$

The higher Schwinger functions are defined by Wick's theorem applied to ψ , together with the constant shift $\langle E \rangle$.

Remark 10.4 (The kernel and the multiplicity sum). The multiplicity-only sum $\sum_l d_l \Delta_{M_l}(x - x')$ is not a distribution. Its momentum-space general term $d_l/(p^2 + M_l^2)$ tends to the positive constant L^2 as $l \rightarrow \infty$, so the series diverges linearly, as stated in Lemma 8.21. Minlos' theorem cannot be applied to it as it stands. The kernel itself is well defined. By the completeness relation $\sum_l K_l(y, y') = \delta(y - y')$, the eigenfunction oscillation regularises the sum for smooth internal test functions. The kernel is the Green function of $\Delta_{\mathbb{R}^4} + \Delta_{RP^3} + m_\delta^2$ on $\mathbb{R}^4 \times RP^3$. It is the two-point function of the massive free scalar field on the product manifold. The field therefore lives on $\mathbb{R}^4 \times RP^3$. The factor $\mathbb{R}^4$ carries the Euclidean and reflection structures of spacetime, and the compact factor RP^3 is the internal space of the framework, whose spectrum generates the mass tower $\{M_l\}$.

10.3.2 Step 2: The standard OS axioms

Lemma 10.5 (Reflection positivity of the shifted free field on the product space). *Let ψ be the free scalar field on $\mathbb{R}^4 \times RP^3$ with two-point function*

$$S_{2,\psi}((x, y), (x', y')) = \sum_{l \text{ even}} K_l(y, y') \Delta_{M_l}(x - x'),$$

where K_l is the spectral projector kernel of the l -th eigenspace of Δ_{RP^3} , a positive operator, and each Δ_{M_l} is the free massive propagator on $\mathbb{R}^4$ with $M_l \geq m_\delta > 0$. Then the Gaussian measure of ψ satisfies reflection positivity with respect to time reflection in the $\mathbb{R}^4$ factor, and the constant shift $\phi = \psi + c$ with $c \in \mathbb{R}$ preserves reflection positivity.

Proof. Decompose a test function $f \in \mathcal{S}(\mathbb{R}^4 \times RP^3)$ into internal spectral modes, $f(x, y) = \sum_l f_l(x)$, where f_l is the projection onto the l -th eigenspace. Then

$$\iint \overline{f(x, y)} S_{2, \psi}((x, y), (\theta x', y')) f(x', y') dx dy dx' dy' = \sum_l \iint \overline{f_l(x)} \Delta_{M_l}(x - \theta x') f_l(x') dx dx' \geq 0.$$

The inequality is reflection positivity of the massive free field of mass $M_l \geq m_\delta > 0$ on $\mathbb{R}^4$, applied to f_l supported in $\{\tau > 0\}$. Each term is non-negative and the sum converges. For the shift, let F be a polynomial in the fields $\phi(f_i)$ with $\text{supp } f_i \subset \{\tau > 0\}$. Writing $\psi(f) = \phi(f) - c \int f$, the polynomial $G(\psi) = F(\psi + c)$ is again a polynomial in the $\tau > 0$ fields $\psi(f_i)$, and

$$\mathbb{E}[F(\theta F)] = \mathbb{E}_0[G(\theta G)] \geq 0$$

by reflection positivity of the zero-mean Gaussian, since the constant c is reflection-invariant. The shift therefore preserves reflection positivity. $\square$

The candidate Schwinger functions of Step 1 satisfy the standard OS axioms as follows.

(OS0) Regularity. The two-point function is the resolvent kernel of $\Delta_{\mathbb{R}^4} + \Delta_{RP^3} + m_\delta^2$ on $\mathbb{R}^4 \times RP^3$. It is a positive-definite tempered distribution, continuous as a bilinear form on $\mathcal{S}(\mathbb{R}^4 \times RP^3) \times \mathcal{S}(\mathbb{R}^4 \times RP^3)$. Every higher moment is a finite polynomial in it, hence tempered.

(OS1) Euclidean invariance. Each massive propagator is $O(4)$ -invariant in the $\mathbb{R}^4$ factor. Each K_l is invariant under the isometries of RP^3 . The shift is constant.

(OS2) Reflection positivity. This is Lemma 10.5.

(OS3) Symmetry. Wick's theorem is symmetric.

(OS4) Clustering. The two-point function decays exponentially in $|x - x'|$ with rate at least m_δ , the lightest spectral mass, attained by the zero mode $l = 0$. The truncated functions of the free field vanish.

10.3.3 Step 3: The Euclidean measure

Theorem 10.6 (Standard-formulation realisation of the colour-singlet data). *The candidate Schwinger functions of Step 1 are the moments of a unique Euclidean measure μ_∞ on $\mathcal{S}'(\mathbb{R}^4 \times RP^3)$. The measure is Gaussian with mean $\langle E \rangle$ and covariance $S_{2, \psi}$. It satisfies the standard OS axioms OS0–OS4 with scalar mass $m_\delta > 0$.*

Proof. The covariance $S_{2, \psi}$ is the Green function of $\Delta_{\mathbb{R}^4} + \Delta_{RP^3} + m_\delta^2$. It is the kernel of the resolvent of a positive elliptic operator at the negative spectral value $-m_\delta^2$, which maps $\mathcal{S}(\mathbb{R}^4 \times RP^3)$ to itself and is positive definite as a bilinear form. Hence $S_{2, \psi}$ is a positive-definite tempered distribution, and by Minlos' theorem [14] there exists a unique Gaussian measure on $\mathcal{S}'(\mathbb{R}^4 \times RP^3)$ with this covariance and mean $\langle E \rangle$. Its moments are the candidate Schwinger functions. The standard OS axioms hold by Step 2. Clustering follows from the scalar mass $m_\delta > 0$. The bottom of the spectrum of $\Delta_{\mathbb{R}^4} + \Delta_{RP^3} + m_\delta^2$ is m_δ^2 , attained by the zero mode at $p = 0$. $\square$

10.3.4 Step 4: OS reconstruction

By the standard Osterwalder–Schrader reconstruction theorem, the measure μ_∞ of Theorem 10.6 defines a Wightman field theory of the colour-singlet sector satisfying the standard Wightman axioms W1–W6, with a unique vacuum, a positive scalar mass $m_\delta > 0$ above the vacuum, and vacuum expectation value $\langle E \rangle \neq 0$ for the scalar field. The reconstructed fields are the internal spectral modes. The projection ψ_l of ψ onto the l -th eigenspace of Δ_{RP^3} is a Wightman field on $\mathbb{R}^4$ of mass M_l . The zero mode $l = 0$ is the four-dimensional field of mass m_δ . The theory is a free field. Its truncated functions vanish, and its S -matrix is trivial. The field $\phi = \psi + \langle E \rangle$ is the free field shifted by a constant, a coherent-state transformation. The theory is unitarily equivalent to the free field, and the vacuum is the Fock vacuum of ψ . The shift changes the field, not the Hilbert space. In the spectral-sum definition, by contrast, the spectral data include the non-Gaussian content $\lambda > 0$, and the vacuum of the colour-singlet sector is not unitarily equivalent to the Fock vacuum by Theorem 8.36. This difference is the content of Theorem 10.1. The standard continuum formulation can realise the framework’s colour-singlet data only in their free form. The non-trivial content exists only in the spectral-sum definition.

10.3.5 The recovery theorem

Theorem 10.7 (Recovery of a standard quantum field theory). *The framework’s colour-singlet spectral data $\{\lambda_l, d_l, m_\delta\}$ determine a unique standard-formulation quantum field theory. It is the free scalar field of mass m_δ on $\mathbb{R}^4 \times RP^3$ with product metric and RP^3 of radius L . Its two-point function is the full spectral kernel of Step 1. The correspondence preserves the scalar mass m_δ , the two-point function, the vacuum, and the clustering rate. The reconstructed Wightman theory is the family $\{\psi_l\}$ of free fields on $\mathbb{R}^4$ with masses $M_l = \sqrt{\lambda_l + m_\delta^2}$. The zero mode is the four-dimensional colour-singlet field of mass m_δ . The correspondence is an isomorphism of spectral data. The Kaluza–Klein decomposition of the free field on the product space is exactly the framework’s spectrum.*

Proof. By Theorem 10.6 the spectral data determine the Gaussian measure μ_∞ on $\mathcal{S}'(\mathbb{R}^4 \times RP^3)$ with covariance $S_{2,\psi}$, the two-point function of the massive free field of mass m_δ on the product manifold. The scalar mass is the bottom of the spectrum, m_δ , the zero mode at $p = 0$. Clustering at rate m_δ follows from OS4. The Wightman reconstruction of Step 4 gives the family $\{\psi_l\}$ with masses $\{M_l\}$. The spectral data of the framework are exactly the Kaluza–Klein data of this field by Lemma 8.21. $\square$

10.3.6 What is not translated

Wilson loops and the non-singlet sector are not translated. Their standard-formulation construction belongs to the standard formulation’s own problem, as stated in Proposition 10.2. The framework’s construction of the colour-singlet sector is defined within the spectral-sum definition. The standard continuum formulation can support at most the free content constructed here. The non-trivial construction exists in the spectral-sum definition alone. The complementary correspondence is the product-space formulation of Appendix B. It realises the framework’s spectral data as the internal-mode content of an explicit action

on $\mathbb{R}^4 \times RP^3$. Its one-loop content in the internal-trace order reproduces the framework's spectral sums, without entering the standard four-dimensional continuum formulation. Its Kaluza–Klein decoupling reproduces the four-dimensional effective structure.

10.4 Relation to the standard formulation

This subsection states how the formulation of Sections 7–9 stands in relation to the standard formulation of quantum field theory. On the free sector the two formulations coincide through the recovery theorem. On the interacting sector they are complementary, and the precise locus of the complementarity is the triviality theorem.

10.4.1 Where the two formulations coincide

On the free sector the formulation and the standard formulation coincide. The recovery theorem of Appendix C constructs the free sector of the formulation by the standard Osterwalder–Schrader reconstruction applied to the spectral kernel. The result is the free scalar field of mass m_δ on $\mathbb{R}^4 \times M$, a standard quantum field theory in the sense of Wightman and Osterwalder–Schrader. There is no difference of content here. The same spectral datum produces the same free field, the same mass tower, the same two-point function, and the same clustering.

10.4.2 The triviality theorem and the interacting sector

The interacting content of the formulation, namely the quartic coupling $\lambda > 0$ and the condensate $\langle E \rangle \neq 0$, has no realisation as a standard four-dimensional scalar field theory. This is the content of the triviality theorem. The continuum limits of the four-dimensional lattice ϕ^4 models are Gaussian, so the interacting scalar sector is trivial in the standard continuum formulation. The statement is a theorem of the standard formulation. It fixes the boundary at which the standard formulation stops.

Remark 10.8 (The measure-theoretic locus of triviality). The triviality theorem is a statement about the standard formulation, whose physical content is carried by a measure. What the theorem excludes is the existence of a non-trivial measure with an interacting continuum limit. The only measures that survive the continuum limit are Gaussian. The formulation's interacting content does not proceed through such a measure. The theorem therefore does not touch the formulation's construction, but it does fix the boundary of the standard formulation against which the formulation's interacting content is compared.

10.4.3 The difference is in the constructions, not in the content

The difference between the formulation and the standard formulation on the interacting sector is not a difference between two sets of physical content. It is a difference between the constructions that the two formulations use to reach that content from the same spectral data.

Remark 10.9 (Two constructions from the same data). Both formulations begin from the same spectral datum. The standard formulation maps the spectral data to the physical structure by a measure: the covariance defines a Gaussian measure, and the physical

structure is reconstructed from that measure. The formulation maps the same spectral data to the physical structure by a formal correspondence, the mapping dictionary and the Weyl displacement of Theorem 9.2, which invoke only the closed forms of the spectral data and the standard algebra of the Fock space, without a measure. The two constructions coincide on the free sector, where the measure-theoretic construction exists and the formal correspondence reduces to it; they diverge exactly where the measure-theoretic construction stops, at the interacting content excluded by triviality. The spectral datum therefore fixes one theory, with two representations: the free sector represented by the Gaussian measure and its reconstruction, the interacting sector by the formal correspondence, sharing the same free content and differing only in the construction.

10.4.4 The classification of the remaining items

The boundaries of the formulation fall into three classes.

Remark 10.10 (Three classes of boundary). (a) *Theorem-excluded boundaries.* The standard-formulation continuum realisation of the interacting scalar sector is excluded by the triviality theorem. This is a theorem of the standard formulation, not an open problem of the formulation. The formulation's interacting content is defined within its own formal correspondence.

(b) *Conceptual gaps, closed in the formulation.* The notion of a non-perturbative quantity that is not defined by a series, the spectral-sum representation of Theorem 9.4, and the formal correspondence of Theorem 9.2 are constructions introduced in the present paper. They are closed by their definitions and theorems.

(c) *Technical boundaries.* The ultraviolet completion of the four-dimensional field content at the operator level, beyond the spectral-sum definition of the physical quantities, is a technical boundary of the standard formulation. The formulation's physical quantities do not depend on it, since they are spectral sums from the outset.

Remark 10.11 (Summary of the relation). The relation of the formulation to the standard formulation is as follows. On the free sector the two coincide. On the interacting sector the formulation supplies a formal correspondence where the standard formulation has no measure-theoretic construction, because triviality cuts the latter at the free sector. The interacting content of the formulation is defined within the spectral-sum definition and is non-perturbative in the spectral-sum sense. Nothing is left open by the absence of the standard construction. The formulation is complementary to the standard formulation in the constructions it uses, not in the physical content it carries.

11 Conclusion

This paper has developed the first step of a spectral framework. It does not attempt to settle the full problem of gauge structure or the mass gap in the standard sense. What it provides is a foundation on which later work can build.

The starting point is a discrete spectrum with multiplicities. Physical quantities are defined as convergent spectral sums over this spectrum. The spectrum itself is not chosen

arbitrarily. It follows from the disorder axiom, together with several explicit framework conventions. From these ingredients the internal space, the gauge algebra, and the space-time dimension are constructed or selected within the framework. The gauge field built on this geometry has a colour-singlet sector. In that sector a scalar order parameter acquires a positive mass through condensation. This mass is the framework’s mass-gap parameter, the positive spectral lower bound of the colour-singlet sector.

The spectral framework was stated as a set of axioms in Section 5. The quantum kinematics was located within this framework in Section 6. The mapping statements of Section 7 show how the spectral datum is related to the field-theoretic structure. Section 8 constructed the gauge field and derived the positive scalar mass parameter. Section 9 recorded the formal correspondence that carries the interacting content without using a measure. Section 10 compared the framework with the standard path integral formulation.

The relation to the standard measure-theoretic formulation is now clear. On the free sector the two approaches coincide. The free sector is recovered by the standard Osterwalder–Schrader reconstruction. On the interacting sector the standard continuum formulation faces a known difficulty. The path-integral measure over field configurations is not defined in the strict mathematical sense, because no translation-invariant Lebesgue measure exists on an infinite-dimensional space. Its construction requires a regularisation and a limiting procedure, and for the relevant scalar models that limit is Gaussian by the triviality theorem.

The present framework supplies a formal correspondence that does not rely on a measure. Its physical quantities are spectral sums. These sums are ordinary convergent series, and the free sector obtained from them coincides with the standard free-field construction. In this way the framework avoids the measure and limit problems of the standard formulation while retaining an interacting content defined within the spectral-sum definition. Whether this content can support a realistic comparison with particle physics remains to be seen.

The disorder principle and the Wightman axioms are logically independent. The present paper constructs a specific spectral framework that carries the content selected by the disorder principle and satisfies the spectral-sum-definition analogue of the Wightman axioms.

Several points are left open in this paper. The existence of a colour factor distinct from the electroweak one is recorded as a structural datum, and the numerical value of the window capacity that fixes the generation count is determined in the numerical evaluation of the framework. The numerical values of the effective couplings are not compared with experiment here. No claim is made that the framework has reproduced the quantitative successes of the Standard Model. These tasks belong to future work.

The present paper is therefore foundational. It establishes the spectral datum, the spectral-sum definition of physical quantities, and the formal correspondence. Whether this foundation can support a realistic comparison with particle physics remains to be seen. The verification of the framework against known physics is left to subsequent papers.

A Spectral analysis: long-root mode, Dirac spectrum, and the emergence window

This appendix collects the harmonic analysis used in the main text. It records the quantum numbers of the long-root mode, the condensation threshold, the spin-2 content, and the emergence window.

A.1 Spin(4) harmonics on S^3 and RP^3

Let S^3 carry the round metric of radius L , and let $RP^3 = S^3/\mathbb{Z}_2$ be the quotient by the antipodal map. The scalar Laplacian spectrum of S^3 is

$$\lambda_l = \frac{l(l+2)}{L^2}, \quad \text{multiplicity } (l+1)^2, \quad l = 0, 1, 2, \dots$$

This is a standard result. The eigenfunctions of degree l transform in the representation $(l/2, l/2)$ of $\text{Spin}(4) \cong SU(2)_L \times SU(2)_R$. The quadratic Casimir of the representation (j_L, j_R) , in the normalisation

$$C_2(j_L, j_R) = 2[j_L(j_L + 1) + j_R(j_R + 1)],$$

satisfies $C_2(l/2, l/2) = l(l+2)$. Hence the Laplacian eigenvalue of a $\text{Spin}(4)$ harmonic is C_2/L^2 in this normalisation. This convention is used throughout the paper.

On RP^3 the antipodal map acts on the harmonic of degree l as $(-1)^l$. The even sector therefore survives. This gives the scalar spectrum used in Lemma 8.21,

$$\lambda_l = \frac{l(l+2)}{L^2}, \quad d_l = (l+1)^2, \quad l = 0, 2, 4, \dots$$

A.2 The long-root mode

Let α be the highest root of $\mathfrak{su}(3)$. In the decomposition $\mathfrak{so}(4) = \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ and the breaking $\mathfrak{su}(2)_R \rightarrow \mathfrak{u}(1)_R$, the unbroken torus assigns to the representation (j_L, j_R) the charge $q = 2j_R$. The long root carries charge $q = 2$, hence $j_R = 1$.

The modes of the R -sector connection are $\mathfrak{su}(2)_R$ -valued 1-forms on RP^3 . The harmonic decomposition of 1-forms on S^3 is

$$\Omega^1(S^3) = \bigoplus_{l \geq 1} \left[\left(\frac{l}{2} + \frac{1}{2}, \frac{l}{2} - \frac{1}{2} \right) \oplus \left(\frac{l}{2} - \frac{1}{2}, \frac{l}{2} + \frac{1}{2} \right) \right].$$

On the quotient RP^3 the antipodal map acts on a 1-form of degree l with an additional sign from the differential. The retained sector therefore has odd l . The $l = 1$ sector is the Killing zero-mode sector, which carries the gauge structure itself.

For the non-zero modes, the first one with $j_R = 1$ is the $l = 3$ harmonic $(j_L, j_R) = (2, 1)$, since $j_R = l/2 - 1/2 = 1$ and $j_L = l/2 + 1/2 = 2$. Its Casimir is

$$C_2(2, 1) = 2[2 \cdot 3 + 1 \cdot 2] = 16.$$

By the normalisation of Section A, the internal eigenvalue is

$$\lambda_{\text{long}} = \frac{C_2(2, 1)}{L^2} = \frac{16}{L^2}.$$

This is the content of Lemma 8.25.

A.3 The Dirac spectrum of RP^3

The Dirac operator on the round S^3 of radius L has spectrum

$$\pm \frac{j+1}{L}, \quad j = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$$

with the standard multiplicities (for lens spaces see [31]). The manifold $RP^3 = S^3/\mathbb{Z}_2$ admits two spin structures. On each, the Dirac spectrum is the restriction of the S^3 spectrum to the sector of definite antipodal parity. For the spin structure selected by the chiral structure of clause (A2), the retained spectrum is

$$\pm \frac{j+1}{L}, \quad j = \frac{1}{2}, \frac{5}{2}, \frac{9}{2}, \dots$$

Equivalently, in the extrusion index n of the generation ladder, $\lambda_n = (n + 3/2)/L$ with $n = 0, 2, 4, \dots$

A.4 The emergence window and the generation count

The scale semigroup of Section 8.4 removes modes whose momentum exceeds $1/\sigma$. The internal spectrum is resolved by a different scale. The internal geometry is a statistical object defined by the ensemble of level sets of the Gaussian scale field. Its coherence scale is the internal radius L , defined as the Gaussian integral width of the scale field. The coarse graining at the breakdown scale retains internal modes whose wavelength exceeds this coherence scale. Shorter-wavelength modes oscillate within the coherence length of the statistics and are averaged out.

The retained window is fixed by the window capacity $(kL)^2$ of the scale flow, where k is the flow momentum scale and L is the internal radius. The retained spinor modes are those with dimensionless Dirac eigenvalue $n + 3/2$ below $(kL)^2$. The window capacity is fixed self-consistently by the spectral sums of the framework, and its numerical value determines the generation count. The determination of this value belongs to the numerical evaluation of the framework. The framework assigns to the retained modes the extrusion order $n = \{0, 2, 4\}$, following the order of their appearance along the scale flow. The identification of these modes with the Standard Model generations is a phenomenological hypothesis recorded in Proposition 8.16.

B The product-space action: one-loop matching and Kaluza–Klein decoupling

Section 10 records the boundary of the standard four-dimensional continuum formulation for the non-trivial scalar sector. This appendix constructs a complementary correspondence at the level of an action. The spectral data of the framework are organised as the internal-mode content of an explicit action S_{prod} on the product space $\mathbb{R}^4 \times RP^3$. The one-loop effective potential of S_{prod} , computed in the internal-trace order, matches the framework's spectral sums term by term. The Kaluza–Klein decoupling of the massive internal modes gives a four-dimensional effective theory.

The purpose of this appendix is accordingly limited. The product-space action is not presented as a renormalisable quantum field theory in seven dimensions; it is an auxiliary object, a generating device whose internal-trace one-loop content reproduces the framework's spectral sums. It is sufficient for this role even though it is power-counting non-renormalisable, because the framework's physical quantities are the spectral sums themselves, defined independently of any seven-dimensional loop expansion.

Two limitations are stated at the outset. First, S_{prod} is a seven-dimensional theory. The triviality theorem for four-dimensional scalar fields does not directly govern it, and no four-dimensional continuum limit is taken here. The internal radius L is the physical endpoint scale, not a removable cutoff. Second, this appendix does not construct the full renormalised theory of S_{prod} in seven-dimensional loop order. The matching theorem is proved in the internal-trace order, where every matched coefficient is a convergent spectral sum. The full loop-order renormalisation lies outside the spectral-sum definition.

B.1 The product-space action

Throughout, $M = \mathbb{R}^4 \times RP^3$ carries the product metric, with RP^3 the round projective space of radius L and volume $\text{Vol}(RP^3) = \pi^2 L^3$. The internal spectra are those of Appendix A. The gauge group is the emergent group $G_{\text{SM}} = SU(3) \times SU(2)_L \times U(1)_Y$ of Theorem 4.12.

Definition B.1 (The product-space action). *The product-space action $S_{\text{prod}} = S_g + S_f + S_s$ on $M = \mathbb{R}^4 \times RP^3$ is defined by three sectors.*

1. Gauge sector. *For the G_{SM} connection one-form A on M ,*

$$S_g[A] = \frac{1}{4g_7^2} \int_M d^7 z \sqrt{g} g^{MN} g^{PQ} \text{Tr}(F_{MP} F_{NQ}),$$

with the seven-dimensional coupling g_7^2 of mass dimension -3 .

2. Fermion sector. *For the spinor field ψ on M carrying the internal parity of clause (A2),*

$$S_f[\psi] = \int_M d^7 z \sqrt{g} \bar{\psi} (\gamma^\mu (\partial_\mu + A_\mu) + D_{\text{int}}) \psi,$$

where γ^μ are the $\mathbb{R}^4$ generators of Theorem 4.13 and D_{int} is the RP^3 Dirac operator of Appendix A.

3. Scalar sector. *For the colour-singlet scalar ϕ on M with zero-mode amplitude*

$$E(x) = \int_{RP^3} \phi(x, y) d\text{vol}(y),$$

the action is

$$S_s[\phi] = \int_M d^7 z \sqrt{g} \left(\frac{1}{2} g^{MN} \partial_M \phi \partial_N \phi + \frac{1}{2} (m_\delta^2 + gE(x)^2) \phi^2 \right).$$

The scalar sector is the recovery-theorem tower of Theorem 10.7 together with the coupling $(g/2)E^2\phi^2$. This is the action-level form of the quadratic coupling entering the spectrum of Theorem 8.29, $\lambda_l + \xi R \mapsto \lambda_l + \xi R + gE^2$.

Remark B.2 (Power counting and conventions). The seven-dimensional gauge sector has $[g_7^2] = L^3$, so the theory is power-counting non-renormalisable in seven dimensions. The framework does not perform the seven-dimensional loop expansion beyond the internal-trace order. The four-dimensional effective sector is obtained by the exact Kaluza–Klein projection of Section B.3, not by a continuum limit.

The internal radius L is the physical endpoint scale. No limit $L \rightarrow \infty$ and no lattice spacing are involved. The coupling g is fixed by the spectral-shift convention $\lambda_l \rightarrow \lambda_l + gE^2$ of Theorem 8.29. The Higgs sector is not part of the present paper. The scalar content carried by S_{prod} is the colour-singlet sector, the glueball field E of Section 8. The electroweak Higgs is outside the present construction.

B.2 The one-loop matching theorem

Define the internal trace family

$$\zeta(s; m^2) = \sum_{l \text{ even}} d_l (\lambda_l + m^2)^{-s},$$

convergent for $\Re s > 3/2$, by Lemma 8.21. The internal-trace one-loop potential of the scalar sector around the background E_0 is the spectral sum

$$F(E_0) = \frac{1}{2} \sum_{l \text{ even}} d_l h(\lambda_l + m_\delta^2 + gE_0^2),$$

where h is the one-loop function of the framework. In the product-space computation the one-loop function is the logarithm, $h(\lambda) = -\ln \lambda$, in the framework’s sign convention.

Theorem B.3 (One-loop matching). *The one-loop effective potential of the scalar sector of S_{prod} , computed in the internal-trace order, matches the framework’s effective potential of Theorem 8.33 term by term, with the physical mass shift m_δ^2 in place of the conformal shift ξR . Its E_0^{2k} coefficient for $k \geq 2$ is the convergent spectral sum*

$$c_k = \frac{(-1)^{k+1} g^k}{2k} \zeta(k; m_\delta^2).$$

This matches the framework’s coefficient $c_k \propto \sum_l d_l (\lambda_l + \xi R)^{-k}$ of Theorem 8.33 up to the stated sign and normalisation conventions, and up to the replacement of the conformal shift ξR by the physical mass shift m_δ^2 . Its E_0^2 coefficient is $-(g/2)\zeta(1; m_\delta^2)$, the $p = 1$ divergent diagonal of Lemma 8.21. Its renormalised value is the framework’s flow-defined long-mode mass $m_{\text{long}}^2 = 2g \sum_l d_l f'(\lambda_l + \xi R)$ of Theorem 8.29.

Proof. Expand the one-loop function with $h = -\ln$. This gives

$$h(\lambda_l + m_\delta^2 + gE_0^2) = h(\lambda_l + m_\delta^2) + \sum_{k \geq 1} \frac{(-1)^{k+1} g^k}{k} \frac{E_0^{2k}}{(\lambda_l + m_\delta^2)^k},$$

a convergent expansion for $gE_0^2 < m_\delta^2$. Summing over l with the multiplicities d_l gives

$$F(E_0) = F(0) + \sum_{k \geq 1} \frac{(-1)^{k+1} g^k}{2k} \zeta(k; m_\delta^2) E_0^{2k}.$$

For $k \geq 2$ each sum $\zeta(k; m_\delta^2)$ converges absolutely by Lemma 8.21. The coefficients match the framework's c_k . The $k = 2$ coefficient $(g^2/2)\zeta(2; m_\delta^2) > 0$ is a positive convergent quartic spectral sum; with the conformal shift ξR in place of the physical mass shift m_δ^2 it reproduces the quartic coupling $\lambda = \kappa \sum_l d_l (\lambda_l + \xi R)^{-2}$ of Proposition 8.31. The $k = 1$ coefficient is the divergent diagonal. Its renormalised value is the framework's long-mode mass of Theorem 8.29. $\square$

Remark B.4 (Gauge and fermion sectors). The one-loop content of the gauge and fermion sectors is structurally the same. The internal-trace potential of the gauge sector is a spectral sum over the one-form and ghost spectra of Appendix A. The zero modes of Proposition B.6 are excluded from the fluctuation trace. The fermion sector contributes the spectral sum over the Dirac spectrum, which is bounded away from zero. The matching statement of Theorem B.3 concerns the colour-singlet scalar sector. The gauge and fermion sectors match at the level of their spectra and zero modes.

Remark B.5 (The full loop order). The full seven-dimensional one-loop computation requires completing the $\mathbb{R}^4$ momentum integration. This corresponds to the completed zeta function

$$\zeta_{\text{full}}(s; E_0) = \sum_{l \text{ even}} d_l \int \frac{d^4 p}{(2\pi)^4} (p^2 + \lambda_l + m_\delta^2 + gE_0^2)^{-s},$$

with poles at $s = 2, 1, 0$. The E_0 -dependence of the full loop order therefore requires the internal zeta family continued to the non-positive integers, together with the corresponding seven-dimensional counterterms. This continuation is standard-formulation renormalisation data. It is not part of the spectral-sum definition. The present paper does not perform the full seven-dimensional loop-order renormalisation of S_{prod} . The matching theorem is exact within the internal-trace order.

B.3 Kaluza–Klein decoupling

Proposition B.6 (Zero-mode content). *The zero-mode content of the product space $M = \mathbb{R}^4 \times RP^3$ matches the framework's four-dimensional content.*

1. Scalars. *The unique zero mode is the constant mode, $l = 0$, $d_0 = 1$. It carries the colour-singlet field E .*
2. One-forms. *The first de Rham cohomology vanishes, $H_{dR}^1(RP^3; \mathbb{R}) = 0$. There are no harmonic one-forms. The gauge zero modes are the symmetry directions. The electroweak sector has four isometric Killing modes, and the colour sector has eight zero modes of $\mathfrak{su}(3)$. This gives twelve gauge bosons, the dimension of G_{SM} .*
3. Spinors. *The Dirac spectrum of Appendix A is bounded away from zero. There are no Dirac zero modes on RP^3 . The four-dimensional chiral content is the emergence-window retention of the three lowest modes of Theorem 8.16.*

Proof. For scalars, the explicit spectrum of Appendix A gives $\lambda_0 = 0$ with multiplicity one. For one-forms, the vanishing of the first cohomology is Theorem 8.34. The gauge zero modes are modes of vanishing internal momentum, the Killing modes of the isometry algebra and the constant colour modes. They are not harmonic one-forms. For spinors, the spectrum is given in Appendix A, and the three-generation content is the emergence-window convention of Theorem 8.16. $\square$

Proposition B.7 (Zero-mode effective action). *The zero-mode projection of S_{prod} is the four-dimensional effective action*

$$S_4^{(0)} = \int d^4x \left(\frac{1}{4g_4^2} \text{Tr } F_{\mu\nu} F^{\mu\nu} + \bar{\psi}_{\text{win}} \gamma^\mu (\partial_\mu + A_\mu) \psi_{\text{win}} + \frac{1}{2} (\partial_\mu E)^2 + V(E) \right),$$

where ψ_{win} is the three-generation window content of Theorem 8.16, and $V(E)$ is the framework's effective potential of Theorems 8.29 and 8.33. The four-dimensional gauge coupling is the metric-normalised Kaluza–Klein coupling

$$g_4^2 = \frac{g_7^2}{\text{Vol}(RP^3)} = \frac{g_7^2}{\pi^2 L^3}.$$

Proof. Restrict the fields of Definition B.1 to the zero modes of Proposition B.6. In the gauge sector, write $A = A_\mu(x) dx^\mu + \omega(y)$, where A_μ is the four-dimensional connection. The zero-mode part of $F_{\mu\nu} F^{\mu\nu}$ integrates over the internal space with weight $\text{Vol}(RP^3)$, giving the coupling relation. The fermion sector projects onto the window modes. The scalar sector projects the $l = 0$ mode onto the kinetic term of E , and the potential is the framework's $V(E)$. $\square$

Proposition B.8 (Decoupling of the massive modes). *Let*

$$M_{KK} = \sqrt{\frac{8}{L^2} + m_\delta^2}$$

be the lowest non-zero scalar mode, corresponding to $l = 2$. Integrating the massive internal modes with $l \geq 2$ in the internal-trace order modifies each four-dimensional coupling by a correction series whose coefficients are the restricted spectral sums

$$\sum_{l \geq 2} d_l (\lambda_l + m^2)^{-k}$$

and their gauge and Dirac analogues. Each correction to a coupling of mass dimension d is of order $(\Lambda/M_{KK})^{2k}$ for the low-energy scale Λ . The massive propagators cluster in coordinate space as $e^{-M_l |x-x'|}$ with $M_l \geq M_{KK}$. The decoupling is exact in the internal-trace order. The framework's endpoint quantities are these spectral sums by Theorem B.3, so the four-dimensional effective theory matches the framework's physical content.

Proof. The restricted sums converge absolutely for $k \geq 2$ by Lemma 8.21. The suppression factors follow from the mass ratios. The zero mode is always lighter than the tower, $m_\delta^2 < 8/L^2 + m_\delta^2$. The colour-singlet decoupling is therefore structural, independent of the value of L . The identification of the correction sums with the framework's spectral quantities is Theorem B.3. $\square$

B.4 The three technical conditions

The decoupling of Proposition B.8 rests on three technical conditions.

(a) *Coupling normalisation.* The relation $g_4^2 = g_7^2/\text{Vol}(RP^3)$ is the metric-normalisation formula. The absolute value of the coupling is not fixed by the present paper.

(b) *Scale separation.* The decoupling requires all low-energy scales Λ_{low} of the four-dimensional sector to satisfy $\Lambda_{\text{low}} \ll 1/L$. The colour-singlet separation is structural. The quantitative separation for the gauge and fermion sectors is an open condition, tied to the absolute value of L , which the present paper does not fix.

(c) *Chiral content.* The three-generation fermion content is the emergence-window convention of Theorem 8.16. It is not a zero-mode theorem. No Dirac zero modes exist on RP^3 .

B.5 Status of the correspondence

The correspondence constructed in this appendix connects the spectral data to a product-space action on $\mathbb{R}^4 \times RP^3$, and then to a four-dimensional effective theory. The spectral data are the internal-mode content of the product-space action. The internal-trace one-loop potential matches the framework’s spectral sums. The zero-mode content matches the framework’s four-dimensional content. The decoupling corrections are convergent spectral sums.

This appendix does not claim the full seven-dimensional loop-order renormalisation. It does not claim a standard four-dimensional continuum realisation of the non-trivial scalar sector. It does not fix the absolute values of the couplings or of L . The product-space theory is a seven-dimensional theory whose low-energy sector matches the standard-model structure of the framework. It complements the recovery theorem of Theorem 10.7, which realises the free sector in the standard formulation. The non-trivial sector is carried by the spectral-sum definition. The product-space action S_{prod} provides an action-level realisation of the spectral data that generates the framework’s effective structure at low energies.

C Recovery of the free sector

This appendix constructs the free sector of the formulation from the spectral datum. The result is a standard quantum field theory in the standard measure-theoretic sense. The construction is the standard Osterwalder–Schrader reconstruction applied to the spectral kernel. It is unconditional on the free sector and supplies the Fock-space structure used by the mapping statements of Section 7. The concrete $M = RP^3$ instance, with the explicit candidate Schwinger functions and the Osterwalder–Schrader verification carried out in detail, is given in Section 10 (Part 2).

C.1 The candidate Schwinger functions

Let the spectral datum of Axiom 5.1 be given. It consists of a compact three-manifold M with spectrum $\{\lambda_l\}$ and multiplicities $\{d_l\}$. The positive scalar mass parameter $m_\delta > 0$ is

derived from this datum through Theorem 8.32. The spectral kernel of the datum is the full two-point function on $\mathbb{R}^4 \times M$,

$$W(x, y; x', y') = \sum_l K_l(y, y') W_{M_l}(x - x'),$$

where K_l is the spectral projector of the l -th eigenspace of Δ_M , and W_{M_l} is the Euclidean two-point function of the free scalar field of mass

$$M_l = \sqrt{\lambda_l + m_\delta^2}$$

on $\mathbb{R}^4$. The candidate n -point Schwinger functions are the Gaussian Wick moments of W ,

$$S_n = \sum_{\text{pairings}} \prod W(x_i, y_i; x_j, y_j),$$

where the sum runs over all pairings of the n points. For odd n , one sets $S_n = 0$. Each S_n is determined by the spectral datum alone.

Theorem C.1 (Reconstruction of the free sector). *Let the spectral datum of Axiom 5.1 be given, and let the candidate Schwinger functions be formed from the spectral kernel. Then the following hold.*

- (a) Osterwalder–Schrader axioms. *The candidate Schwinger functions satisfy the standard Osterwalder–Schrader axioms of Euclidean covariance, symmetry, reflection positivity, and clustering [8, 9].*
- (b) Minlos theorem. *By Minlos’ theorem [14], the covariance W determines a Gaussian measure μ on $\mathcal{S}'(\mathbb{R}^4 \times M)$ whose moments are the candidate Schwinger functions.*
- (c) Osterwalder–Schrader reconstruction. *The reconstruction theorem produces from μ a Hilbert space $\mathcal{H}$, a field algebra $\mathcal{A}$ of operator-valued distributions $\phi(f \otimes g)$, a strongly continuous unitary dynamics $U(t)$ with self-adjoint generator H , and a cyclic vacuum vector Ω [8, 10]. The vacuum expectation values of the fields are the candidate Schwinger functions.*
- (d) The mass tower. *The Hamiltonian H has spectrum*

$$\text{spec}(H) = \{0\} \cup \{M_l\},$$

with multiplicities $\{d_l\}$, the mass tower of the datum. The Kaluza–Klein decomposition of the free field on $\mathbb{R}^4 \times M$ matches the mode decomposition of the spectrum.

Proof. (a) The spectral kernel is the two-point function of the free scalar field of mass m_δ on the product $\mathbb{R}^4 \times M$, with product metric and M of fixed volume. The candidate Schwinger functions are its Wick moments. A free-field two-point function is reflection-positive and Euclidean-covariant, and its Wick moments satisfy the Osterwalder–Schrader axioms [8, 9]. (b) A reflection-positive, continuous covariance is the covariance of a Gaussian measure on $\mathcal{S}'(\mathbb{R}^4 \times M)$ by Minlos’ theorem [14]. The moments of that measure are

the candidate Schwinger functions. (c) The Osterwalder–Schrader reconstruction theorem applied to the measure μ produces the physical structure $(\mathcal{H}, \mathcal{A}, U, \Omega)$ [8, 10]. (d) The spectral decomposition of the internal Laplacian,

$$\Delta_M = \sum_l \lambda_l K_l,$$

decomposes the free field into independent modes of mass M_l . Each mode contributes a single-particle state of mass M_l with multiplicity d_l . The spectrum of H is therefore the mass tower $\{M_l\}$ with the stated multiplicities. The lowest mode is $M_0 = m_\delta$. $\square$

Theorem C.2 (The free sector is a standard quantum field theory). *The free-sector spectral datum $S = (\{\lambda_l\}, \{d_l\}, m_\delta)$ determines a unique standard-formulation quantum field theory. It is the free scalar field of mass m_δ on $\mathbb{R}^4 \times M$, with product metric. Its two-point function is the full spectral kernel. The correspondence preserves the scalar mass m_δ , the two-point function, the vacuum, and the clustering rate. The Kaluza–Klein decomposition of the free field matches the mass tower of the datum.*

Proof. The reconstruction of Theorem C.1 produces the free scalar field of mass m_δ on $\mathbb{R}^4 \times M$. The two-point function is the covariance by construction, so the correspondence preserves the two-point function, the vacuum, and the clustering rate. The scalar mass m_δ is preserved by Theorem C.1(d). The Kaluza–Klein decomposition matches the mode decomposition of the datum by the same item. Uniqueness holds within the Gaussian class, since a quasi-free field is determined by its two-point function. $\square$

This records the free-sector instance of the mapping relation stated in Axiom 5.11.

Remark C.3 (The Källén–Lehmann representation). The two-point function of the free sector carries the Källén–Lehmann representation [1, 2] with spectral measure

$$\rho(\mu^2) = \sum_l d_l \delta(\mu^2 - M_l^2),$$

the spectral measure of the mass tower. The atomic part of ρ is the single-particle spectrum of the theory by Theorem 7.18. The positivity of ρ is Theorem 7.4. The reconstruction of Theorem C.1 is the geometric realisation of this measure-theoretic content on the product $\mathbb{R}^4 \times M$. It is the carrier of the structural mapping recorded in Section 7.2.

Declarations

Conflict of interest. The author declares no competing interests.

Data availability. Data sharing is not applicable to this article as no datasets were generated or analysed during the current study.

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